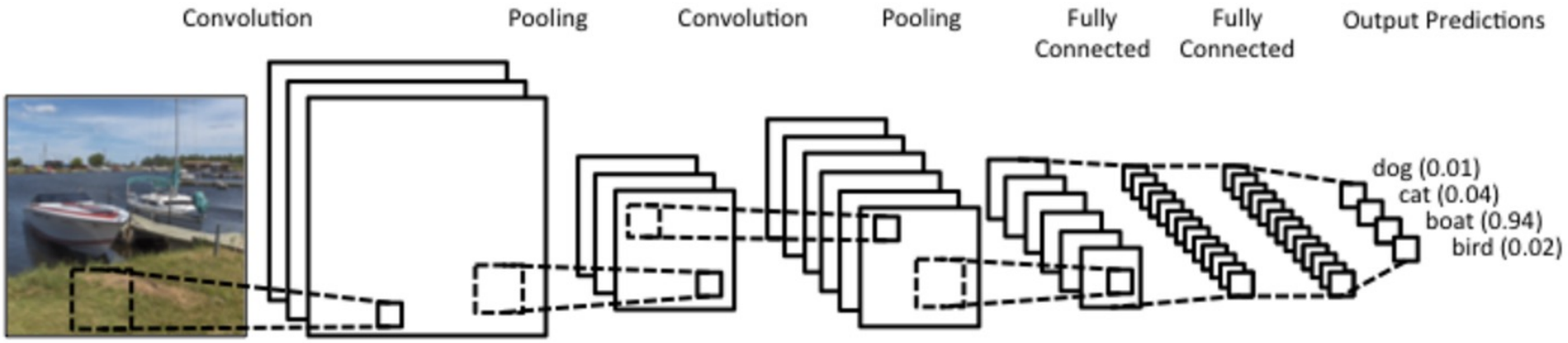
Object Detection

(Part I)

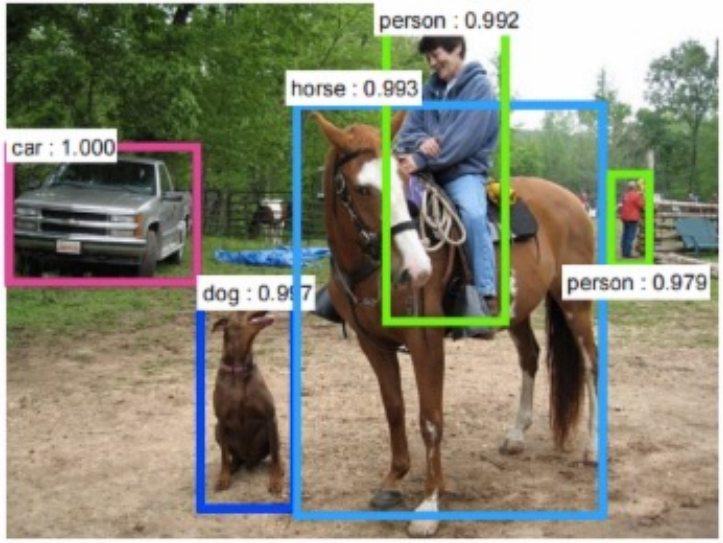
Many slides from Ross Girshick and Lana Lazebnik

From image classification to object detection

Image classification



Object detection



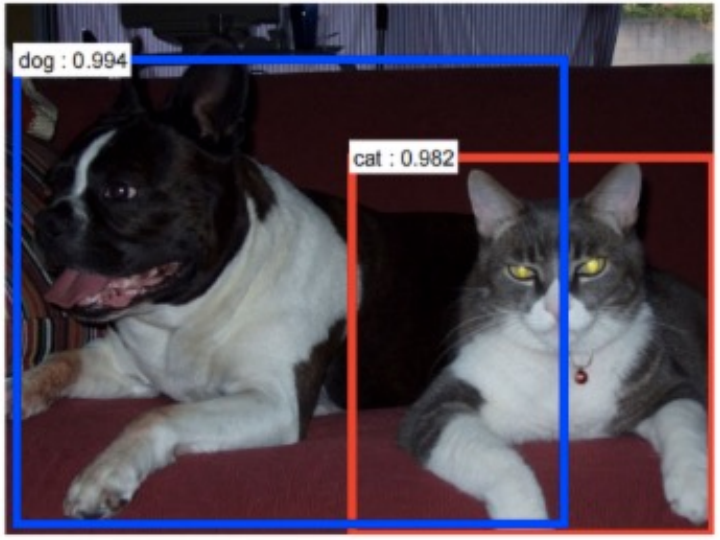


Image source

Deep learning based detectors



Detectron2 from Facebook AI Research https://github.com/facebookresearch/detectron2
Yolo Series from Ultralytics: https://docs.ultralytics.com/models/

Challenges in object detection

- Bounding box localization + classification
- Images may contain more than one class, multiple instances from the same class
- How do we evaluate the performance?

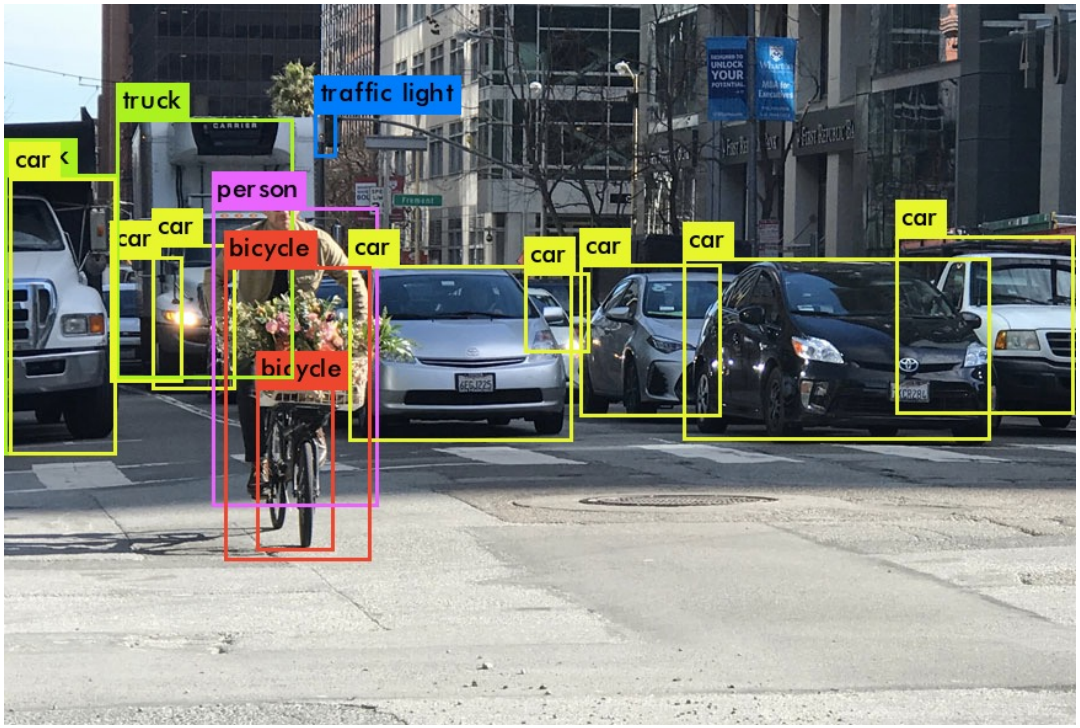
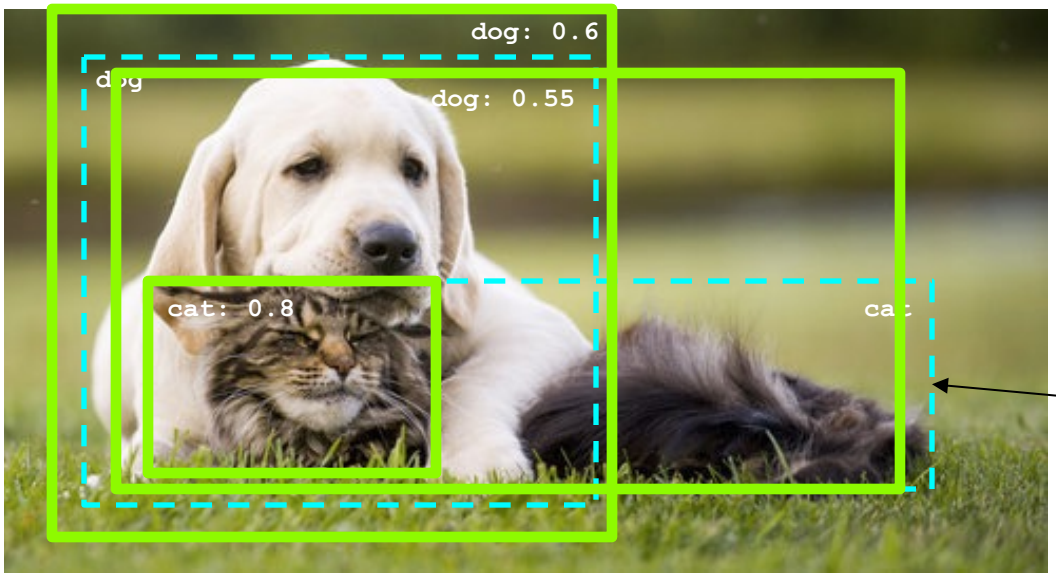


Image source

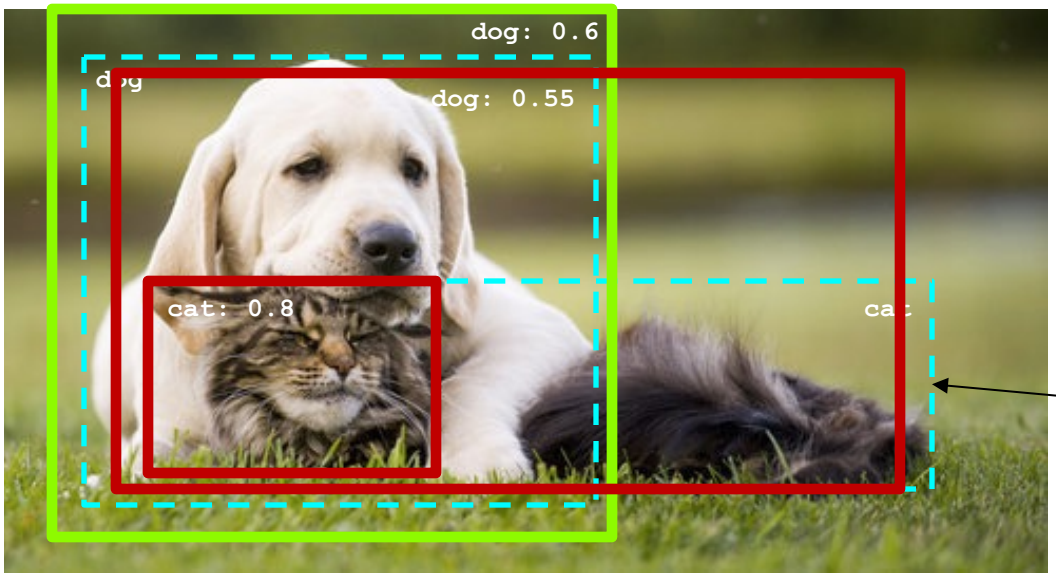
Object detection evaluation

- At test time, given an input image, predict bounding boxes, class labels, and confidence scores
- For each detection, determine whether it is a true or false positive using *intersection over union scores*
  - A valid detection w. confidence score ≥threshold (e.g., 0.5)
  - PASCAL criterion: Area(GT ∩ Det) / Area(GT ∪ Det) ≥0.5



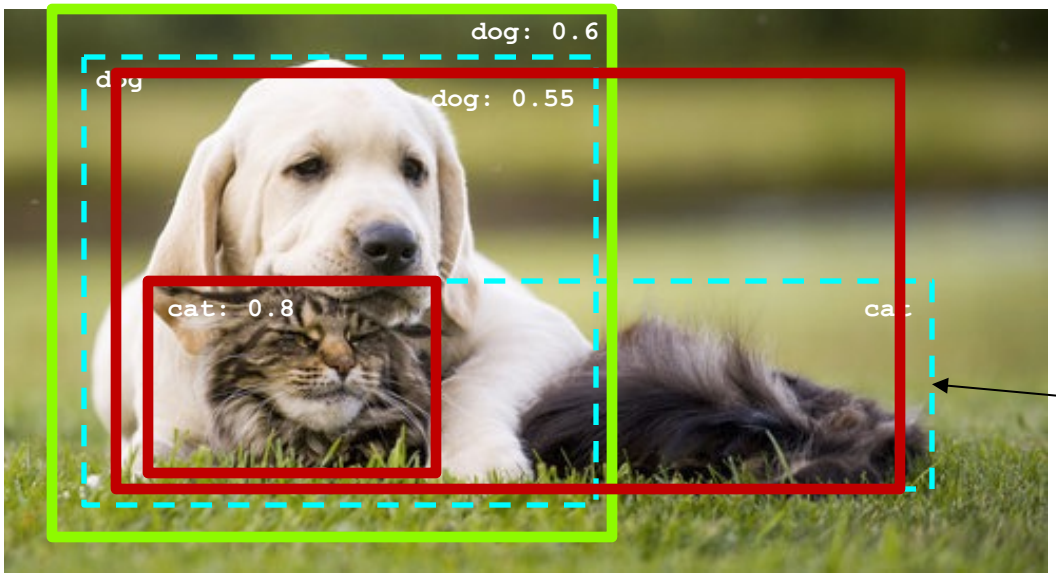
Object detection evaluation

- At test time, given an input image, predict bounding boxes, class labels, and confidence scores
- For each detection, determine whether it is a true or false positive using *intersection over union scores*
  - For multiple detections of the same ground truth box, *only one considered a true positive*



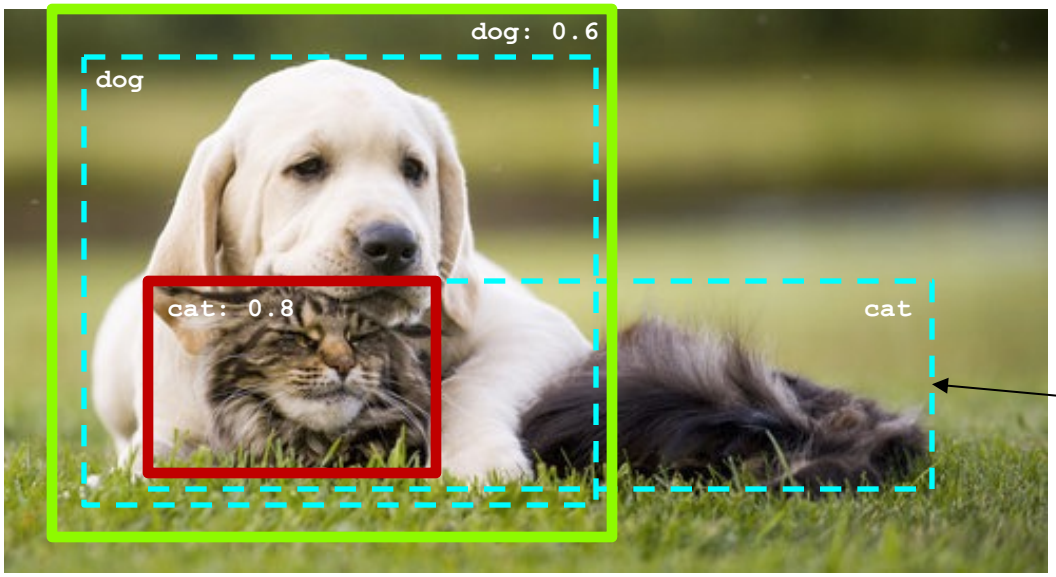
Object detection evaluation

- At test time, given an input image, predict bounding boxes, class labels, and confidence scores
- For each detection, determine if it is a true or false positive using IoU scores at varying confidence thresholds
  - Threshold = 0.5



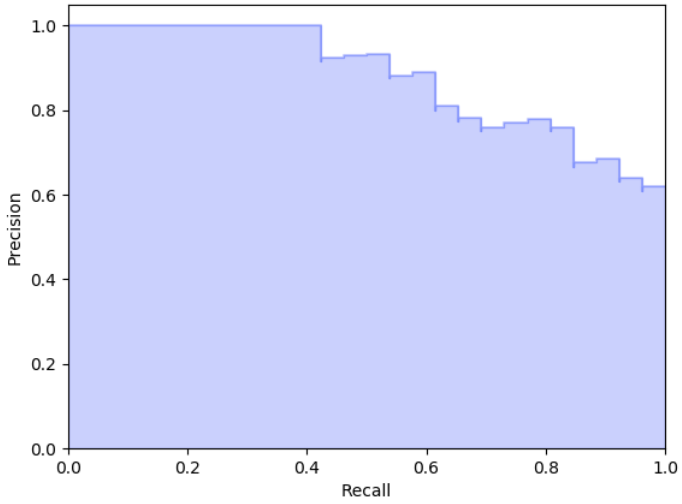
Object detection evaluation

- At test time, given an input image, predict bounding boxes, class labels, and confidence scores
- For each detection, determine if it is a true or false positive using IoU scores at varying confidence thresholds
  - Threshold = 0.6



Object detection evaluation

- At test time, given an input image, predict bounding boxes, class labels, and confidence scores
- For each detection, determine whether it is a true or false positive using IoU scores at varying thresholds
- For each class, plot Recall-Precision curve and compute Average Precision (area under the curve)
- Take mean of AP over classes to compute mAP



Precision:
true positive detections / total detections

Recall:
true positive detections / total positive test instances

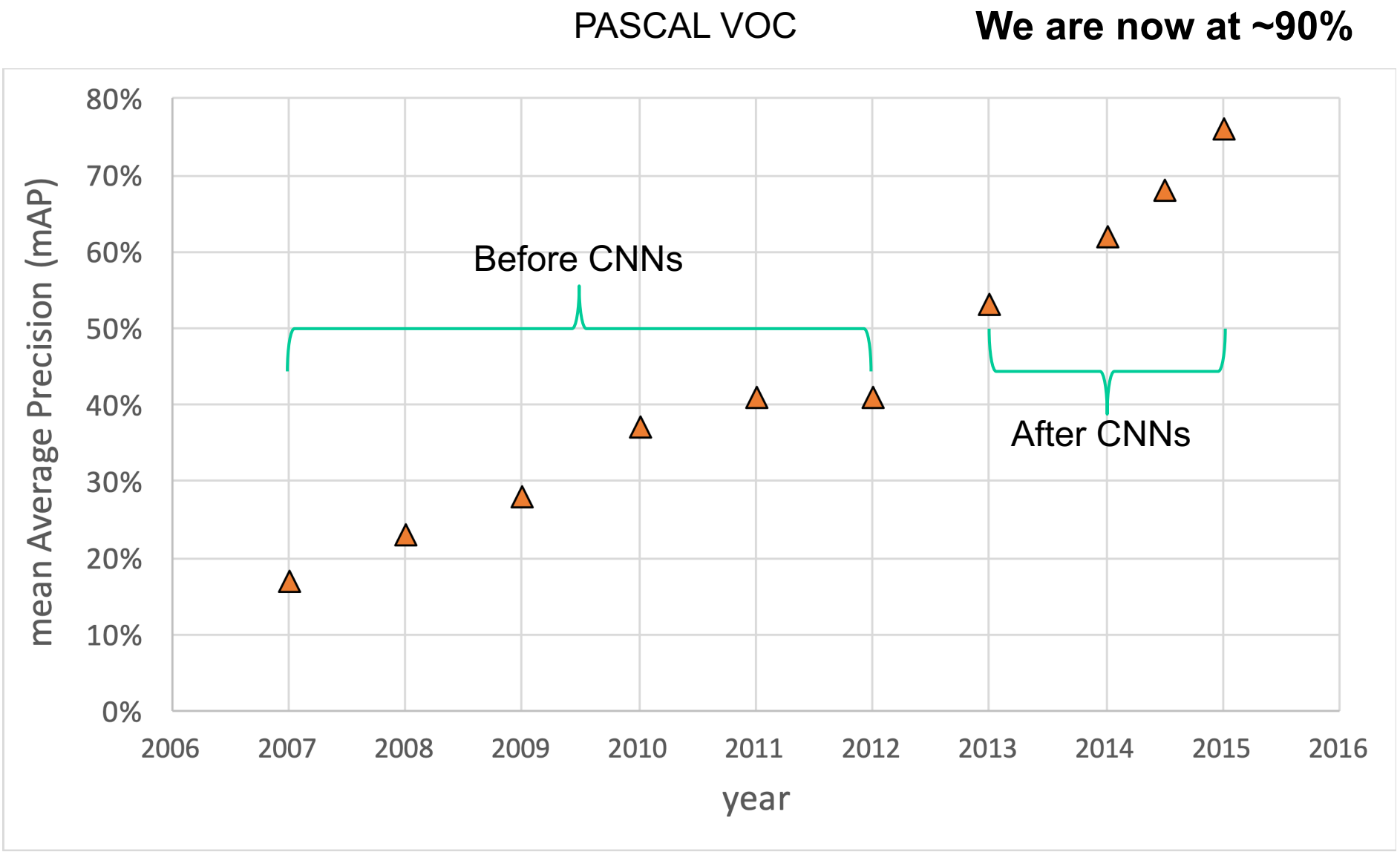
PASCAL VOC Challenge (‘05-’12)



- 20 challenge classes:
  - *Person*
  - *Animals:* bird, cat, cow, dog, horse, sheep
  - *Vehicles:* aeroplane, bicycle, boat, bus, car, motorbike, train
  - *Indoor:* bottle, chair, dining table, potted plant, sofa, tv/monitor
- Dataset size (by 2012): 11.5K training/validation images, 27K bounding boxes, 7K segmentations

http://host.robots.ox.ac.uk/pascal/VOC/

Progress on PASCAL detection



Newer benchmark: COCO

What is COCO?



COCO is a large-scale object detection, segmentation, and captioning dataset. COCO has several features:

- Object segmentation
- Recognition in context
- Superpixel stuff segmentation
- 330K images (>200K labeled)
- 1.5 million object instances
- 80 object categories
- 91 stuff categories
- 5 captions per image
- 250,000 people with keypoints

COCO
Common Objects in Context

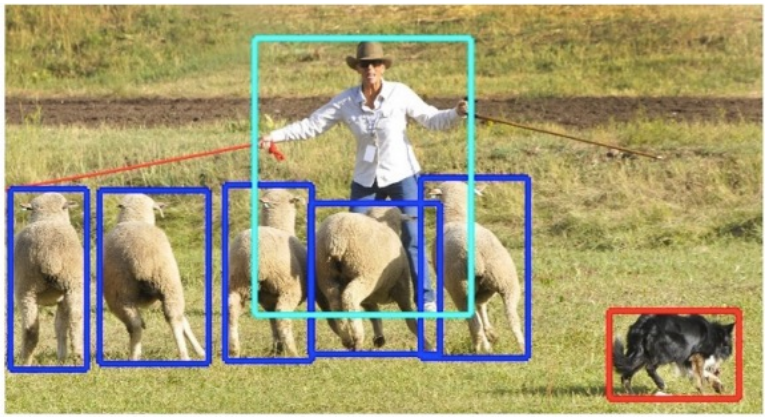


http://cocodataset.org

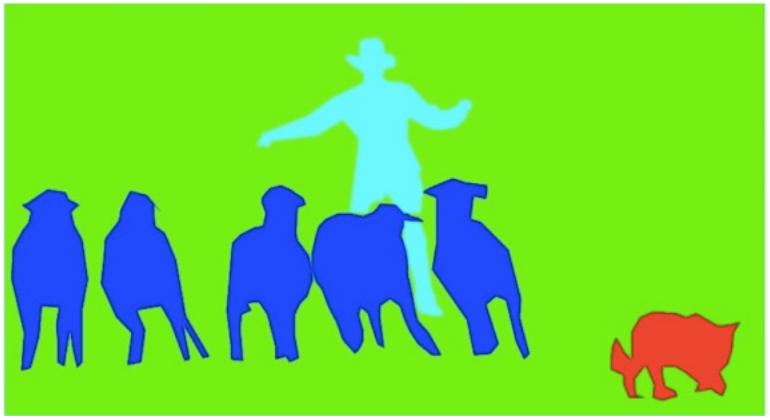
COCO dataset: Tasks



image classification



object detection



semantic segmentation



instance segmentation

- Also: keypoint prediction, captioning, question answering…

COCO detection metrics

```
Average Precision (AP):
   AP                  % AP at IoU=.50:.05:.95 (primary challenge metric)
   AP^IoU=.50          % AP at IoU=.50 (PASCAL VOC metric)
   AP^IoU=.75          % AP at IoU=.75 (strict metric)
AP Across Scales:
   AP^small            % AP for small objects: area < 32^2
   AP^medium           % AP for medium objects: 32^2 < area < 96^2
   AP^large            % AP for large objects: area > 96^2
Average Recall (AR):
   AR^max=1            % AR given 1 detection per image
   AR^max=10           % AR given 10 detections per image
   AR^max=100          % AR given 100 detections per image
AR Across Scales:
   AR^small            % AR for small objects: area < 32^2
   AR^medium           % AR for medium objects: 32^2 < area < 96^2
   AR^large            % AR for large objects: area > 96^2
```

- Leaderboard: http://cocodataset.org/#detection-leaderboard
- Official COCO challenges no longer include detection
  - Emphasis has shifted to instance segmentation and dense semantic segmentation

Conceptual approach: Sliding window

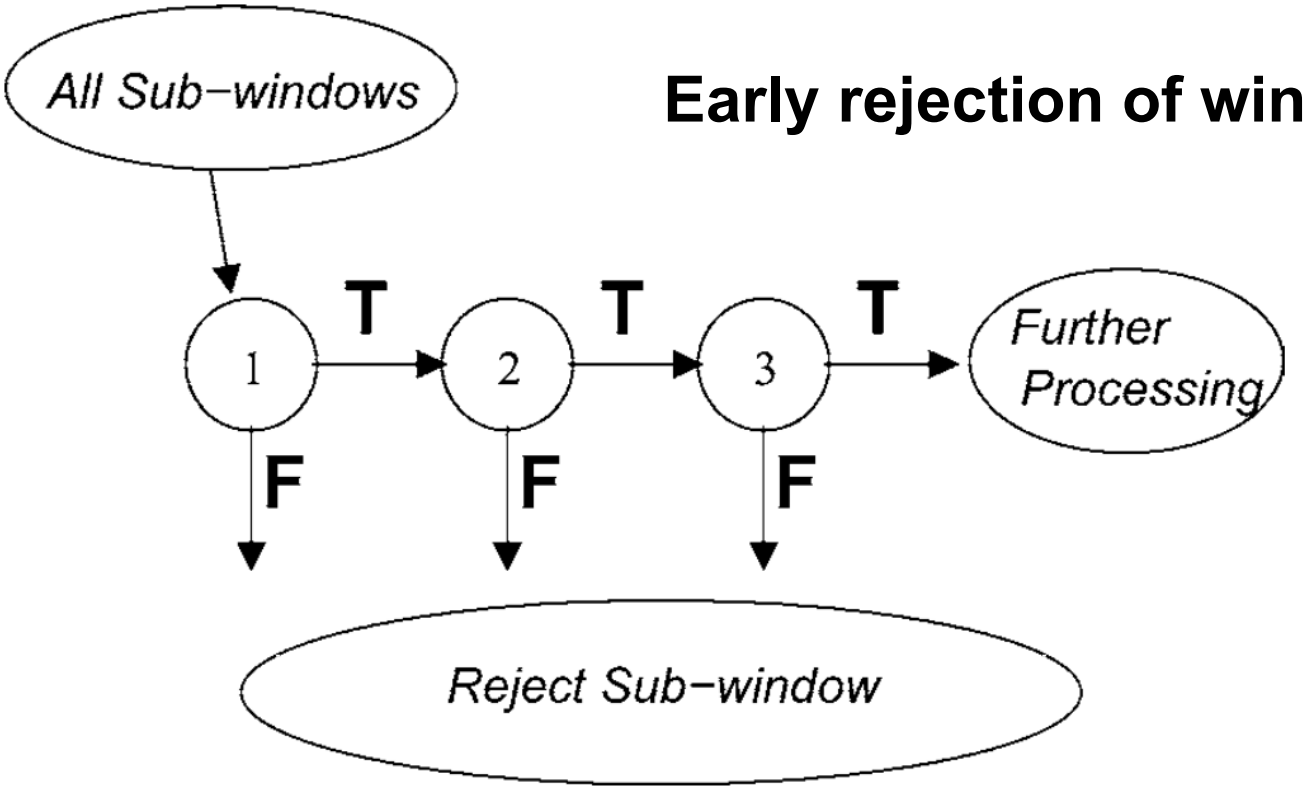


- Slide a window across the image and evaluate a classification model at each location
  - Millions of windows to evaluate; efficiency and low false positive rates are essential
  - Difficult to extend to many scales, aspect ratios

Conceptual approach: Sliding window

Viola–Jones face detector

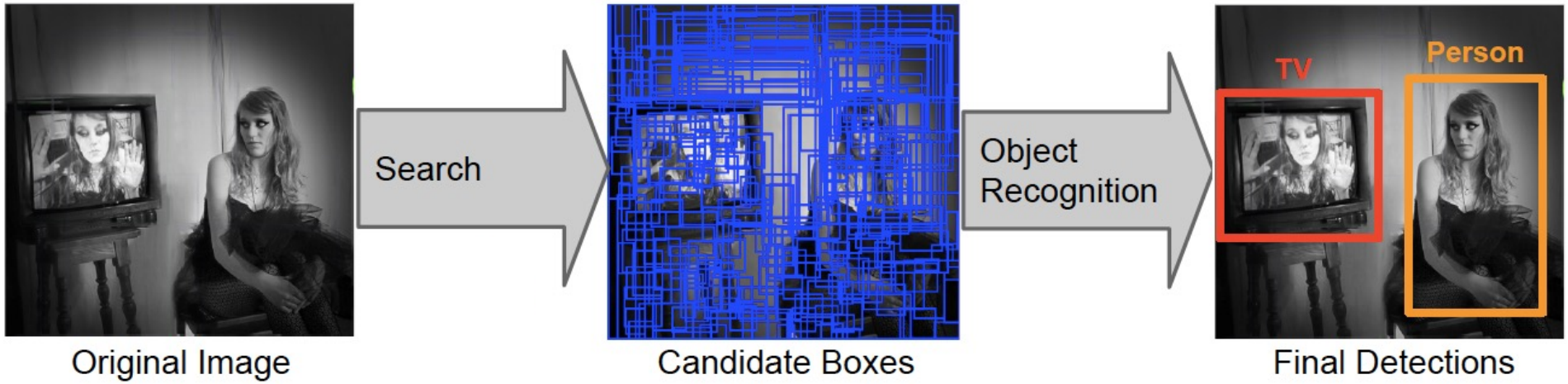
Key idea: Cascaded Boosting



Early rejection of windows

P. Viola and M. Jones.
Rapid Object Detection using a Boosted Cascade of Simple Features, CVPR 2001

Conceptual approach: Proposal-driven



- Generate and evaluate <1000 *region proposals*
  - Proposal leverages low-level *perceptual* cues
  - Proposal mechanism can be category-specific or category-independent, hand-crafted or trained
  - Classifier can be slower but more powerful
  - Can be considered as a two-stage cascaded approach

Selective search for detection

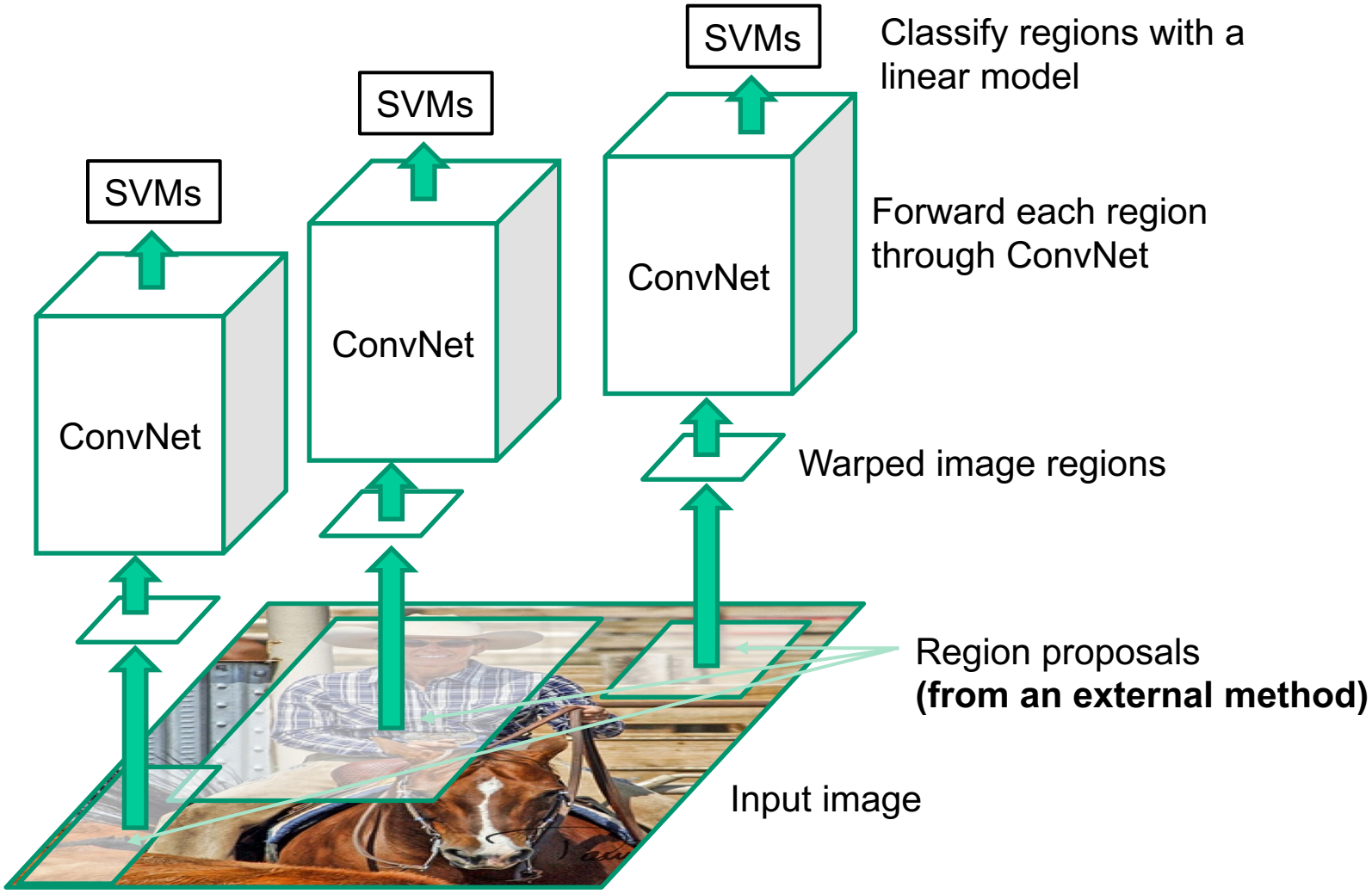
- Use hierarchical segmentation: start with small *superpixels* and merge based on diverse cues



J. Uijlings, K. van de Sande, T. Gevers, and A. Smeulders, Selective Search for Object Recognition, IJCV 2013

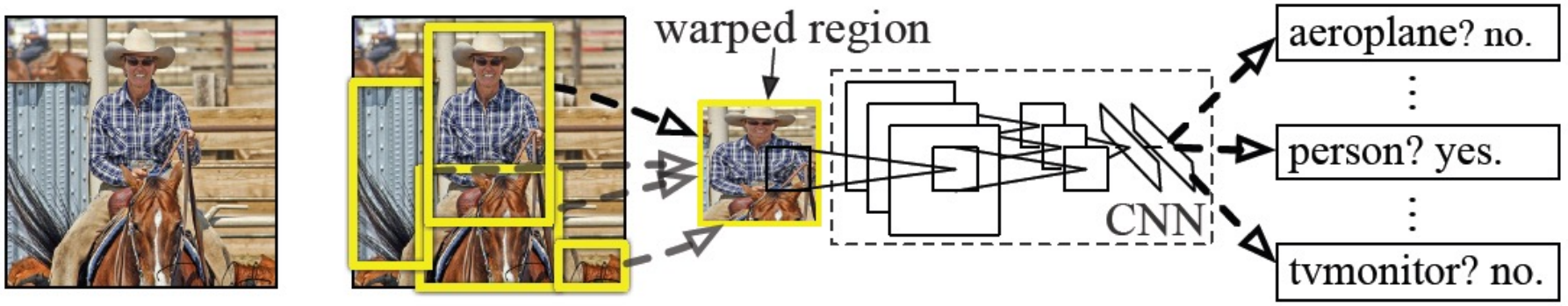
R-CNN: Region proposals + CNN features

Source: R. Girshick



R. Girshick, J. Donahue, T. Darrell, and J. Malik, **Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation**, CVPR 2014.

R-CNN details



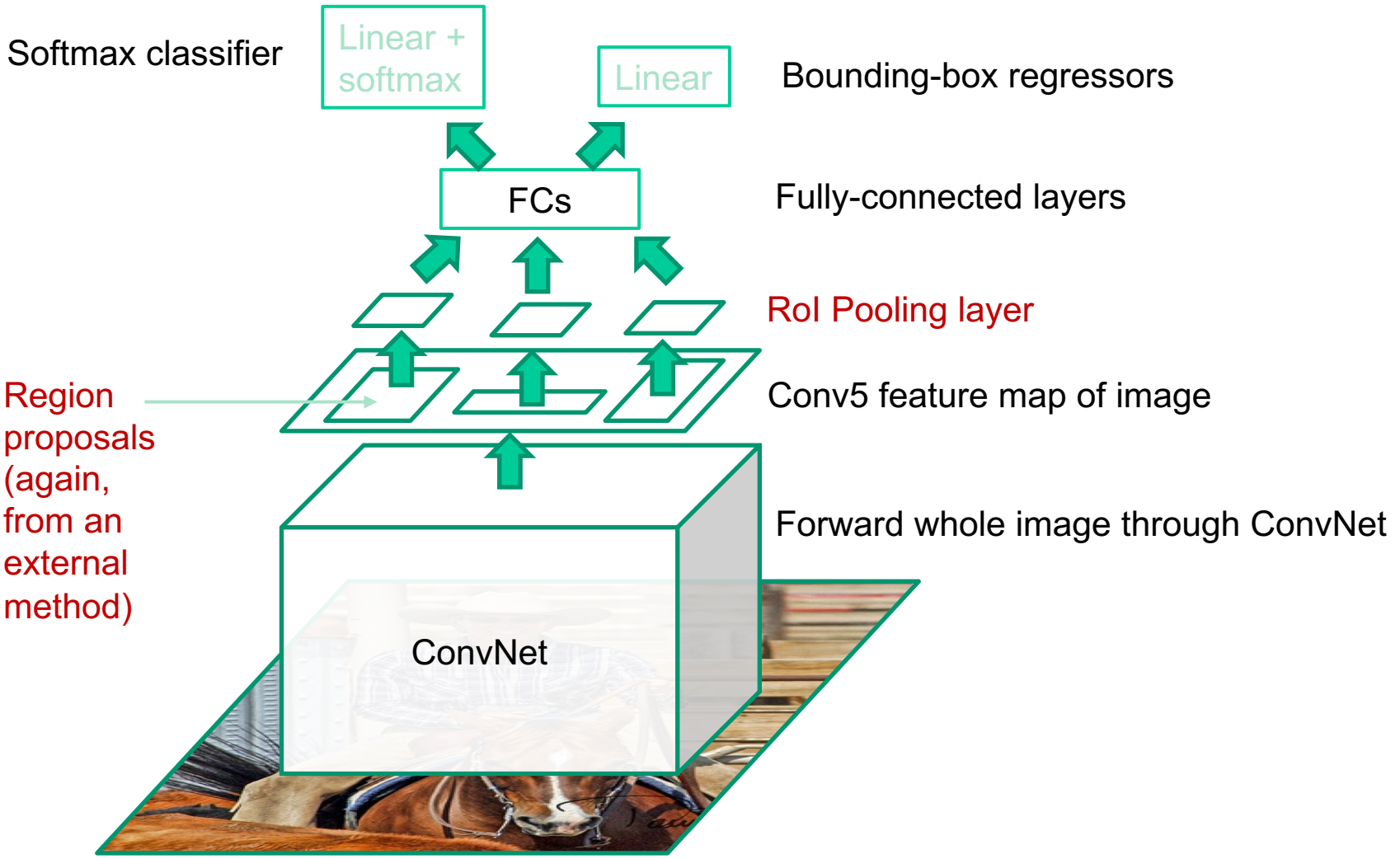
- **Regions**: ~2000 Selective Search proposals
- **Network**: AlexNet *pre-trained* on ImageNet (1000 classes), *fine-tuned* on PASCAL (21 classes)
- **Final detector**: warp proposal regions, extract fc7 network activations (4096 dimensions), classify with linear SVM
- **Bounding box regression** to refine box locations
- **Performance:** mAP of **53.7%** on PASCAL 2010 (vs. **35.1%** for Selective Search and **33.4%** for Deformable Part Models)

R-CNN pros and cons

- **Pros**
  - Accurate (against hand-crafted methods)!
  - Any deep architecture can be “plugged in”

- **Cons**
  - Not a single end-to-end system
    - Fine-tune network with softmax classifier (log loss)
    - Train post-hoc linear SVMs (hinge loss)
    - Train post-hoc bounding-box regressions (least squares)
  - Training is slow (84h), takes a lot of disk space
    - 2000 CNN passes per image
  - Inference (detection) is slow (47s / image with VGG16)

Fast R-CNN

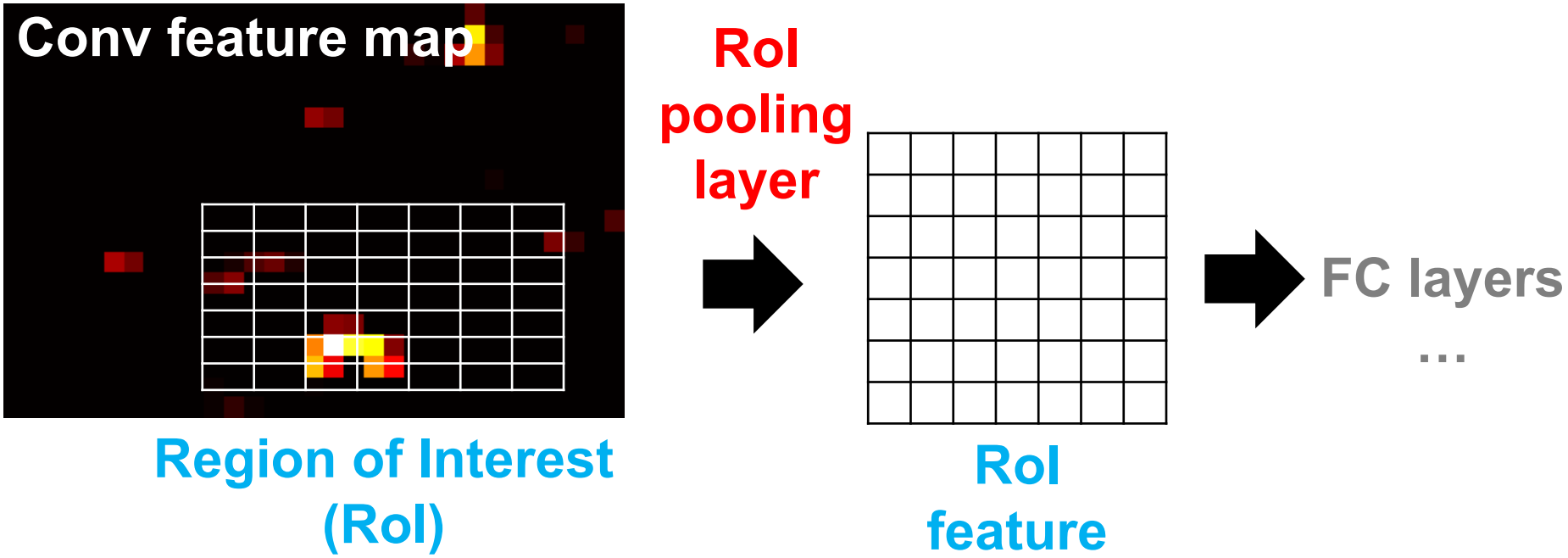


Source: R. Girshick

R. Girshick, Fast R-CNN, ICCV 2015

RoI pooling

- "Crop and resample" a fixed-size feature representing a region of interest out of the outputs of the last conv layer
  - Use nearest-neighbor interpolation of coordinates, max pooling



Source: R. Girshick, K. He

RoI (max) pooling illustration

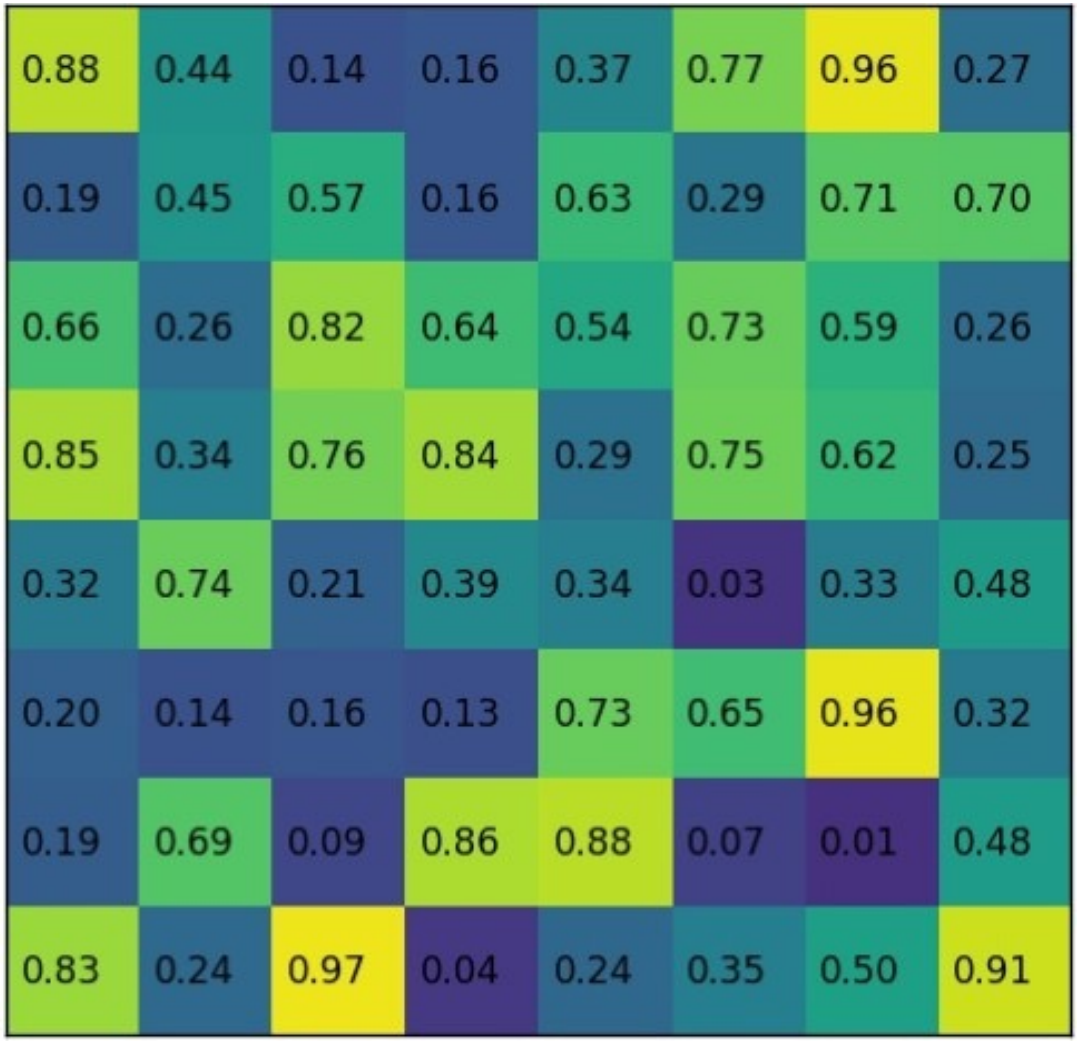
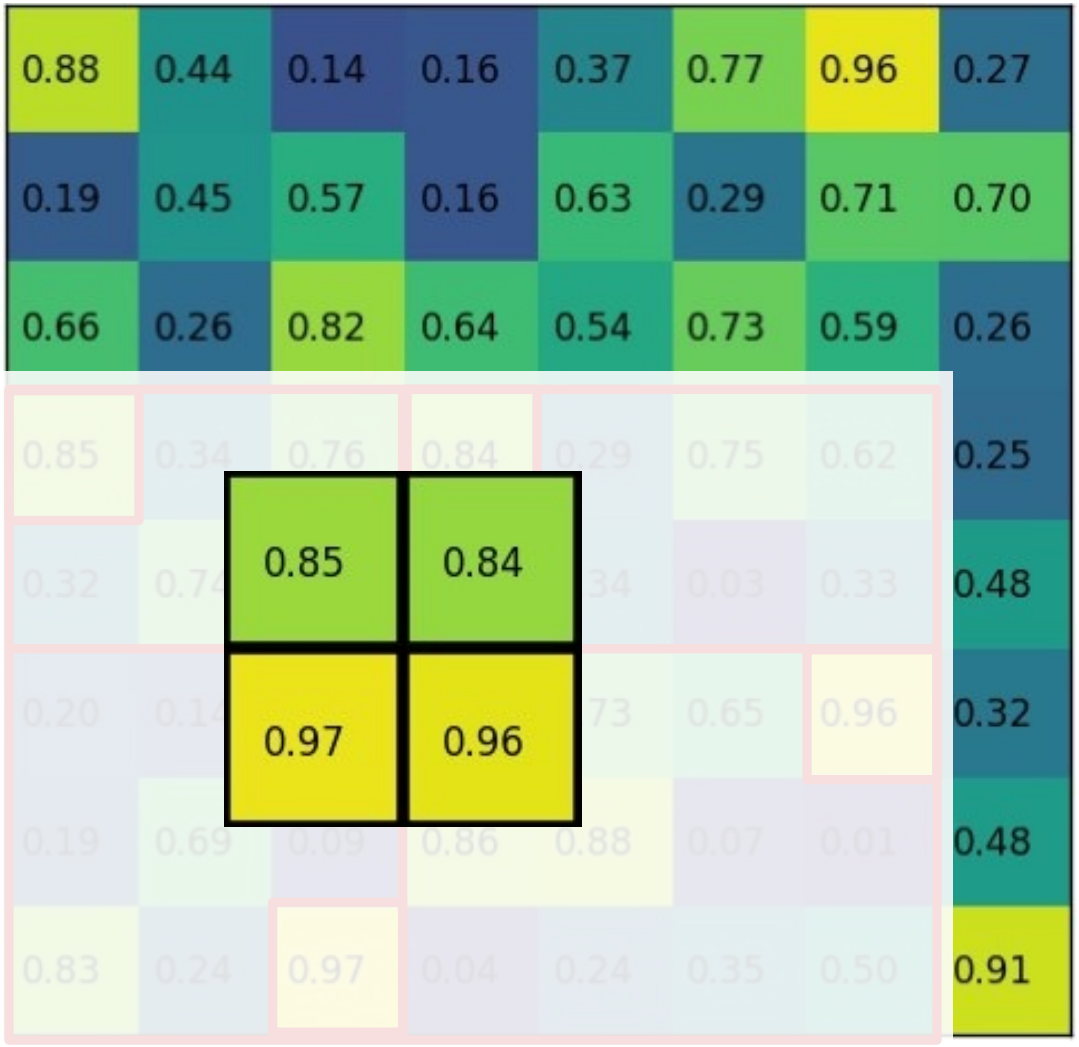


Image source

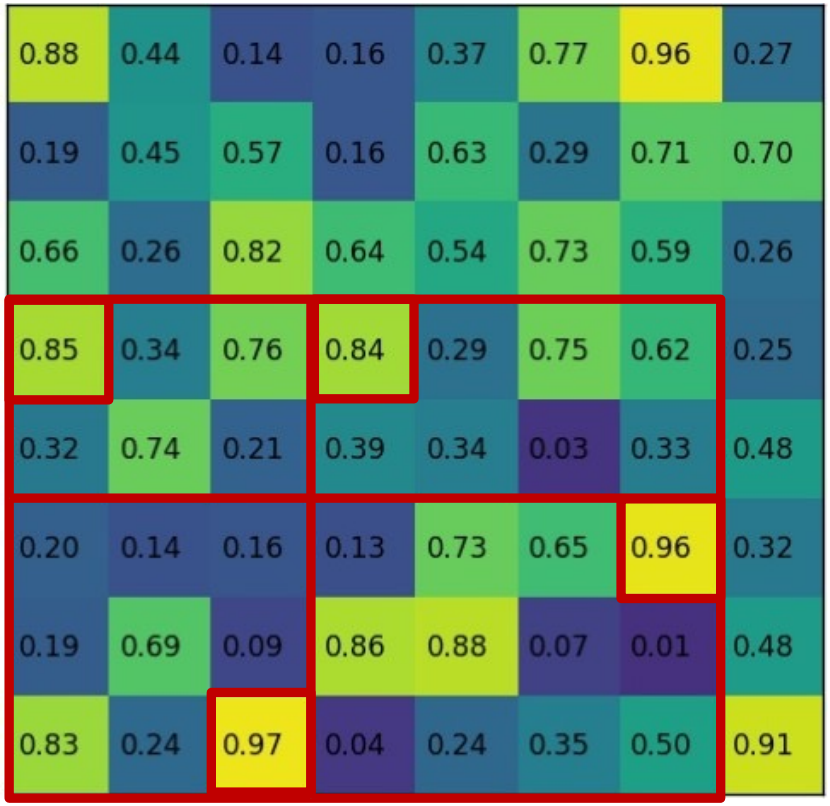
RoI (max) pooling illustration



RoI (max) pooling illustration



RoI (max) pooling: backprop



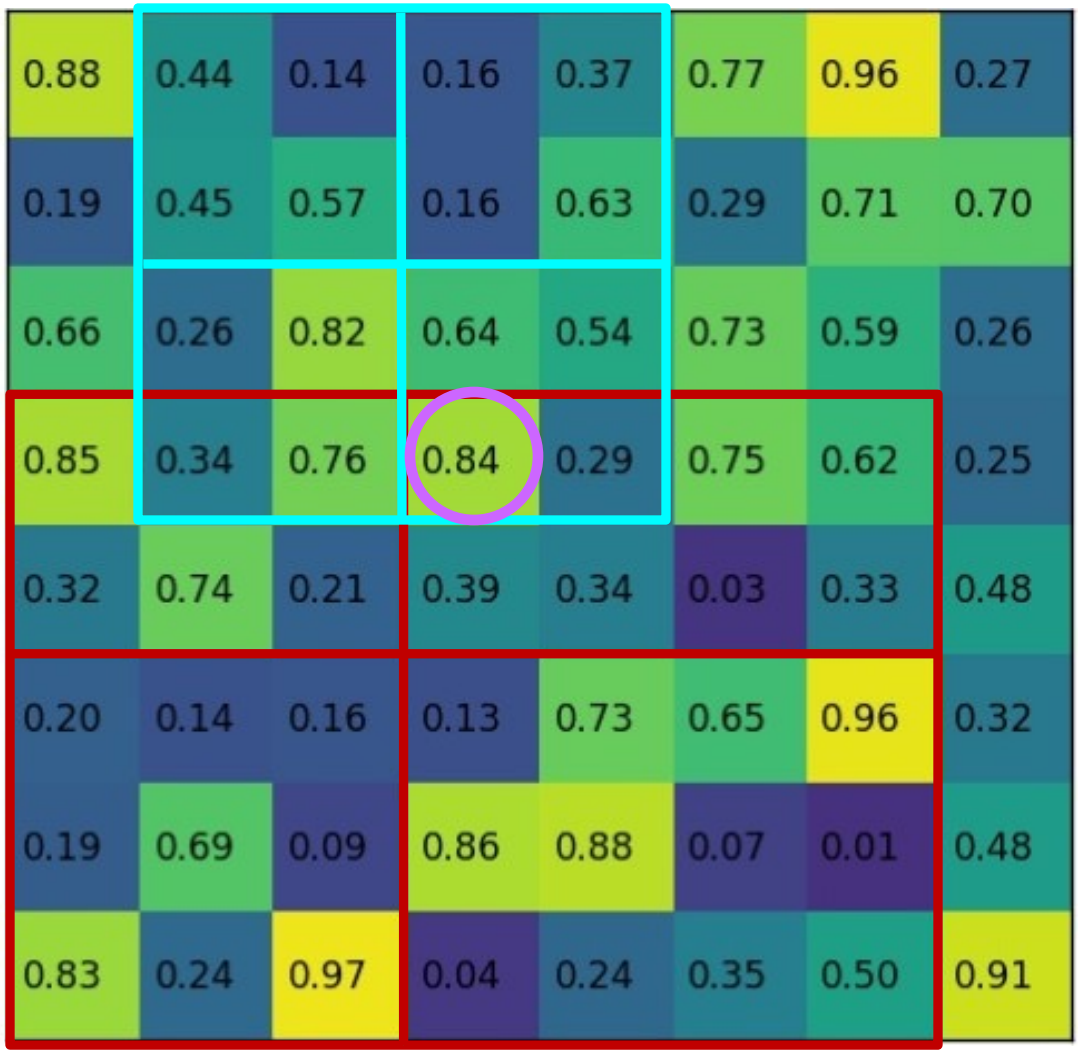
$$\frac{\partial e}{\partial x_i} = \sum_r \sum_j \frac{\partial e}{\partial z_{rj}} \frac{\partial z_{rj}}{\partial x_i} = \sum_r \sum_j \mathbb{I}[i = i^*(r,j)] \frac{\partial e}{\partial z_{rj}}$$

Over regions r, RoI indices j

1 if r, j "pooled" input i; 0 o/w

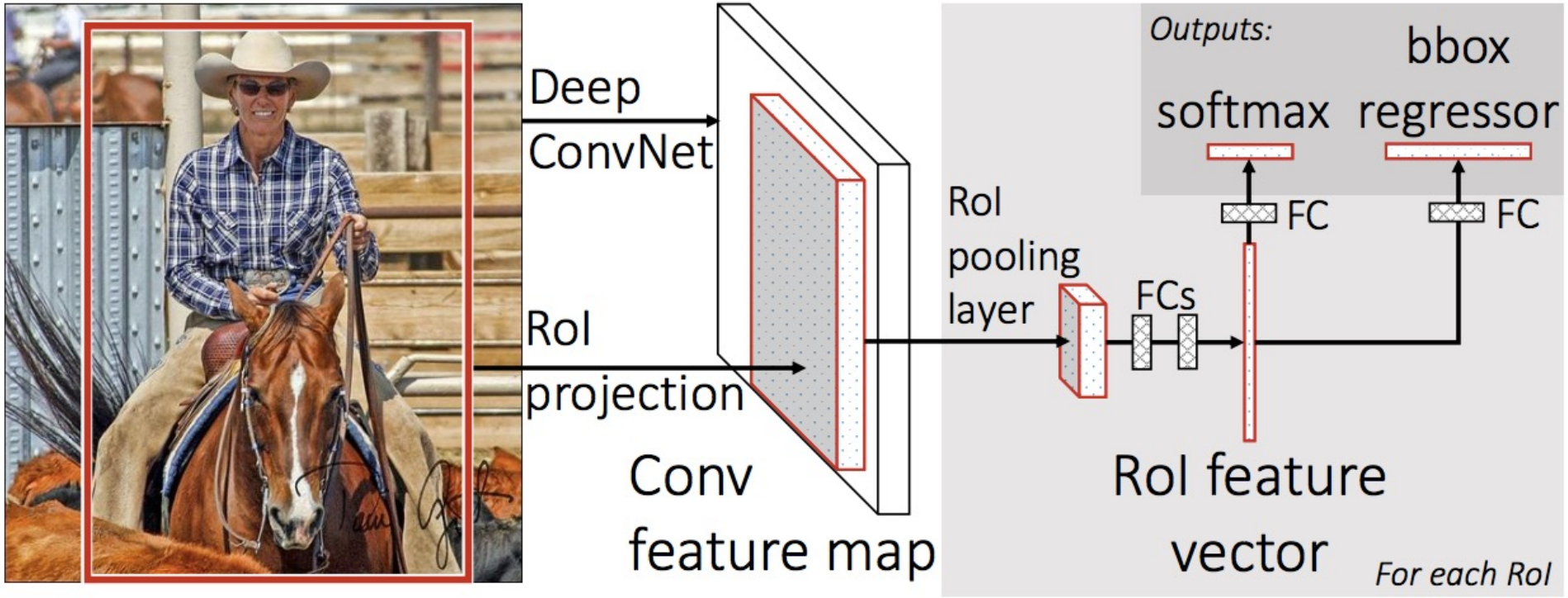
Source: Ross Girshick

RoI (max) pooling: backprop



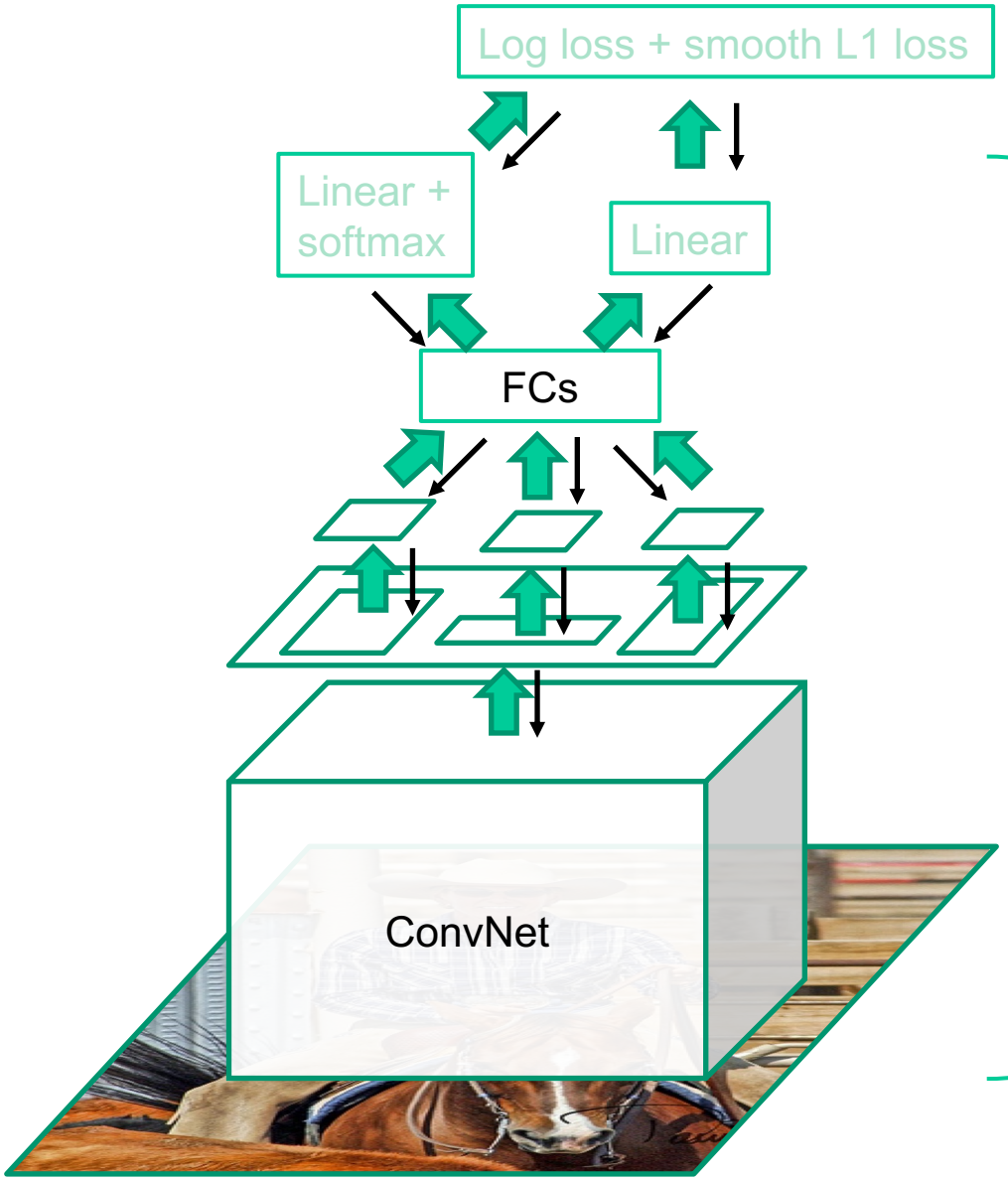
Prediction

- For each RoI, network predicts probabilities for C+1 classes (class 0 is background) and four bounding box offsets for C classes



R. Girshick, Fast R-CNN, ICCV 2015

Fast R-CNN training



Multi-task loss

Trainable

Source: R. Girshick

R. Girshick, Fast R-CNN, ICCV 2015

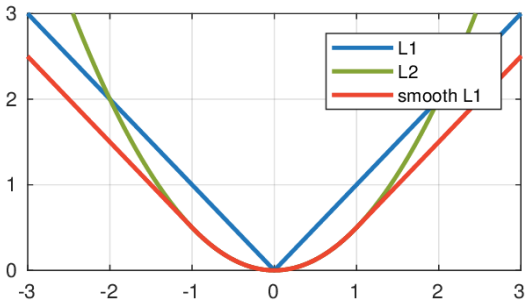
Multi-task loss

- Loss for ground truth class y, predicted class probabilities $P(y)$, ground truth box b, and predicted box $\hat{b}$:

$$L(y, P, b, \hat{b}) = \underbrace{-\log P(y)}_{\text{(softmax) cross entropy loss}} + \lambda \mathbb{I}[y \geq 1] \underbrace{L_{\text{reg}}(b, \hat{b})}_{\text{regression loss}}$$

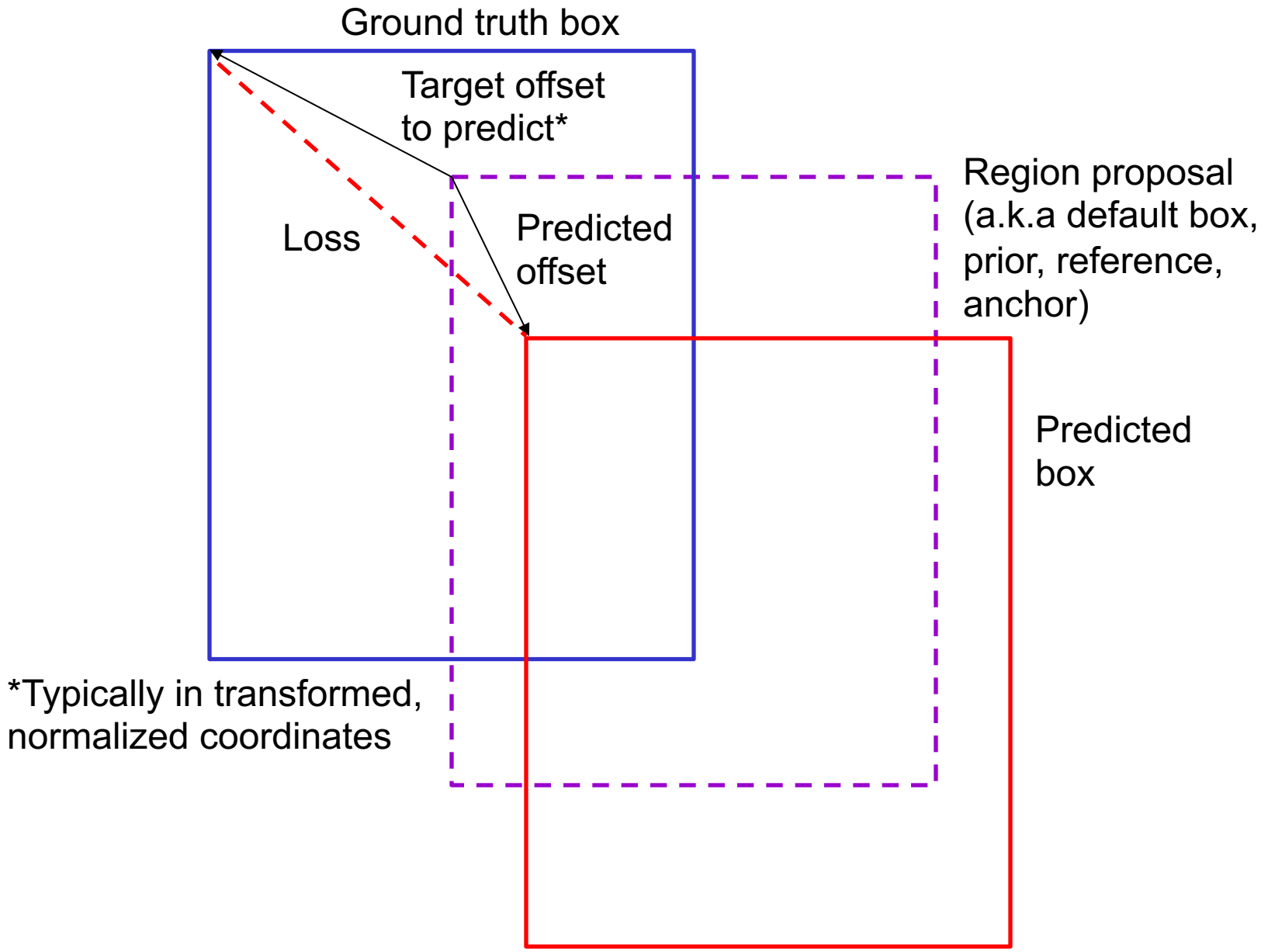
- Regression loss: *smooth L1 loss (a.k.a. Huber loss)* on top of offsets relative to proposals (*L1* will work equally well)

$$L_{\text{reg}}(b, \hat{b}) = \sum_{i=\{x,y,w,h\}} \text{smooth}_{L_1}(b_i - \hat{b}_i)$$



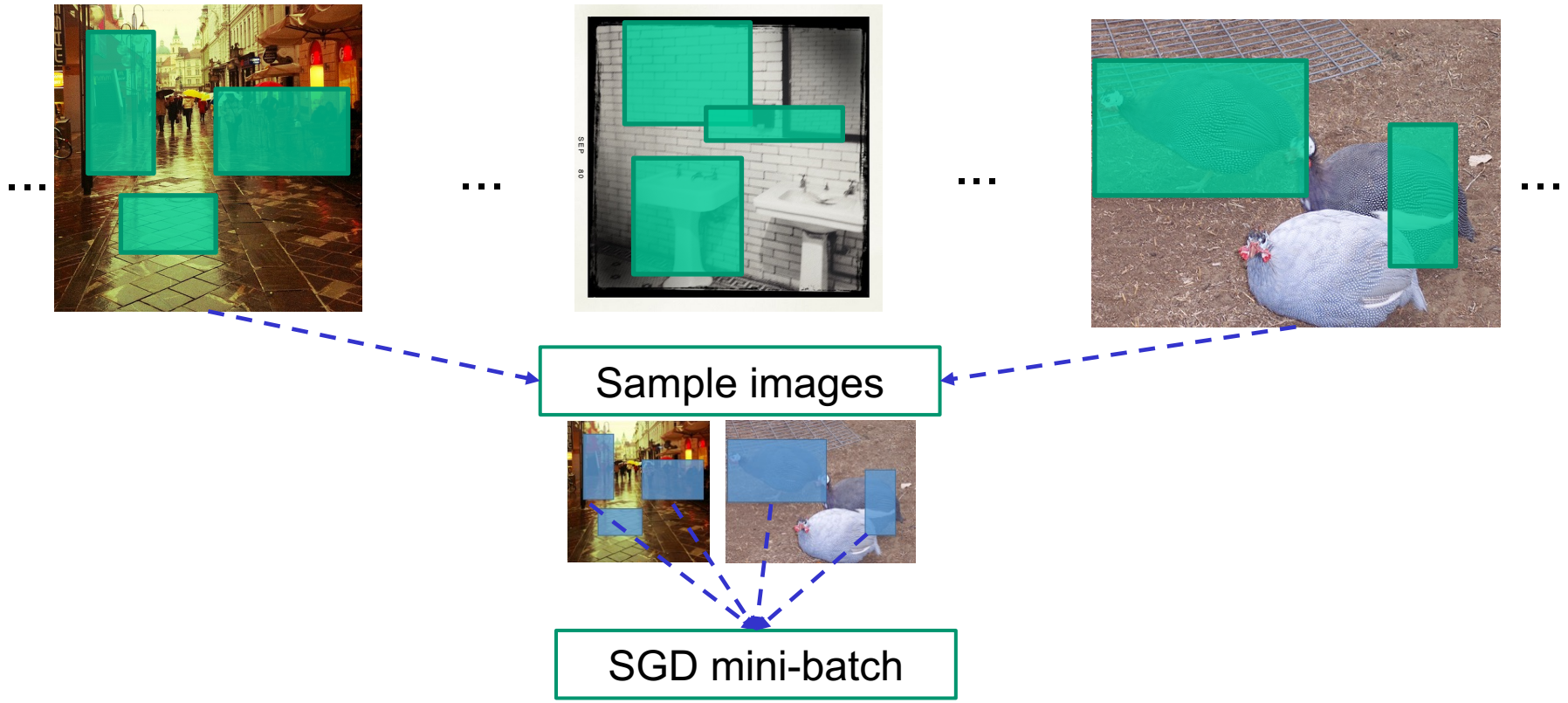
$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5x^2 & \text{if } |x| < 1 \\ |x| - 0.5 & \text{otherwise} \end{cases}$$

Bounding box regression



Mini-batch sampling

- Sample a few images (e.g., 2)
- Sample many regions from each image (e.g.,128)
  - Positive object regions are always included



Source: R. Girshick, K. He

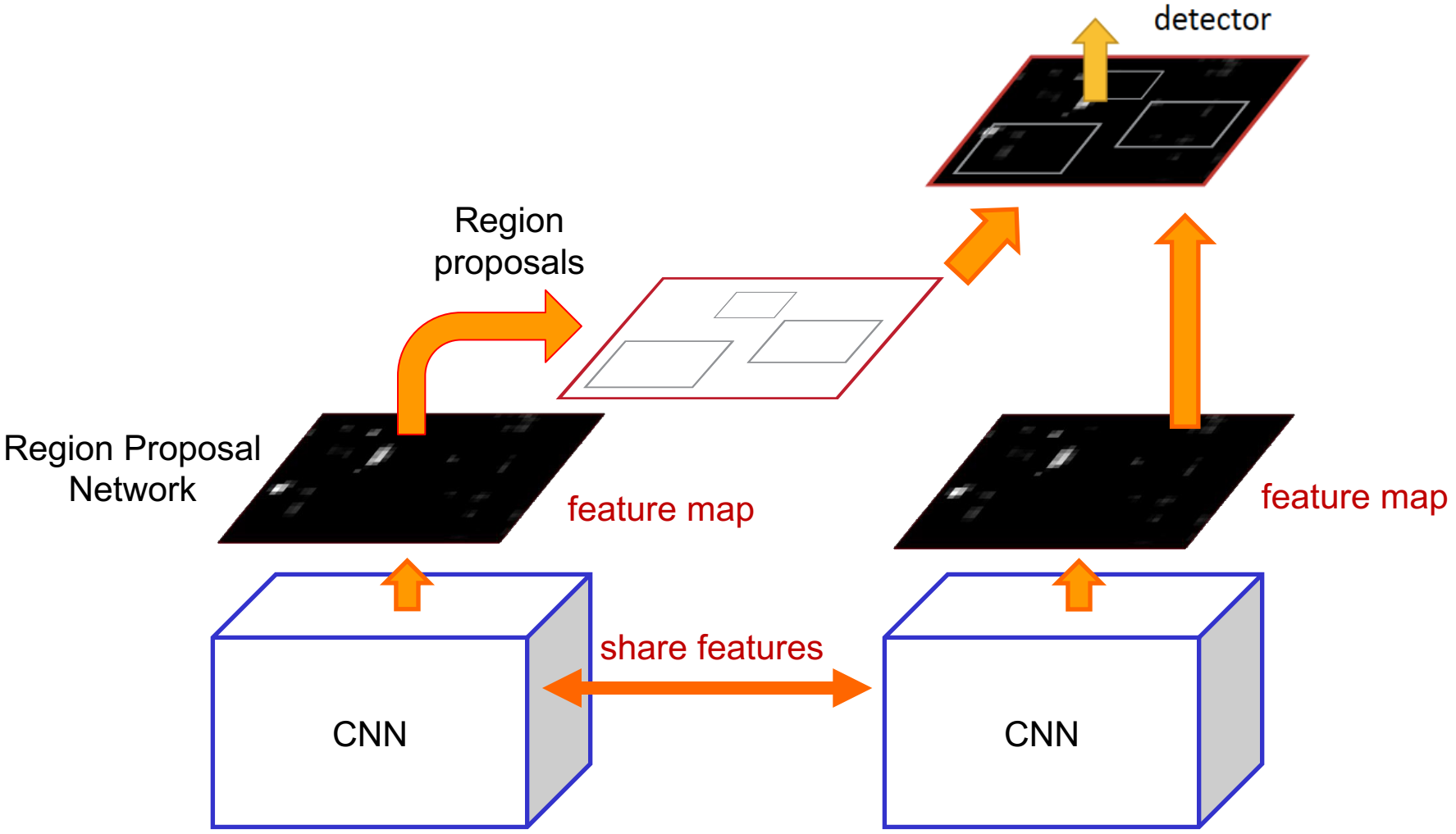
Fast R-CNN results

	Fast R-CNN	R-CNN
Train time (h)	**9.5**	84
- Speedup	**8.8x**	1x
Test time / image	**0.32s**	47.0s
Test speedup	**146x**	1x
mAP	**66.9%**	66.0%

Timings exclude object proposal time, which is equal for all methods. All methods use VGG16 from Simonyan and Zisserman.

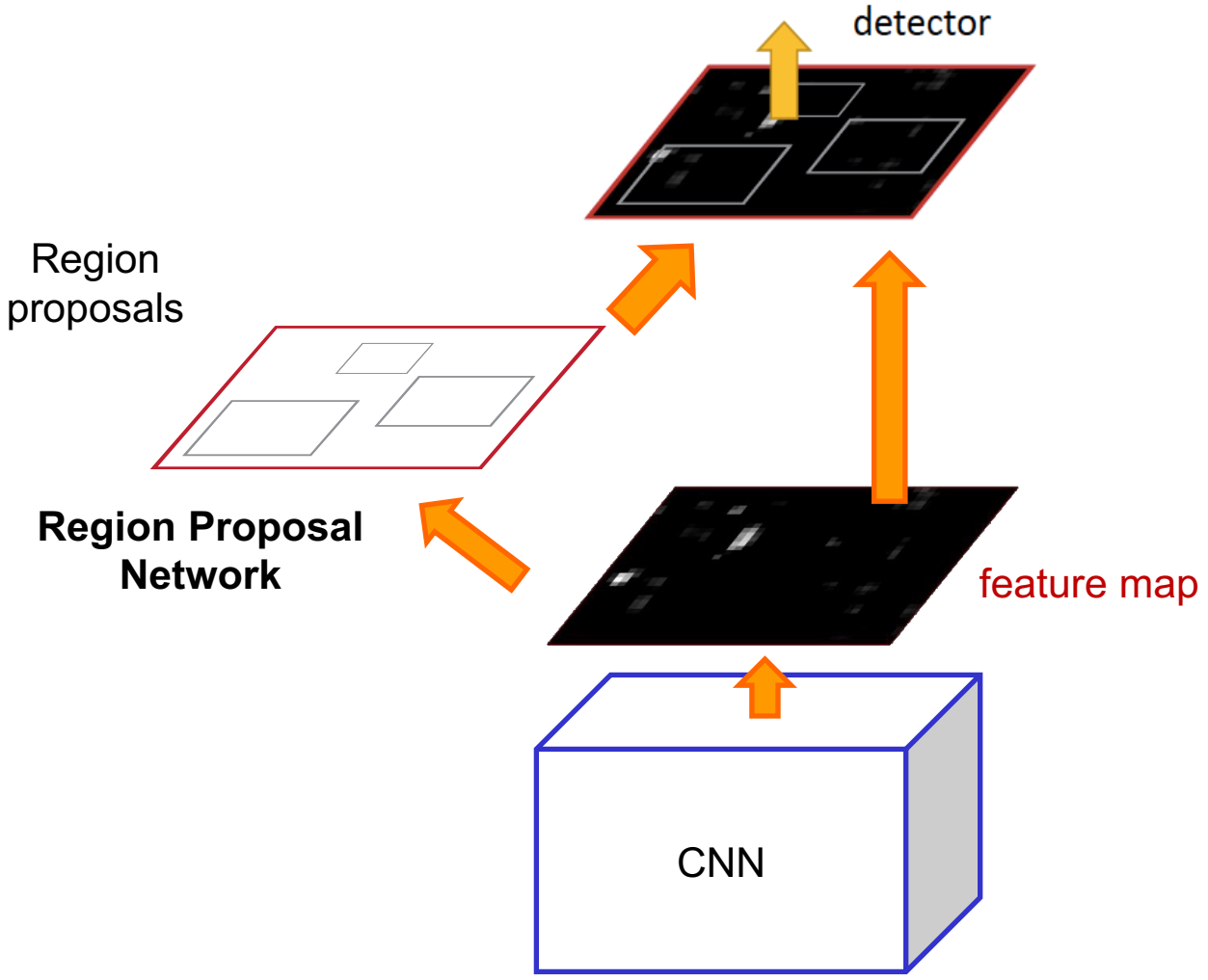
Source: R. Girshick

Faster R-CNN



S. Ren, K. He, R. Girshick, and J. Sun, Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks, NIPS 2015

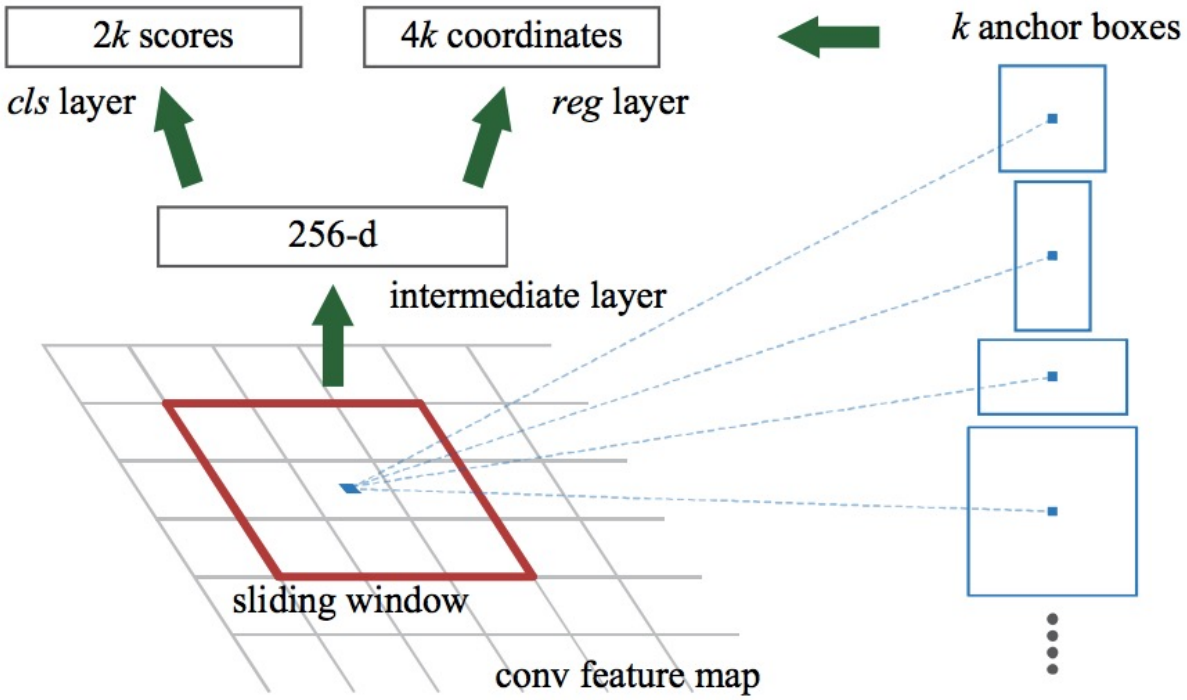
Faster R-CNN



S. Ren, K. He, R. Girshick, and J. Sun, Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks, NIPS 2015

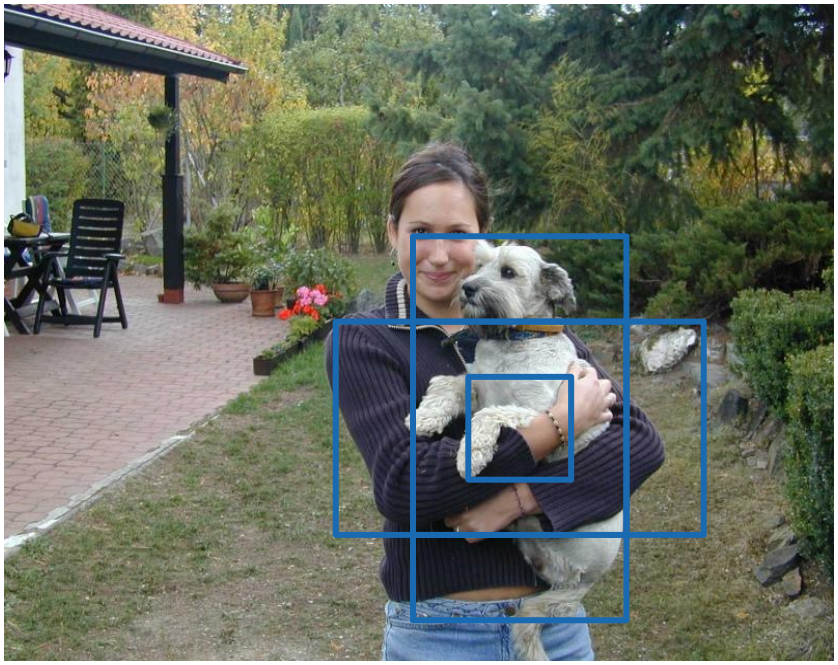
Region proposal network (RPN)

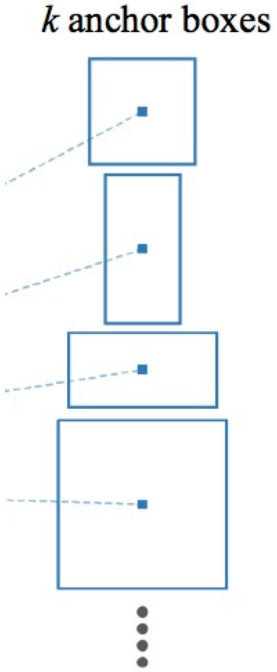
- Slide a small window (3x3) over the conv5 layer
  - Predict object/no object
  - Regress bounding box coordinates with reference to *anchors* (across scales and aspect ratios)
  - Anchors are carefully designed (local) boxes to cover objects at different scales and aspect ratios



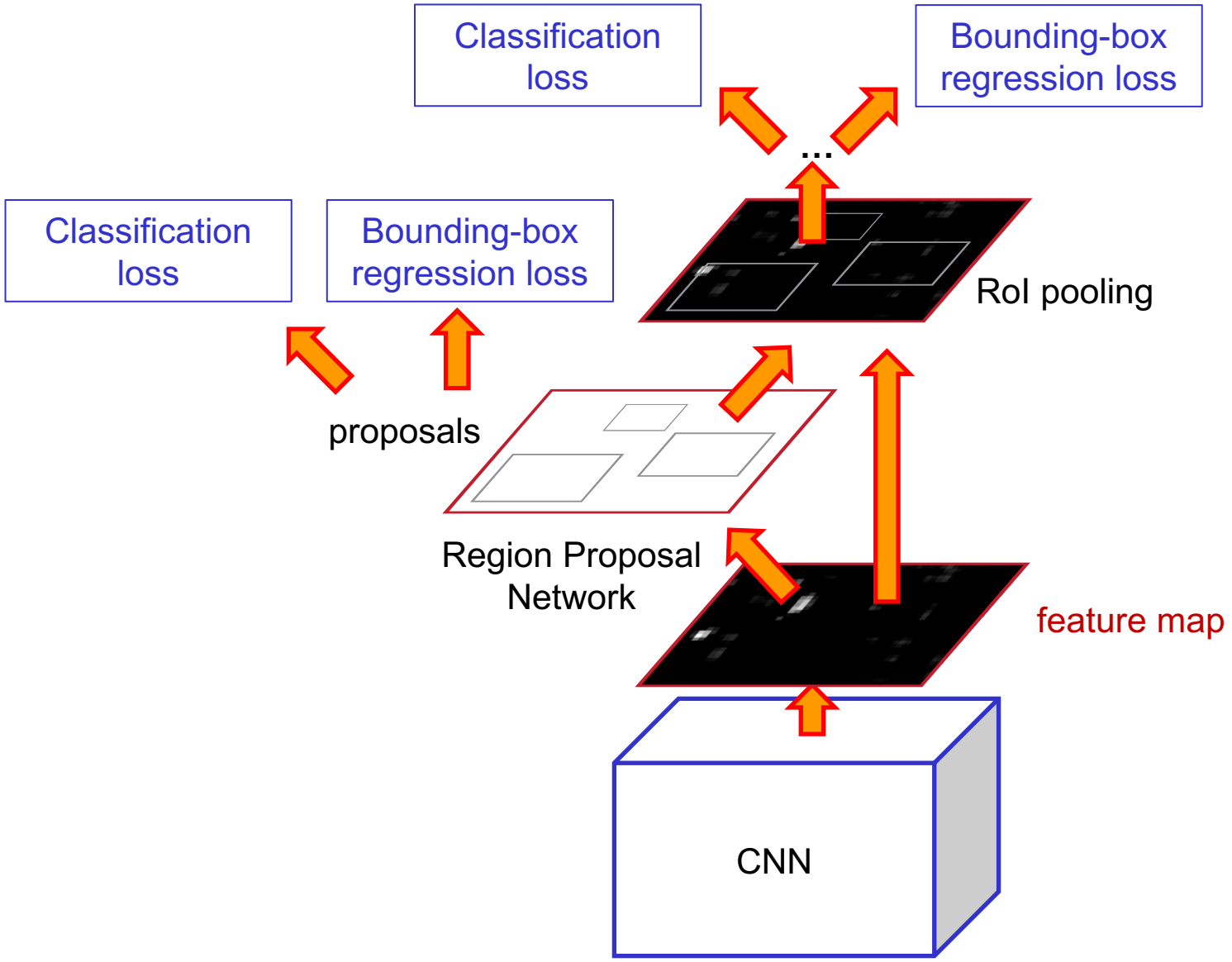
What are anchors, really?

- Local windows of different scales / aspect ratios
- Centered at every local spot
- Enumerating possible object boxes





One network, four losses



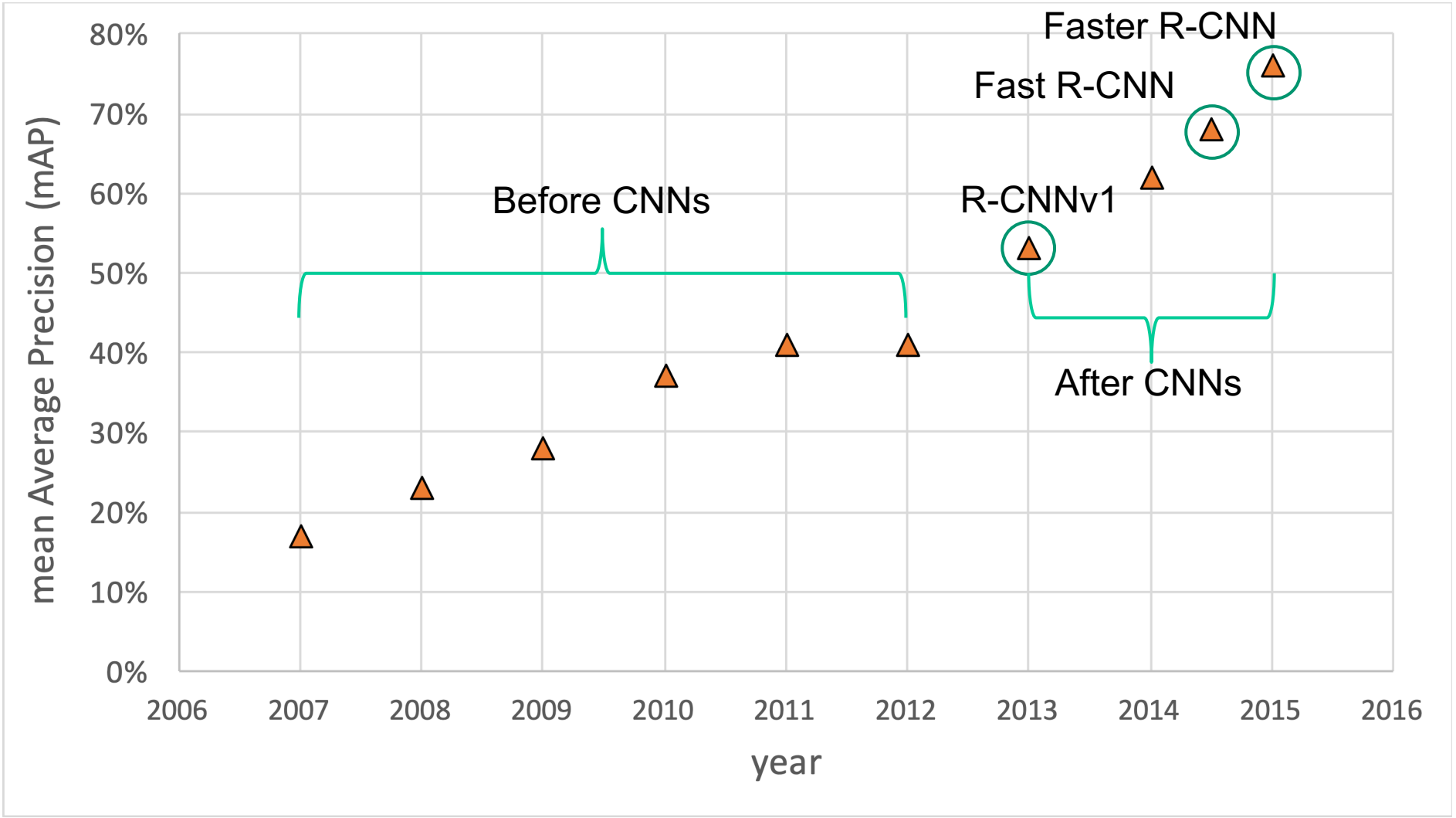
Source: R. Girshick, K. He

Faster R-CNN results

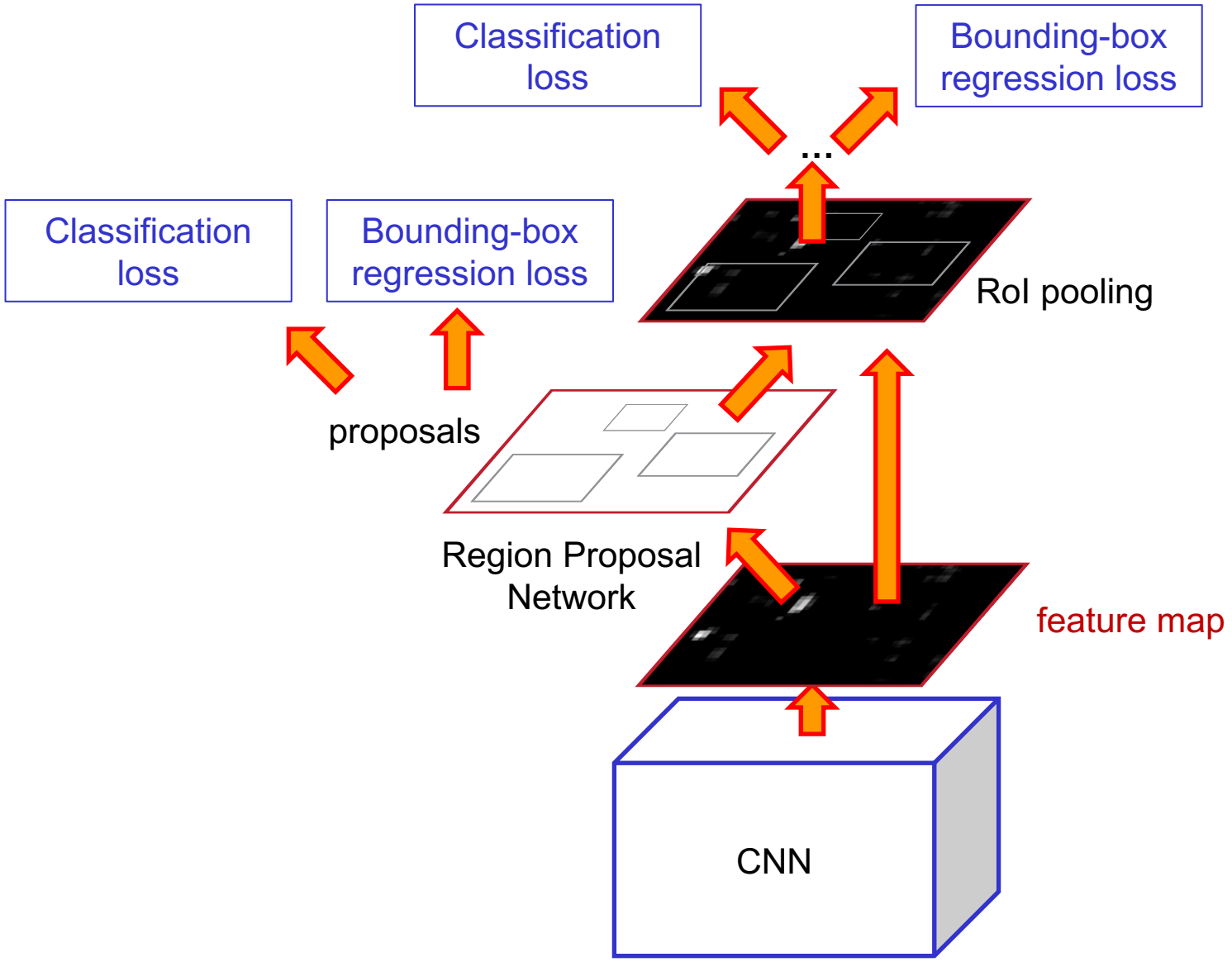
system	time	07 data	07+12 data
R-CNN	~50s	66.0	-
Fast R-CNN	~2s	66.9	70.0
Faster R-CNN	198ms	**69.9**	**73.2**

detection mAP on PASCAL VOC 2007, with VGG-16 pre-trained on ImageNet

Object detection progress

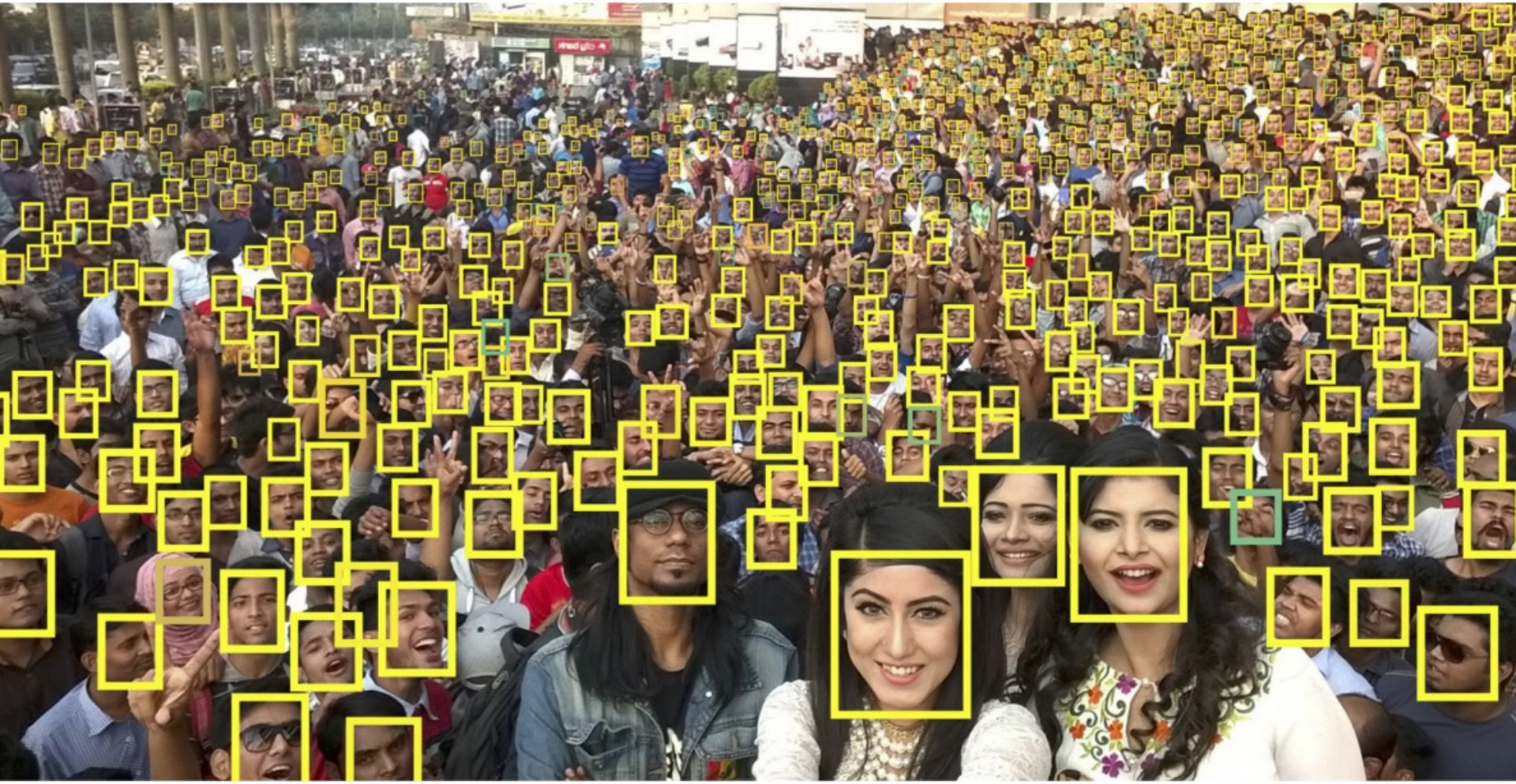


One network, one scale



Source: R. Girshick, K. He

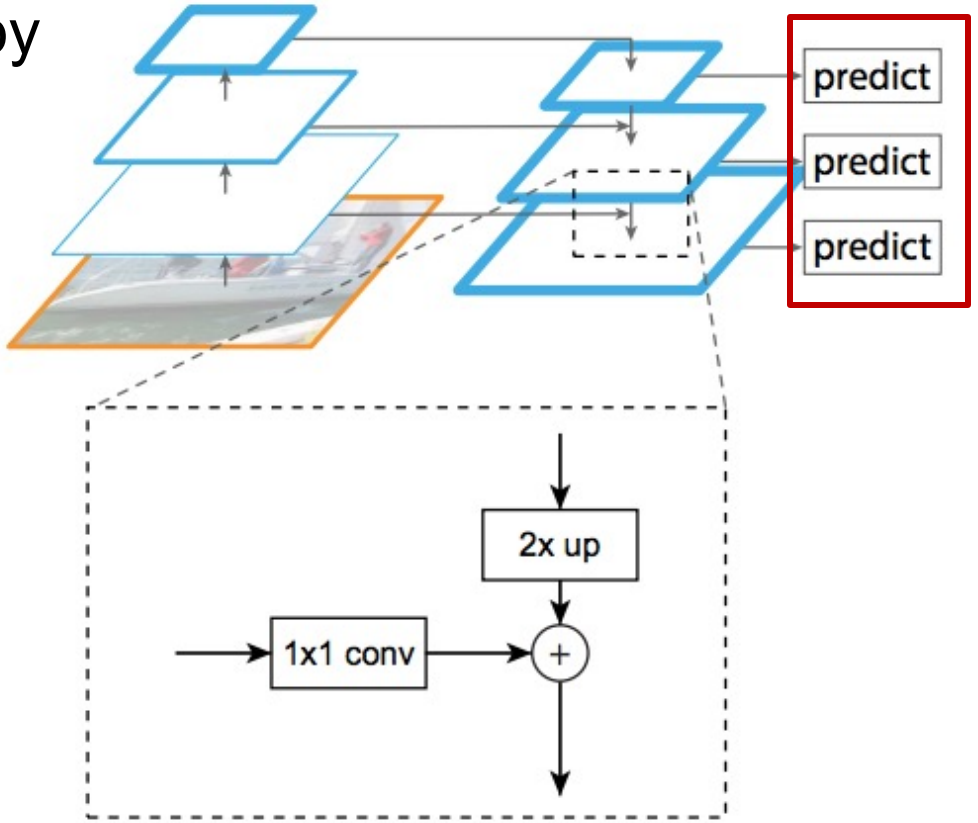
Object detection: Scales



Source: P. Hu, D. Ramanan

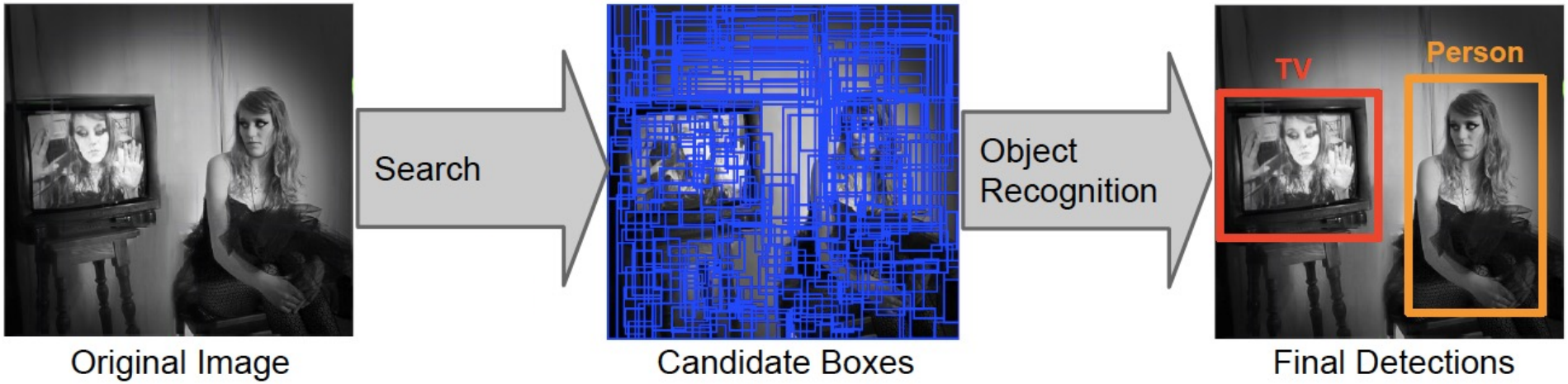
Feature pyramid networks

- Improve predictive power of lower-level feature maps by adding contextual information from higher-level feature maps
- Each feature map (a level on the pyramid) predicts boxes with different sizes (yet shares the same prediction head)
- Used as backbone for Faster R-CNN

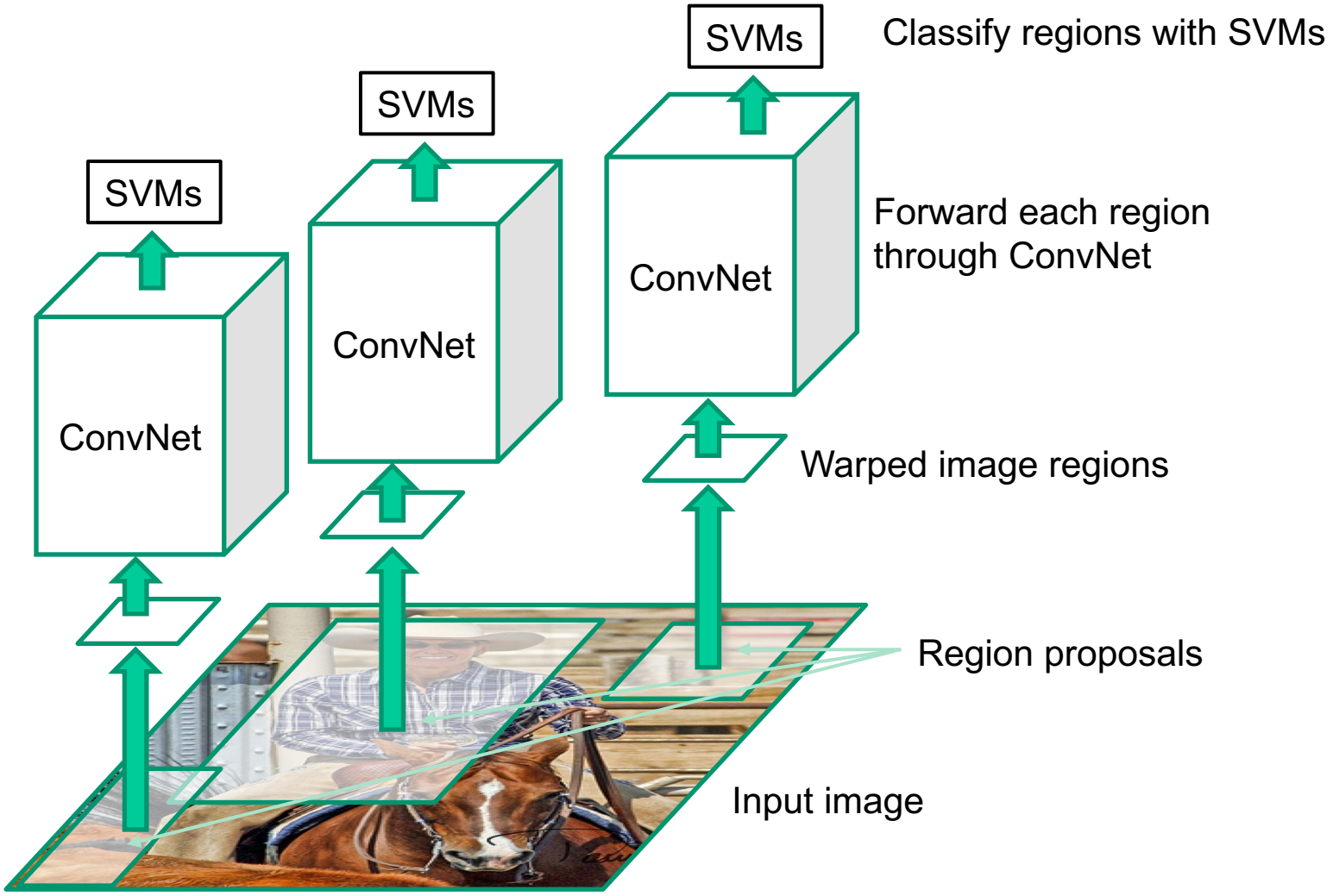


T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, and S. Belongie, Feature pyramid networks for object detection, CVPR 2017.

Review: Two-Stage Detection

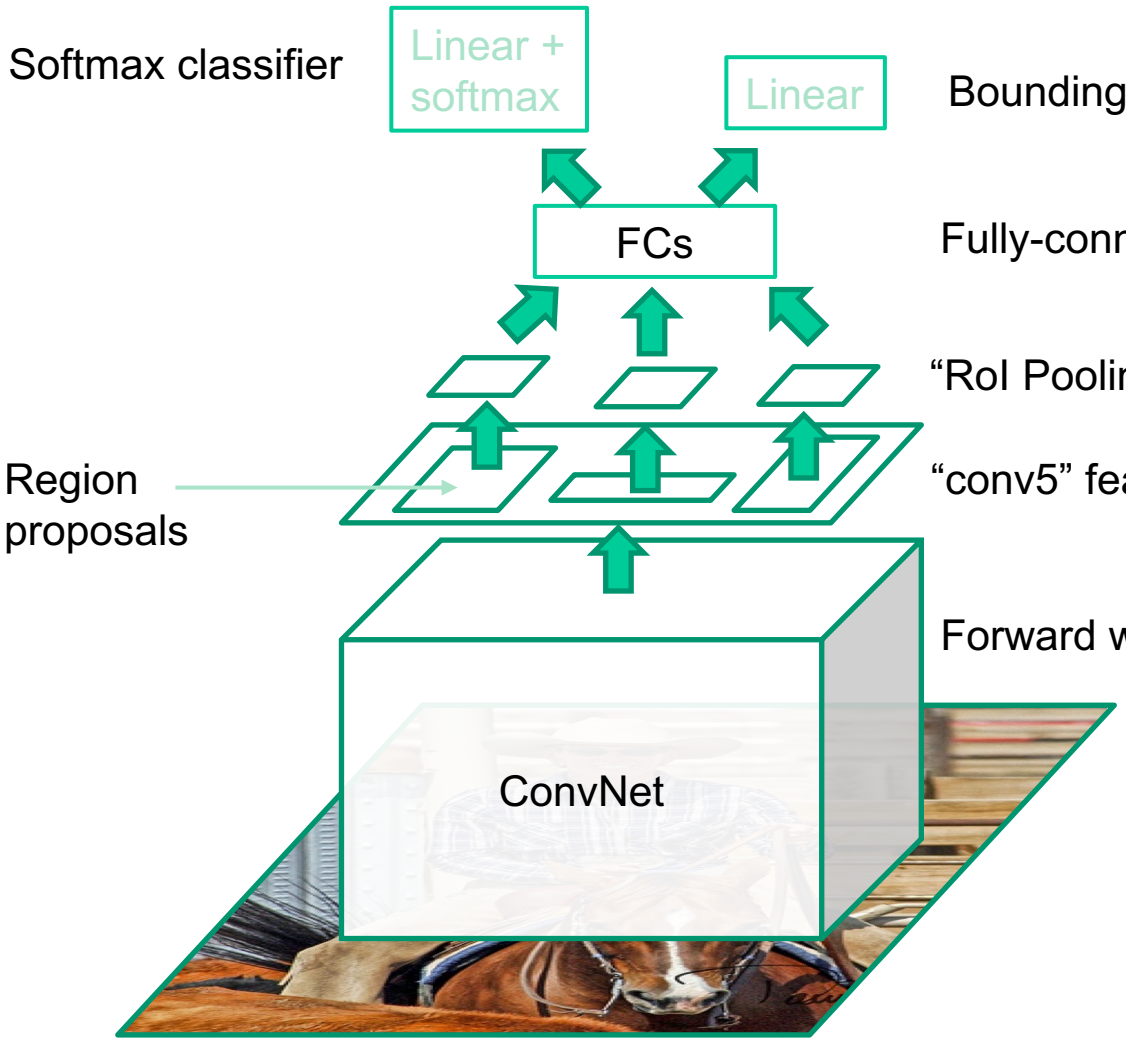


Review: R-CNN



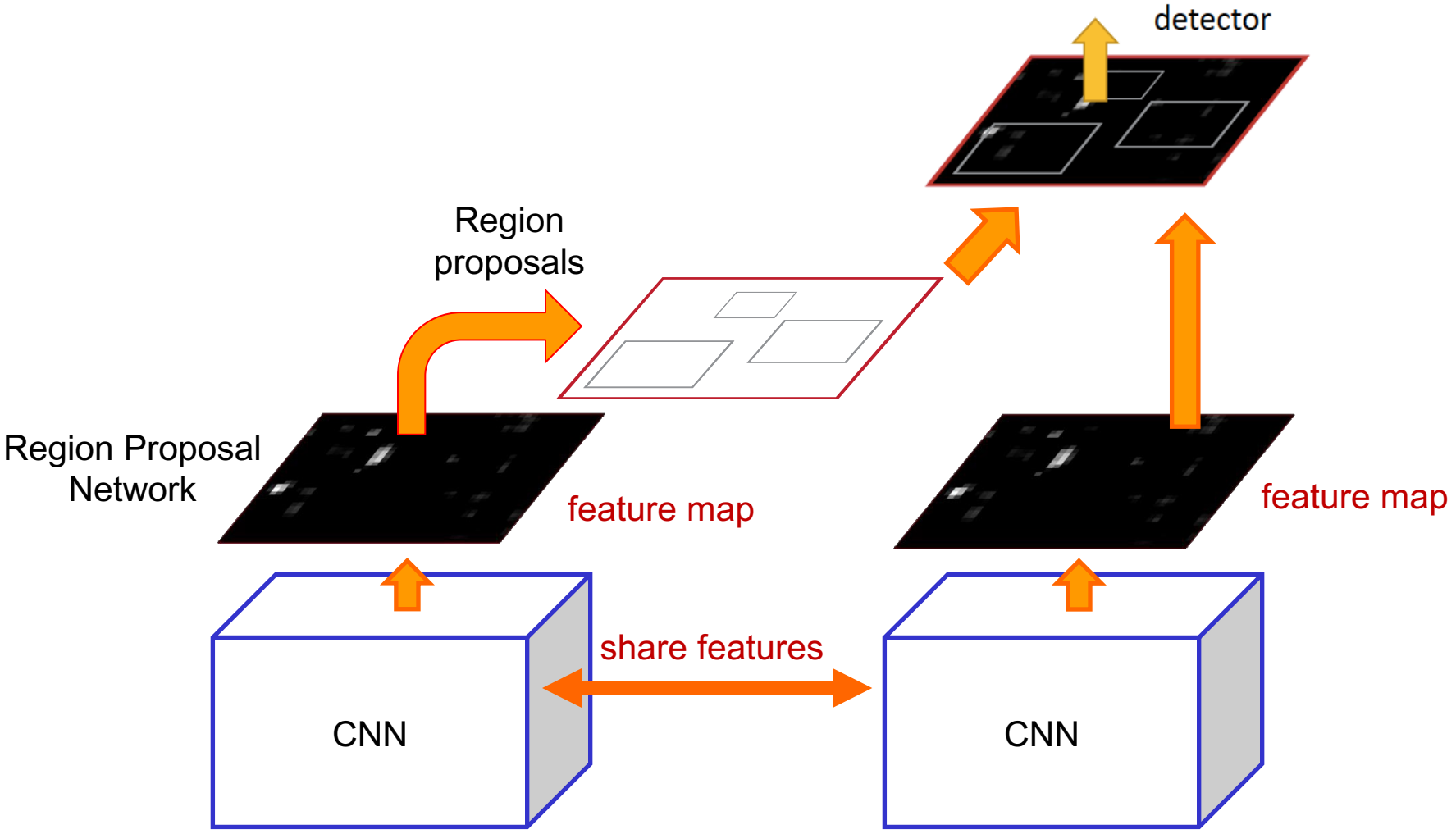
R. Girshick, J. Donahue, T. Darrell, and J. Malik, **Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation**, CVPR 2014.

Review: Fast R-CNN

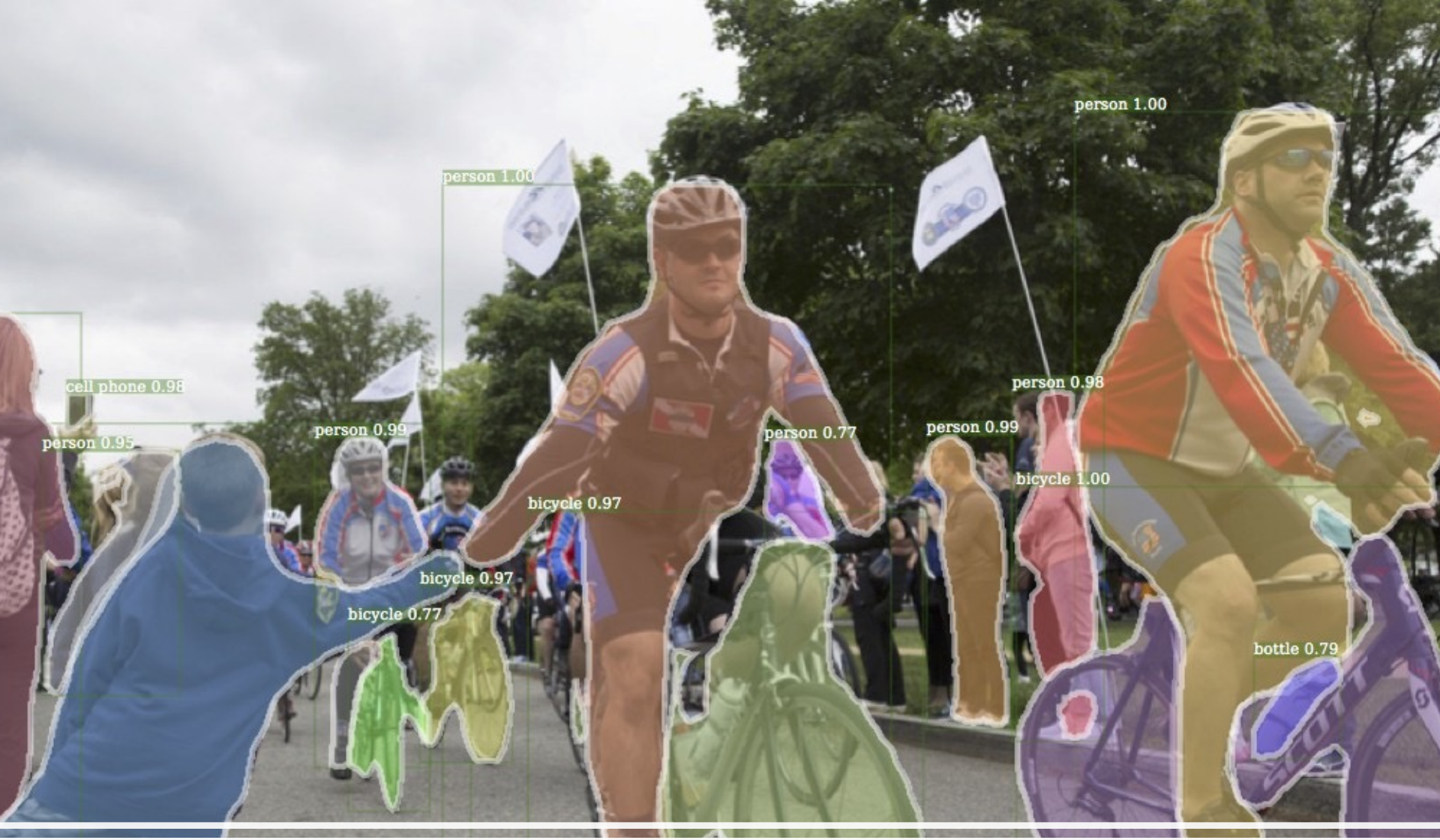


R. Girshick, Fast R-CNN, ICCV 2015

Review: Faster R-CNN

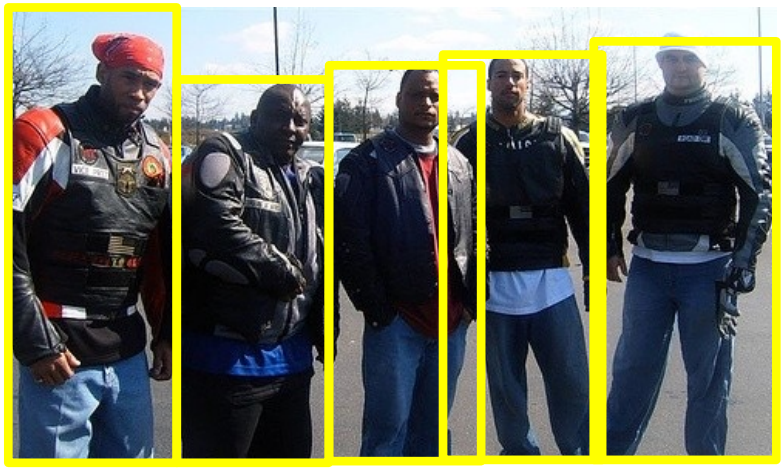


S. Ren, K. He, R. Girshick, and J. Sun, Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks, NIPS 2015



Moving beyond Boxes: Mask R-CNN

Mask R-CNN: Instance Segmentation

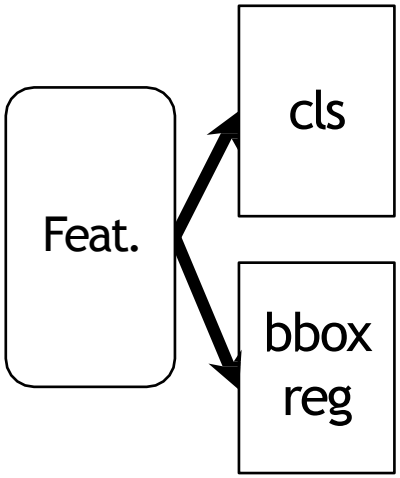


Object Detection

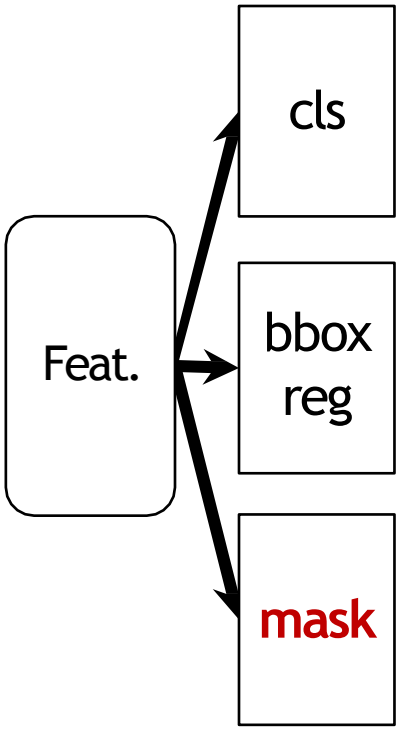


Instance Segmentation

Mask R-CNN: Architecture



Fast/er R-CNN

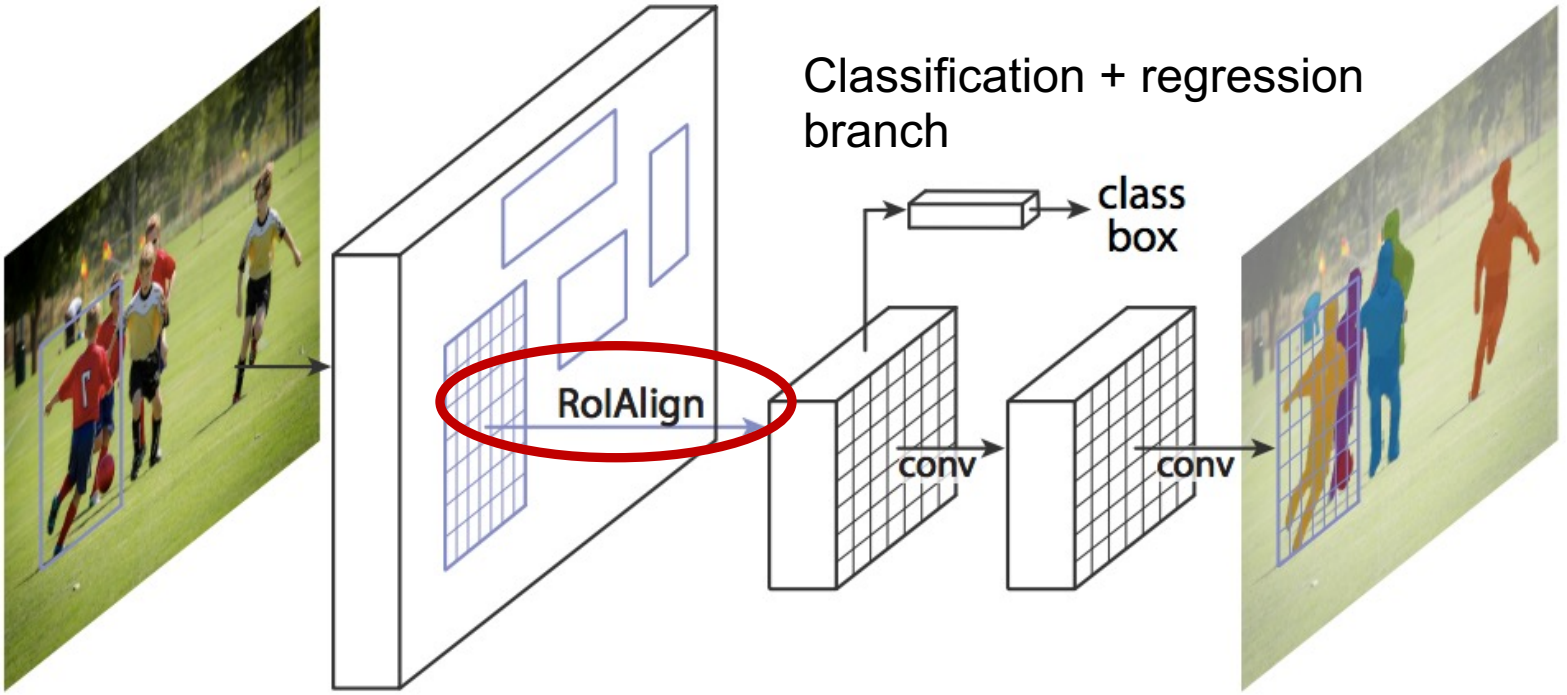


Mask R-CNN

Slide credit: Kaiming He

Mask R-CNN

- Mask R-CNN = Faster R-CNN
  + Fully Convolutional Networks on RoIs

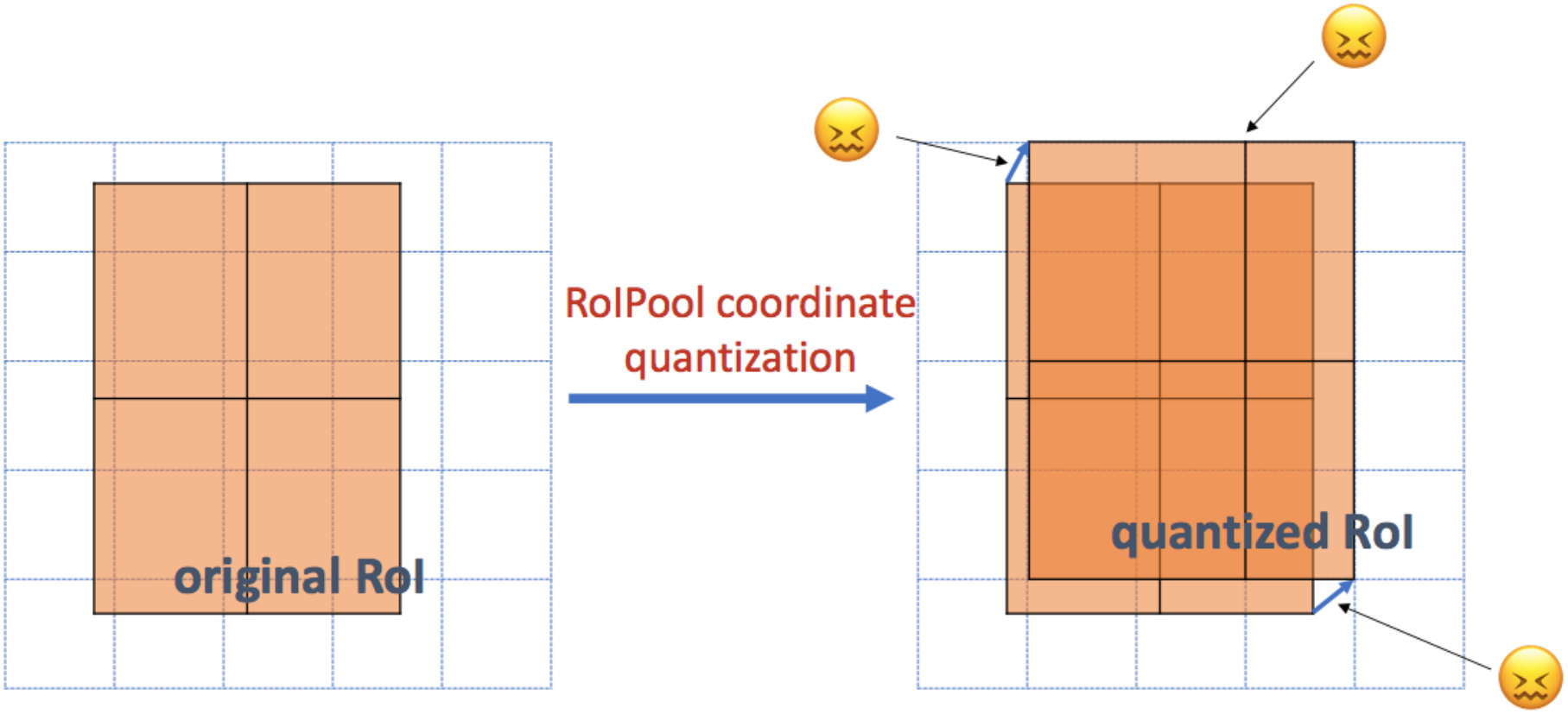


Mask branch: separately predict segmentation for each possible class

K. He, G. Gkioxari, P. Dollar, and R. Girshick, Mask R-CNN, ICCV 2017 (Best Paper Award)

RoIAlign vs. RoIPool

- RoIPool: nearest neighbor quantization

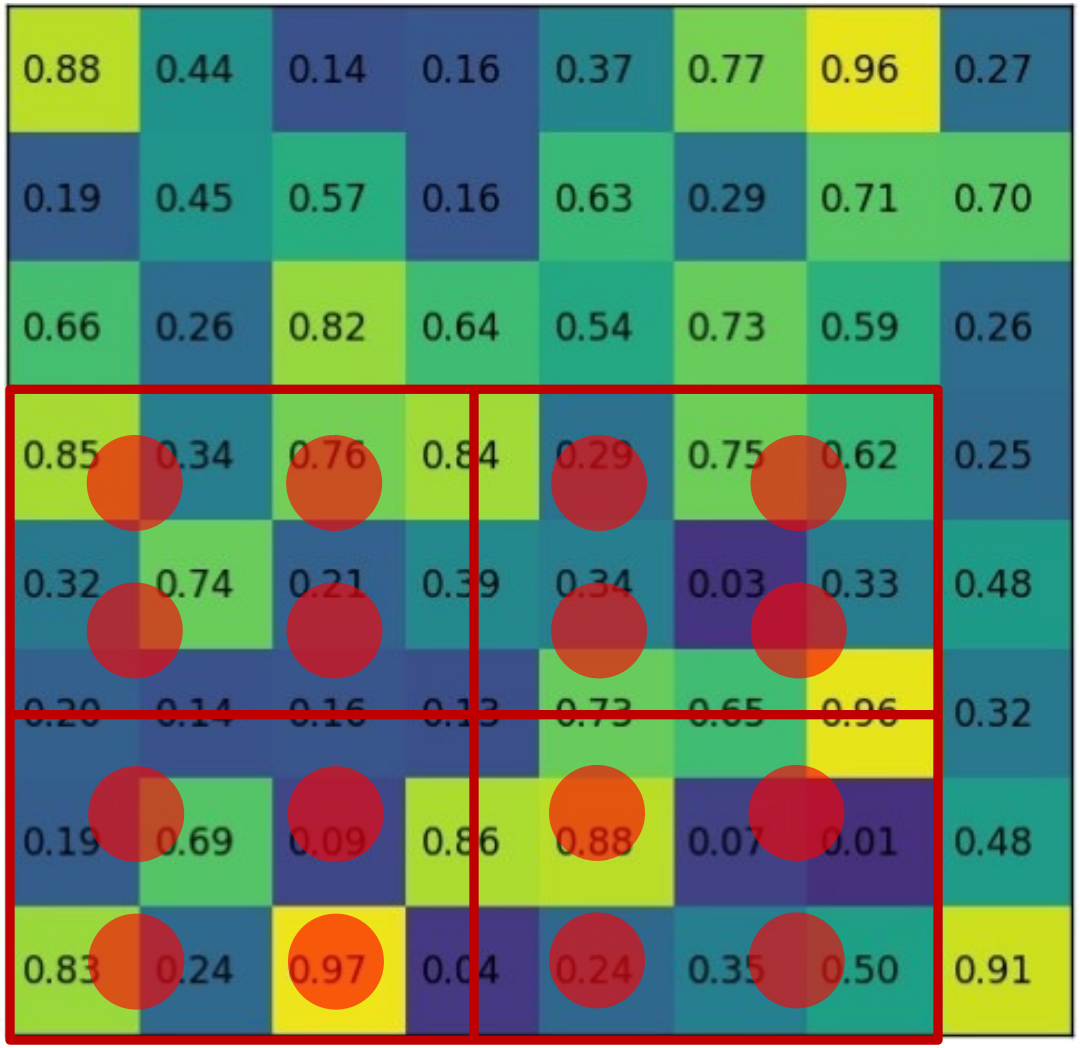


RoI pooling: a revisit



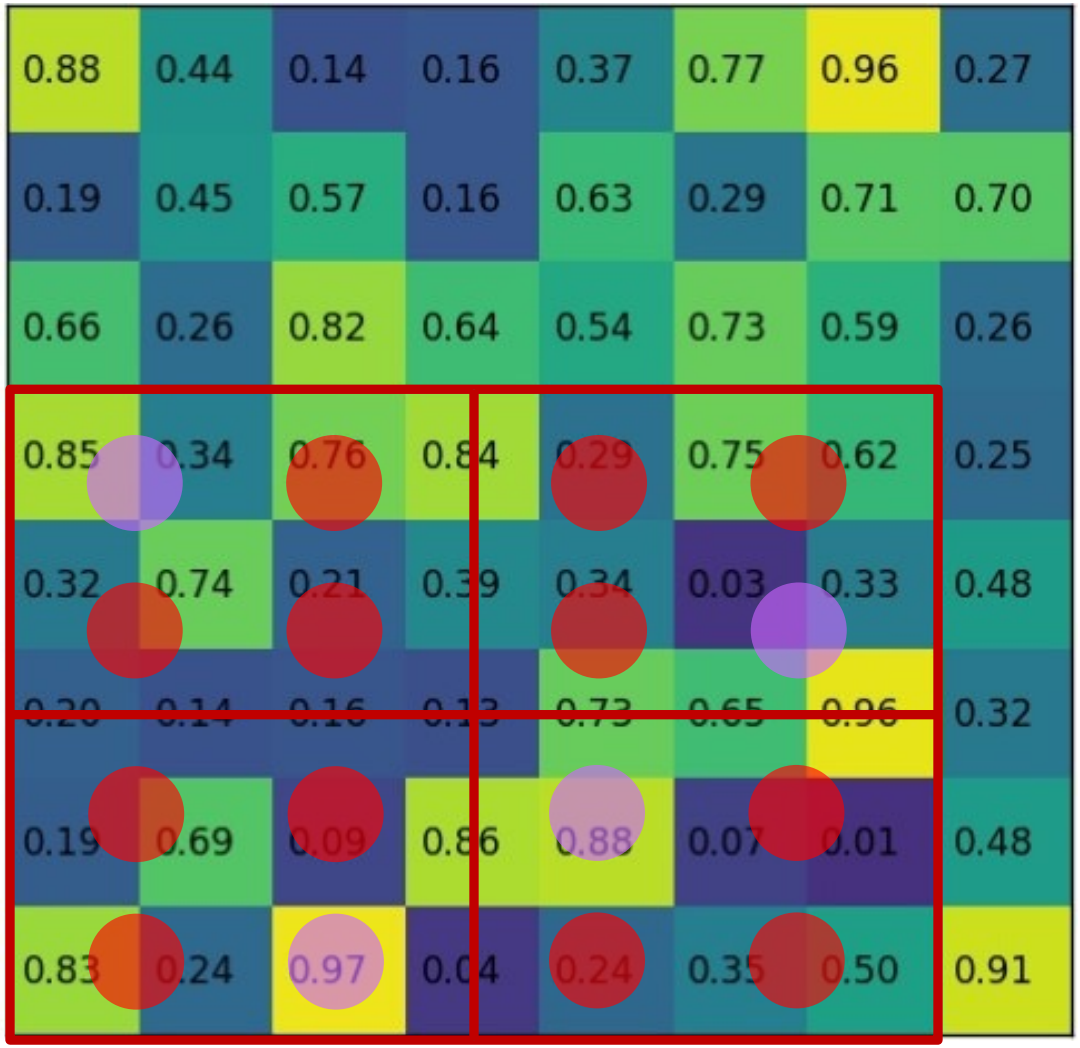
Pooled features are not spatially aligned
(important for mask prediction!)

Mask R-CNN: RoI align



Align the features during pooling
(resample the features based on bilinear interpolation)

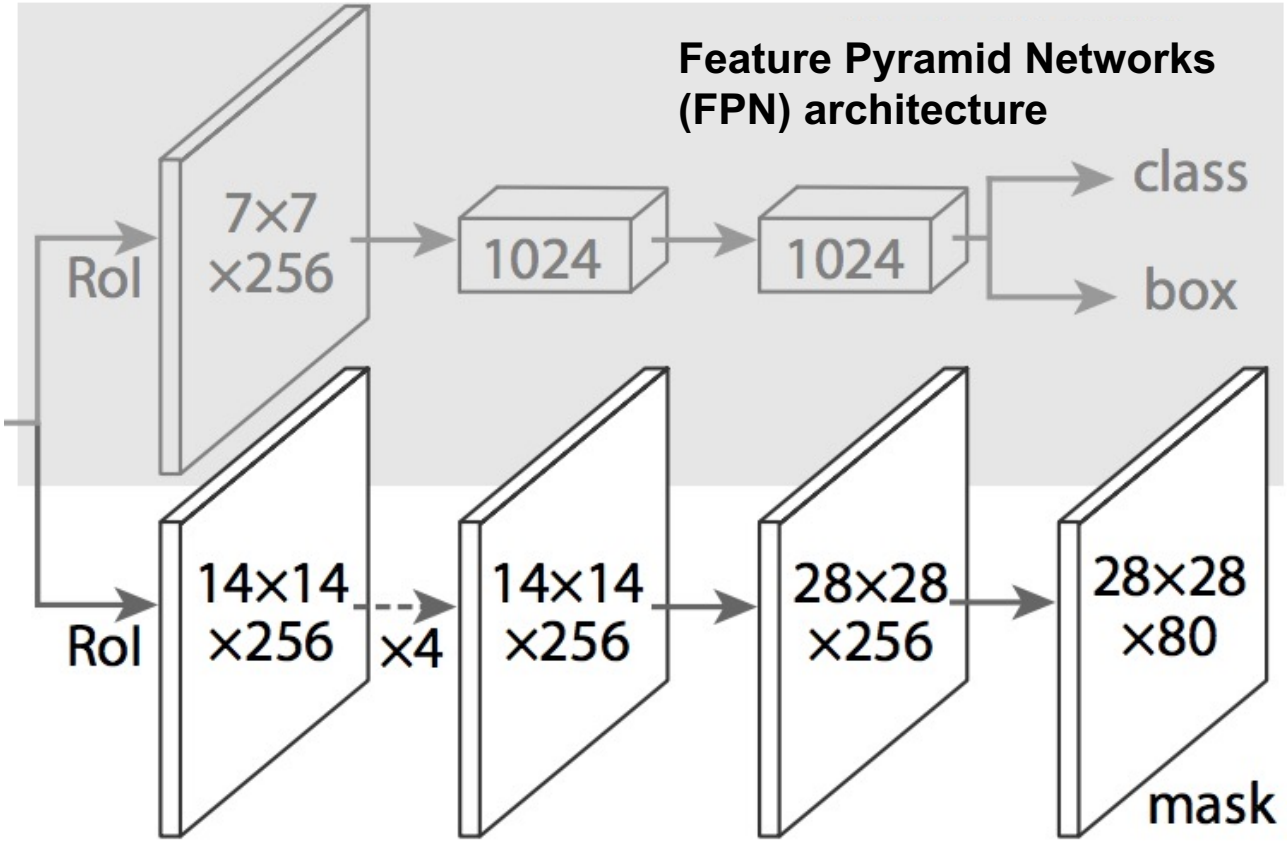
Mask R-CNN: RoI align



Align the features during pooling
(resample the features based on bilinear interpolation)

Mask R-CNN: Mask branch

- From RoIAlign features, predict class label, bounding box, and segmentation mask

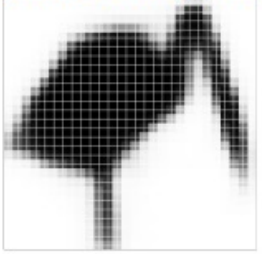


Mask R-CNN: Mask branch



Validation image with box detection shown in red

28x28 soft prediction



Resized Soft prediction



Final mask



K. He, G. Gkioxari, P. Dollar, and R. Girshick, Mask R-CNN,
ICCV 2017 (Best Paper Award)

Mask R-CNN: Results

	backbone	AP^{bb}	AP^{bb}_{50}	AP^{bb}_{75}	AP^{bb}_{S}	AP^{bb}_{M}	AP^{bb}_{L}
Faster R-CNN+++ [15]	ResNet-101-C4	34.9	55.7	37.4	15.6	38.7	50.9
Faster R-CNN w FPN [22]	ResNet-101-FPN	36.2	59.1	39.0	18.2	39.0	48.2
Faster R-CNN by G-RMI [17]	Inception-ResNet-v2 [32]	34.7	55.5	36.7	13.5	38.1	52.0
Faster R-CNN w TDM [31]	Inception-ResNet-v2-TDM	36.8	57.7	39.2	16.2	39.8	**52.1**
Faster R-CNN, RoIAlign	ResNet-101-FPN	37.3	59.6	40.3	19.8	40.2	48.8
Mask R-CNN	ResNet-101-FPN	38.2	60.3	41.7	20.1	41.1	50.2
Mask R-CNN	ResNeXt-101-FPN	**39.8**	**62.3**	**43.4**	**22.1**	**43.2**	51.2

Bounding box detection improved by:

- RoIAlign

Mask R-CNN: Results

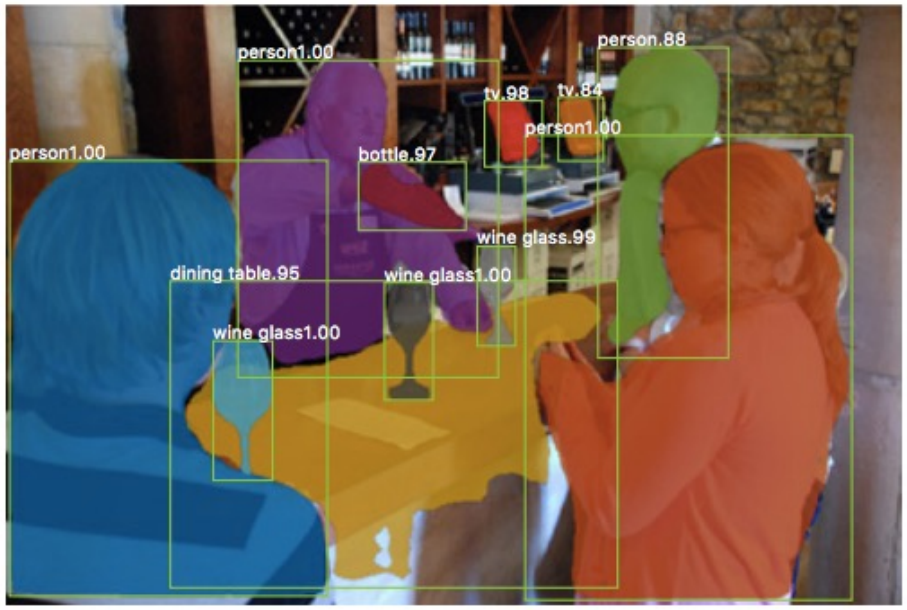
	backbone	AP^{bb}	AP^{bb}_{50}	AP^{bb}_{75}	AP^{bb}_{S}	AP^{bb}_{M}	AP^{bb}_{L}
Faster R-CNN+++ [15]	ResNet-101-C4	34.9	55.7	37.4	15.6	38.7	50.9
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Bounding box detection improved by:

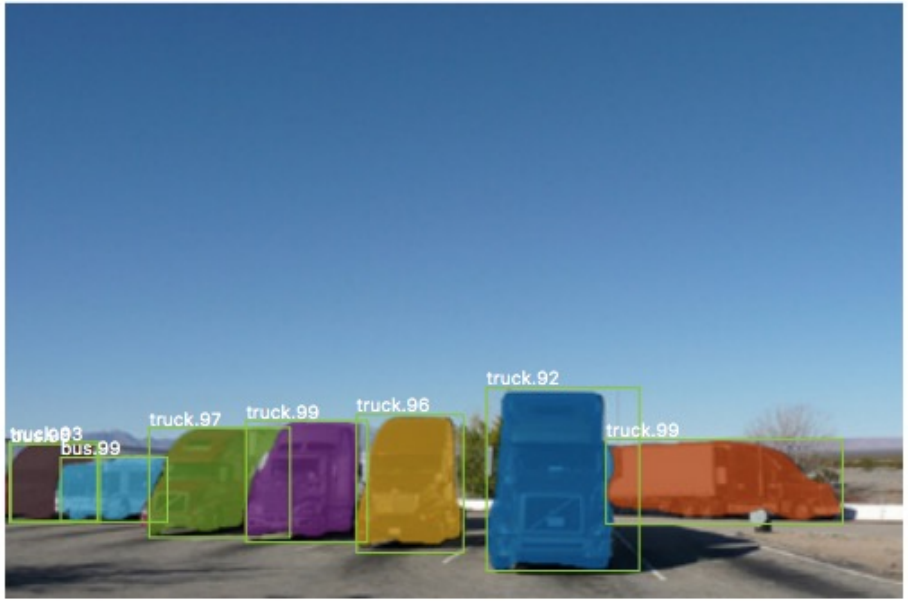
- RoIAlign
- Multi-task training w. mask

Other model designs are similar to Faster RCNN, including the use of FPN

Mask R-CNN: Example results



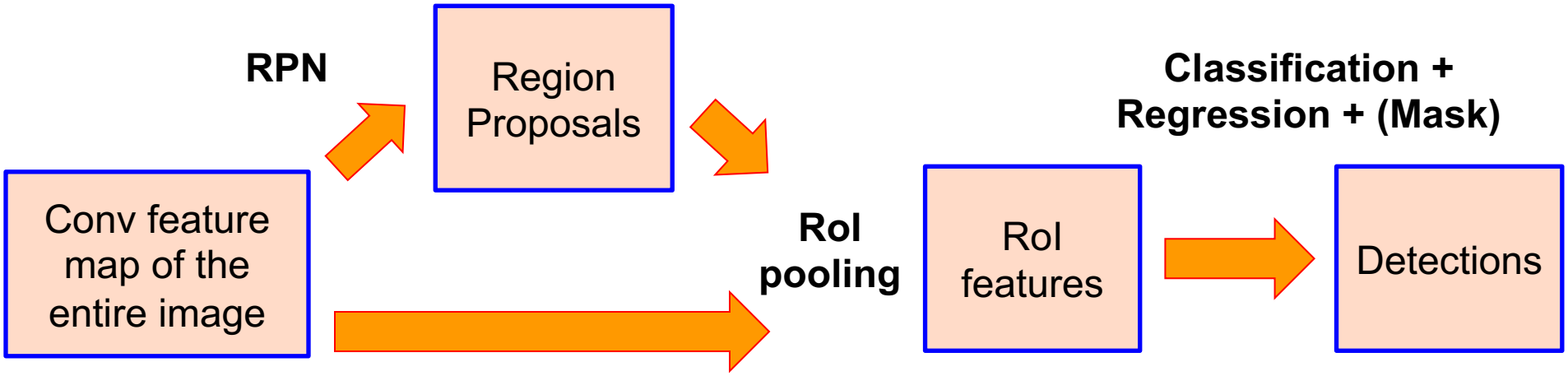






Streamlined detection architectures

- The Faster/Mask R-CNN pipeline separates proposal generation and region classification:



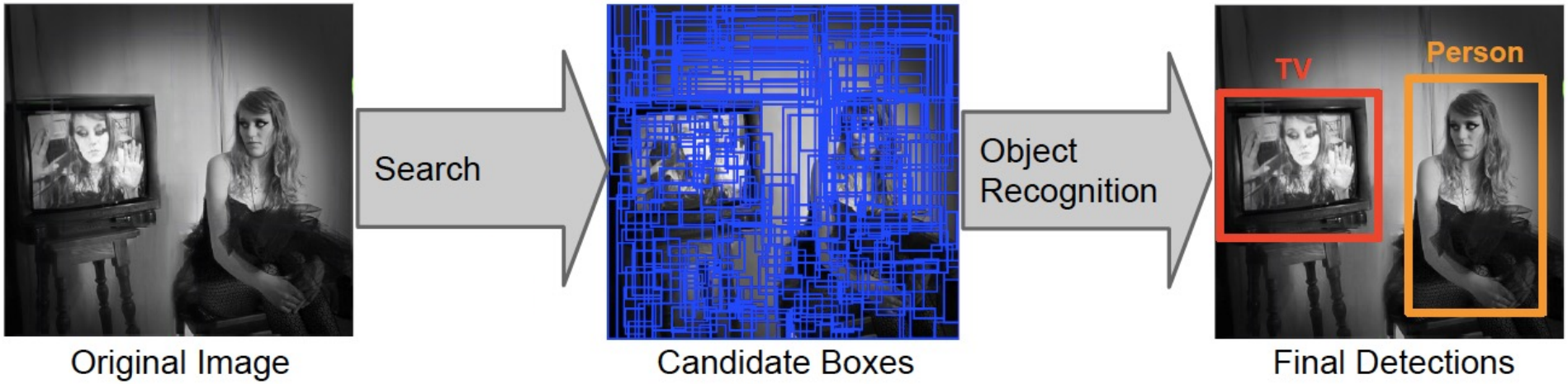
To Be Continued

Object Detection

(Part II)

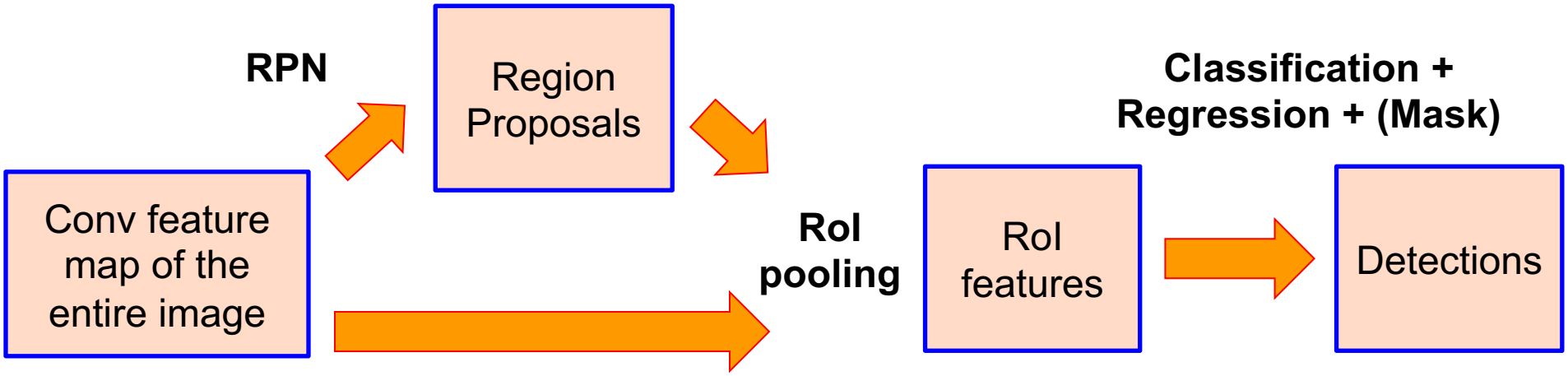
Many slides from Lana Lazebnik

Review: Two-Stage Detection



Streamlined detection architectures

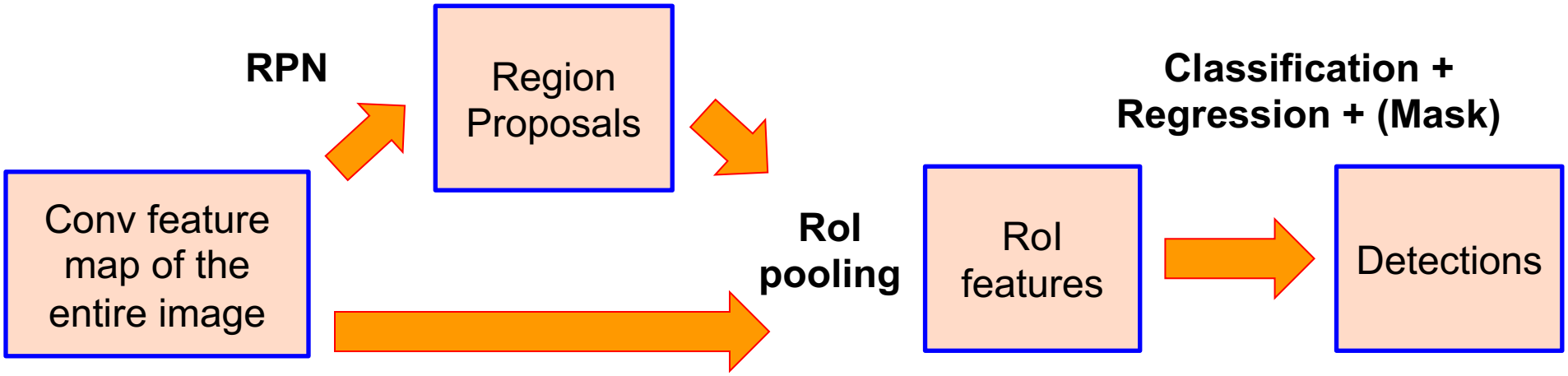
- The Faster/Mask R-CNN pipeline separates proposal generation and region classification:



  - An intuitive idea, yet not very efficient
  - Implementation requires some effort (e.g., Faster RCNN with ResNet50 and FPN: mAP 33.9 in the paper -> 40.2 in Detectron2)

Streamlined detection architectures

- The Faster/Mask R-CNN pipeline separates proposal generation and region classification:

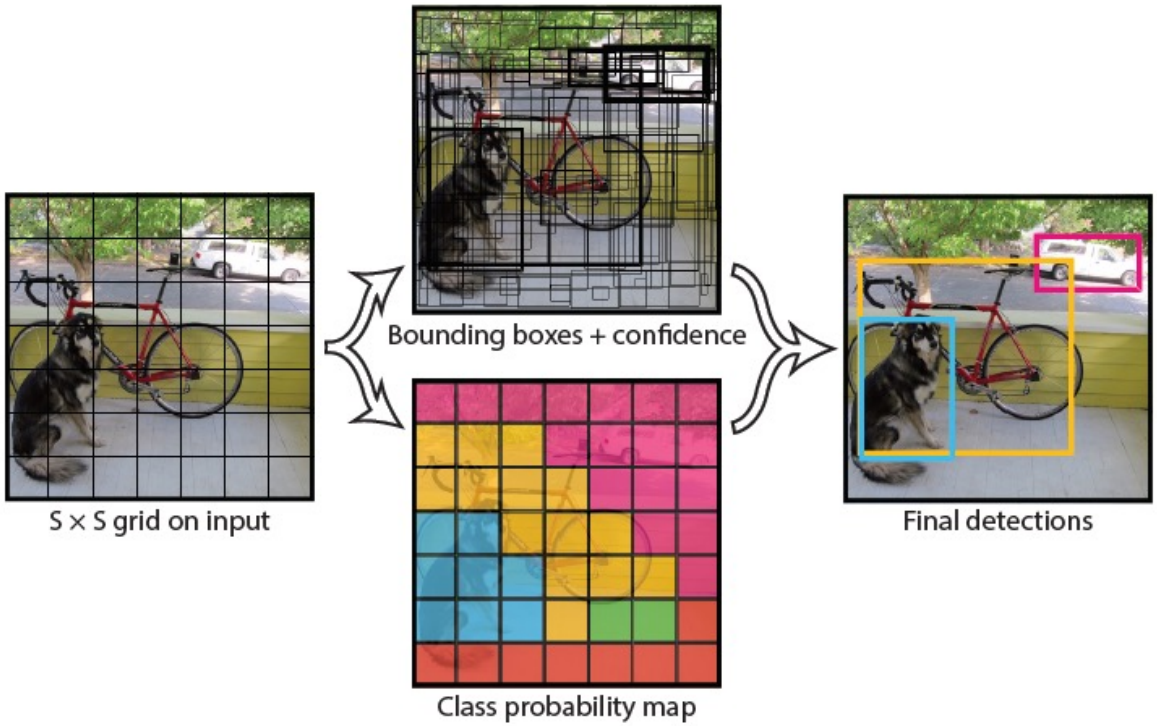


- Is it possible do detection in one shot?



YOLO

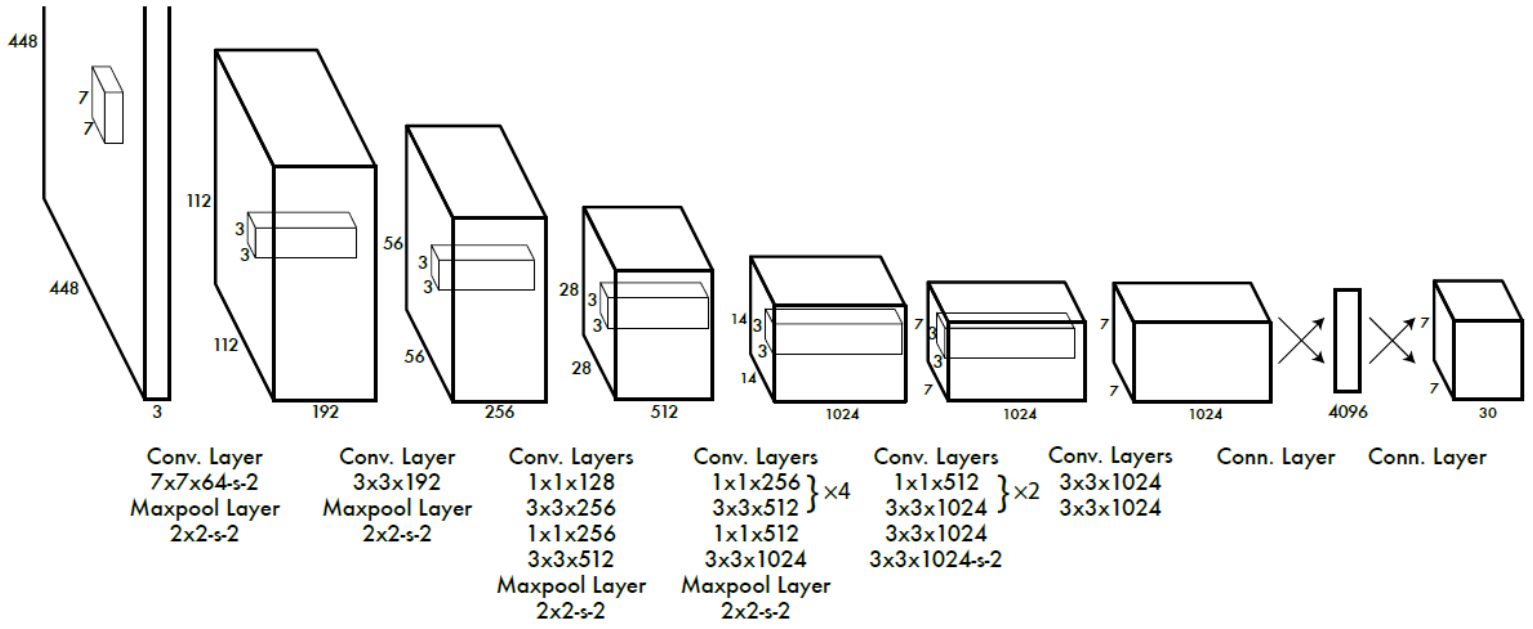
- Divide the image into a coarse grid and directly predict class label and a few candidate boxes for each grid cell



J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, You Only Look Once: Unified, Real-Time Object Detection, CVPR 2016

YOLO: An early attempt

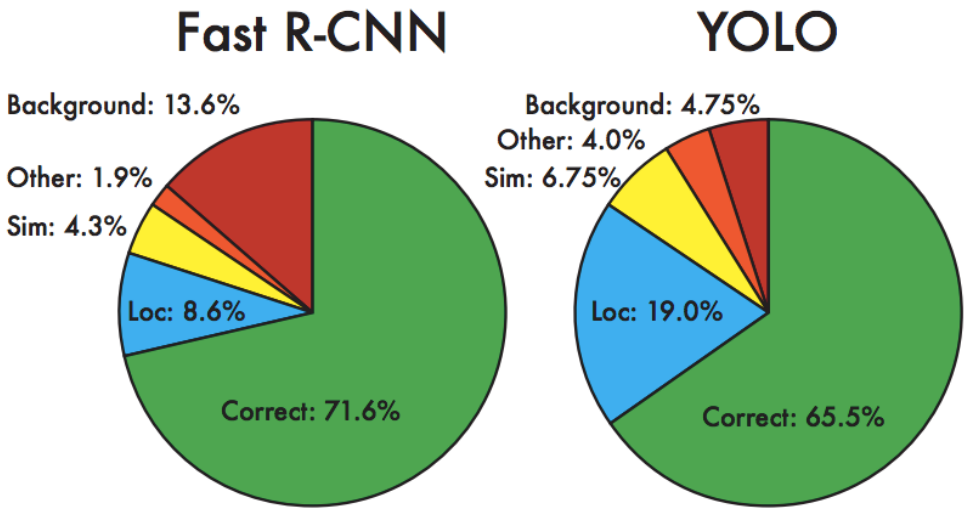
1. Take conv feature maps at 7x7 resolution
2. Add two FC layers to predict, at each location, a score for each class and 2 bboxes w/ confidences
   - For PASCAL, output is 7x7x30 (30 = 20 (classes) + 2*(4+1))



J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, You Only Look Once: Unified, Real-Time Object Detection, CVPR 2016

YOLO: Results

- 7x speedup over Faster R-CNN (45-155 FPS vs. 7-18 FPS)
- Each grid predicts only two boxes with only one class – limited number of nearby objects that can be predicted
- Localization accuracy suffers compared to Fast(er) R-CNN due to coarser features; errors on small boxes

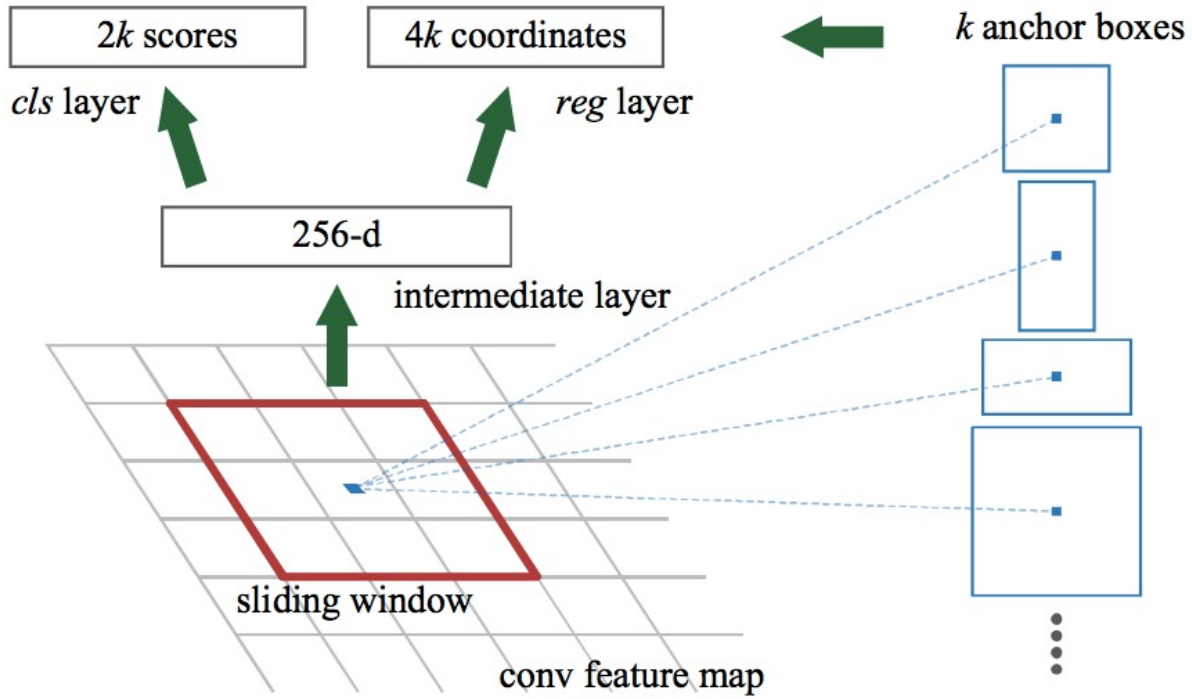


Performance on PASCAL 2007

This is essentially sliding window with a reduced set of windows

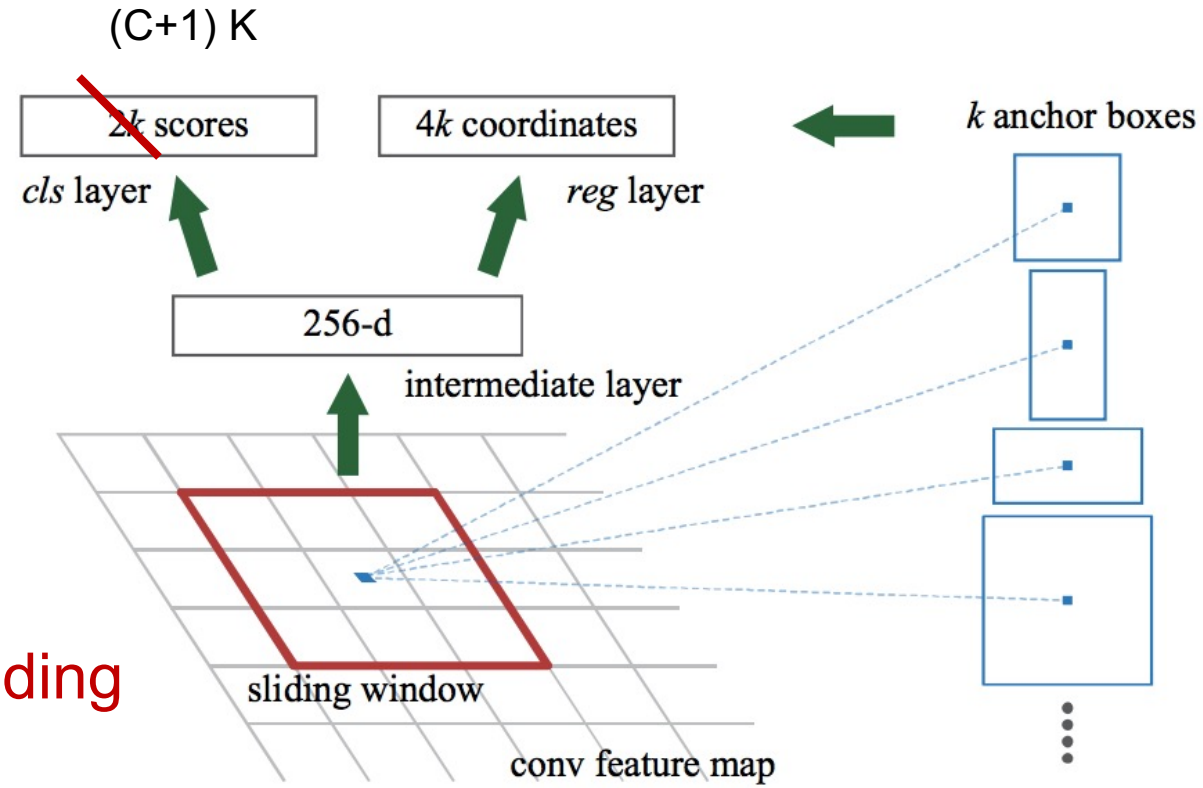
Revisiting region proposal network

- Slide a small window (3x3) over the conv5 layer
  - Predict object/no object
  - Regress bounding box coordinates with reference to *anchors*
  - Anchors are carefully designed (local) boxes to cover objects at different scales and aspect ratios
  - *What if we directly predict object categories?*



Revisiting region proposal network

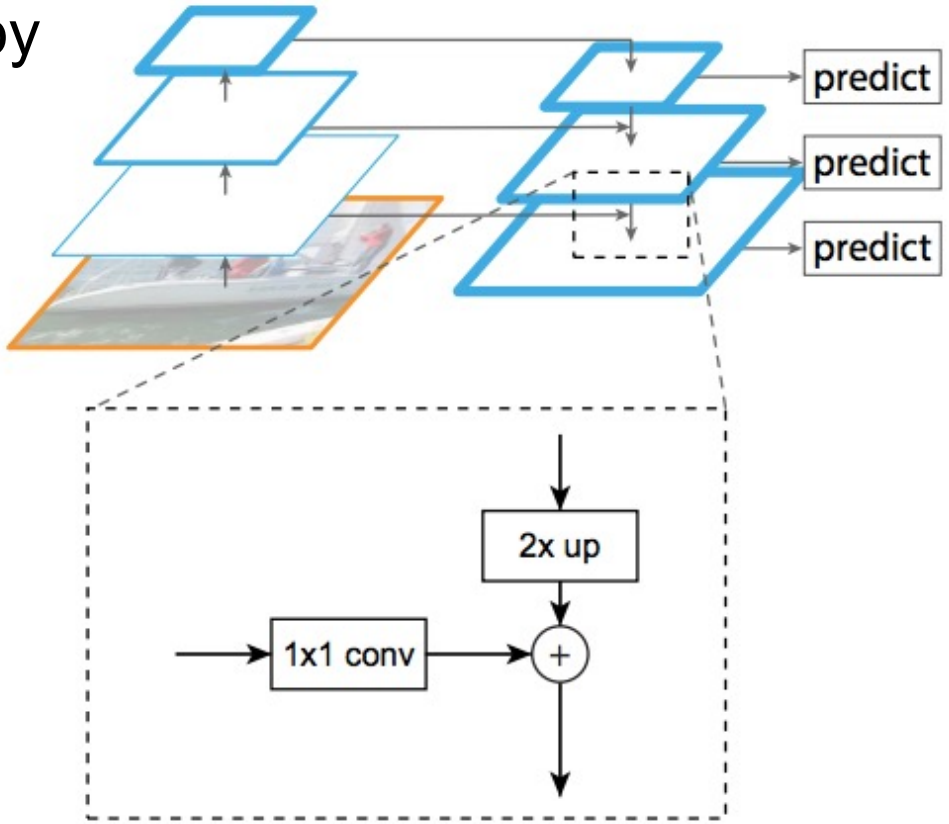
- Slide a small window over a feature map
  - Predict foreground object categories + background
  - Regress K bounding box coordinates (optionally with reference to *anchors)*



Again, sliding window!

Handling scale: FPN revisited

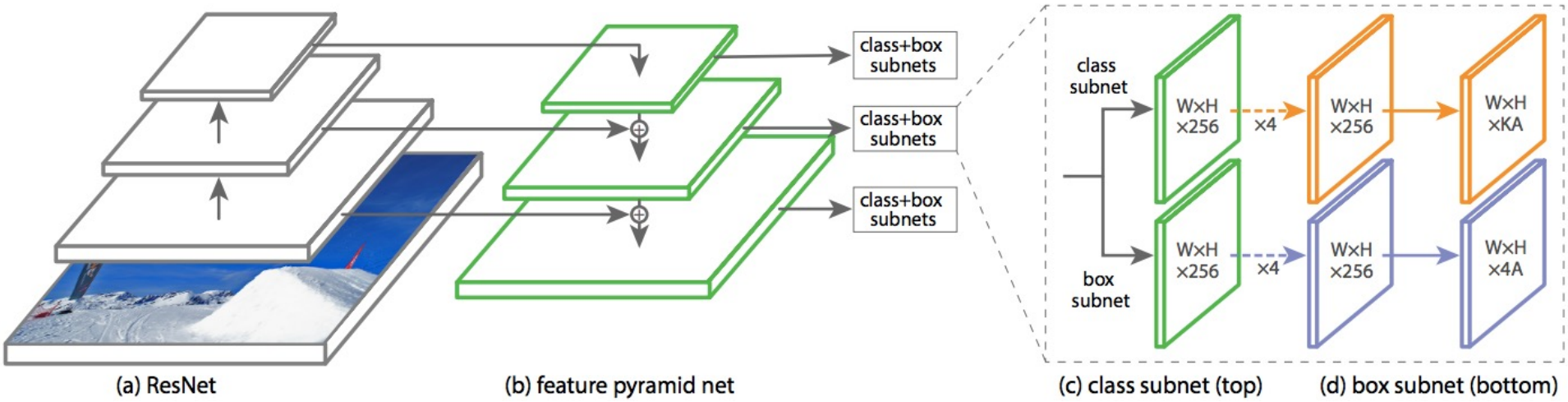
- Improve predictive power of lower-level feature maps by adding contextual information from higher-level feature maps
- Each feature map (a level on the pyramid) predicts boxes with different sizes (yet shares the same prediction head)
- Can be used for single-stage detectors



T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, and S. Belongie, Feature pyramid networks for object detection, CVPR 2017.

RetinaNet: Anchor-based detector

- Combine feature pyramid network with a ResNet backbone (or any network)
- Classify each position on every level of the pyramid and regress a box (with reference to **anchors**)
- Challenge: too many negative samples

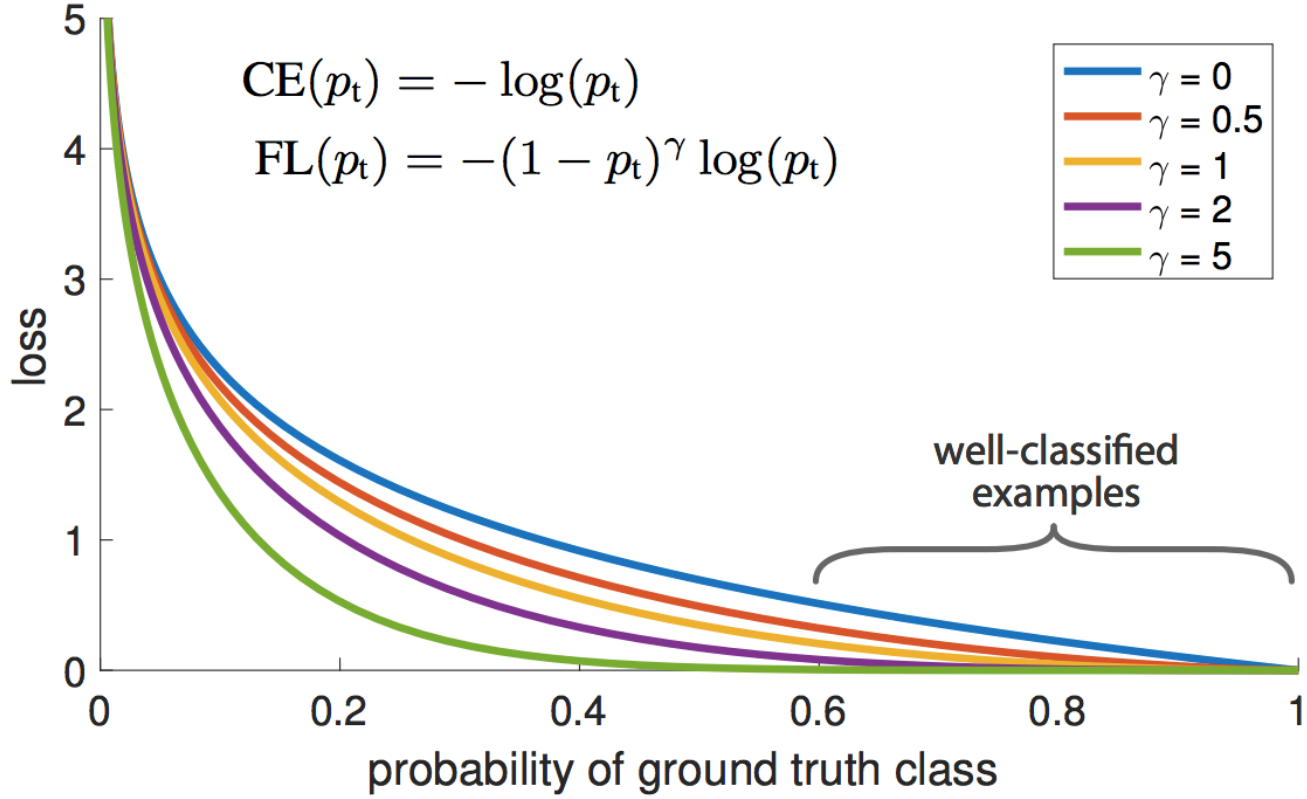


Shared predictors for all levels

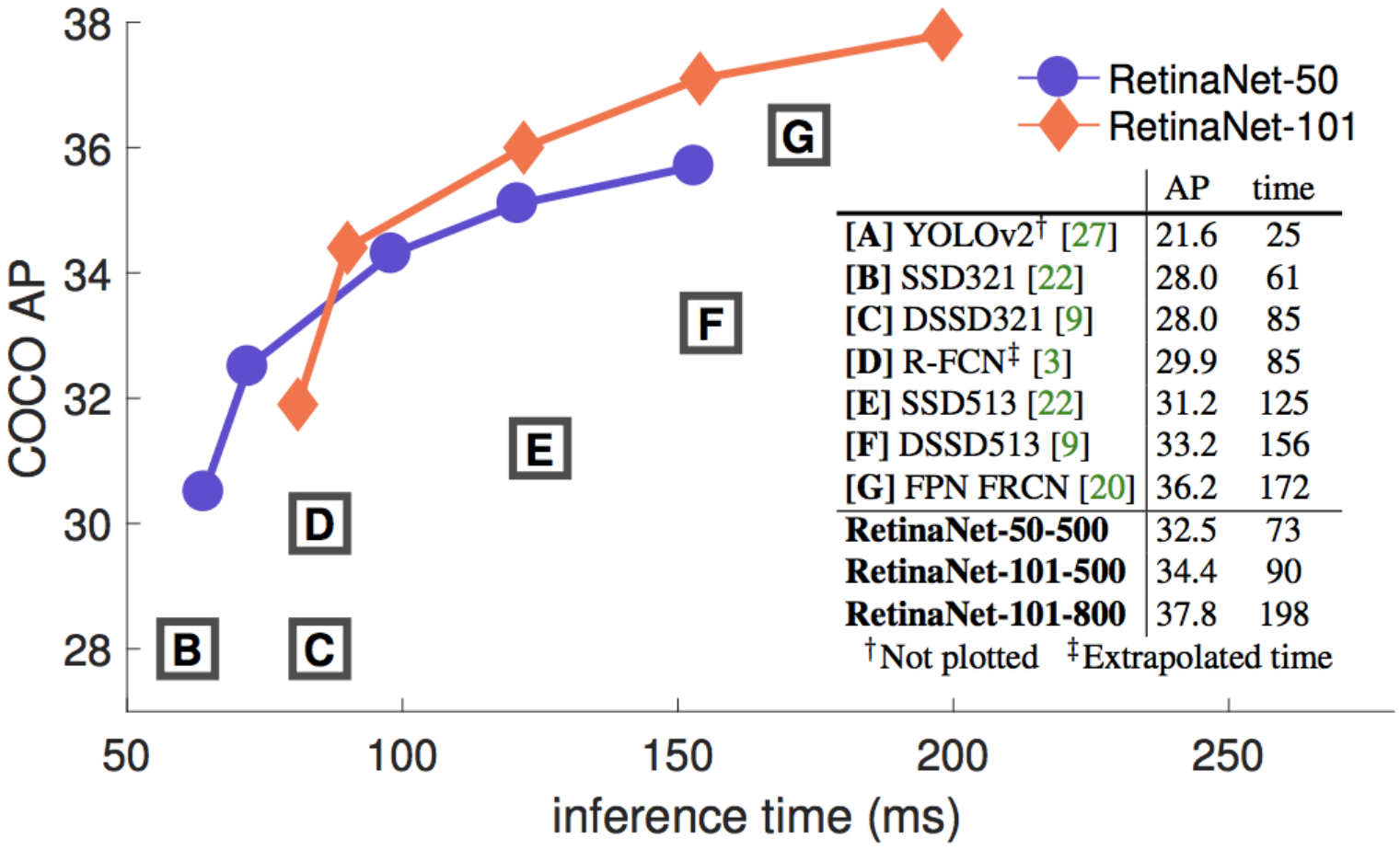
T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollar, Focal loss for dense object detection, ICCV 2017.

RetinaNet

- Consider *focal loss* to reduce the standard cross-entropy loss for well-classified examples
- RetinaNet uses sigmoid + focal loss for each category



RetinaNet: Results



	AP	time
[A] YOLOv2† [27]	21.6	25
[B] SSD321 [22]	28.0	61
[C] DSSD321 [9]	28.0	85
[D] R-FCN‡ [3]	29.9	85
[E] SSD513 [22]	31.2	125
[F] DSSD513 [9]	33.2	156
[G] FPN FRCN [20]	36.2	172
RetinaNet-50-500	32.5	73
RetinaNet-101-500	34.4	90
RetinaNet-101-800	37.8	198

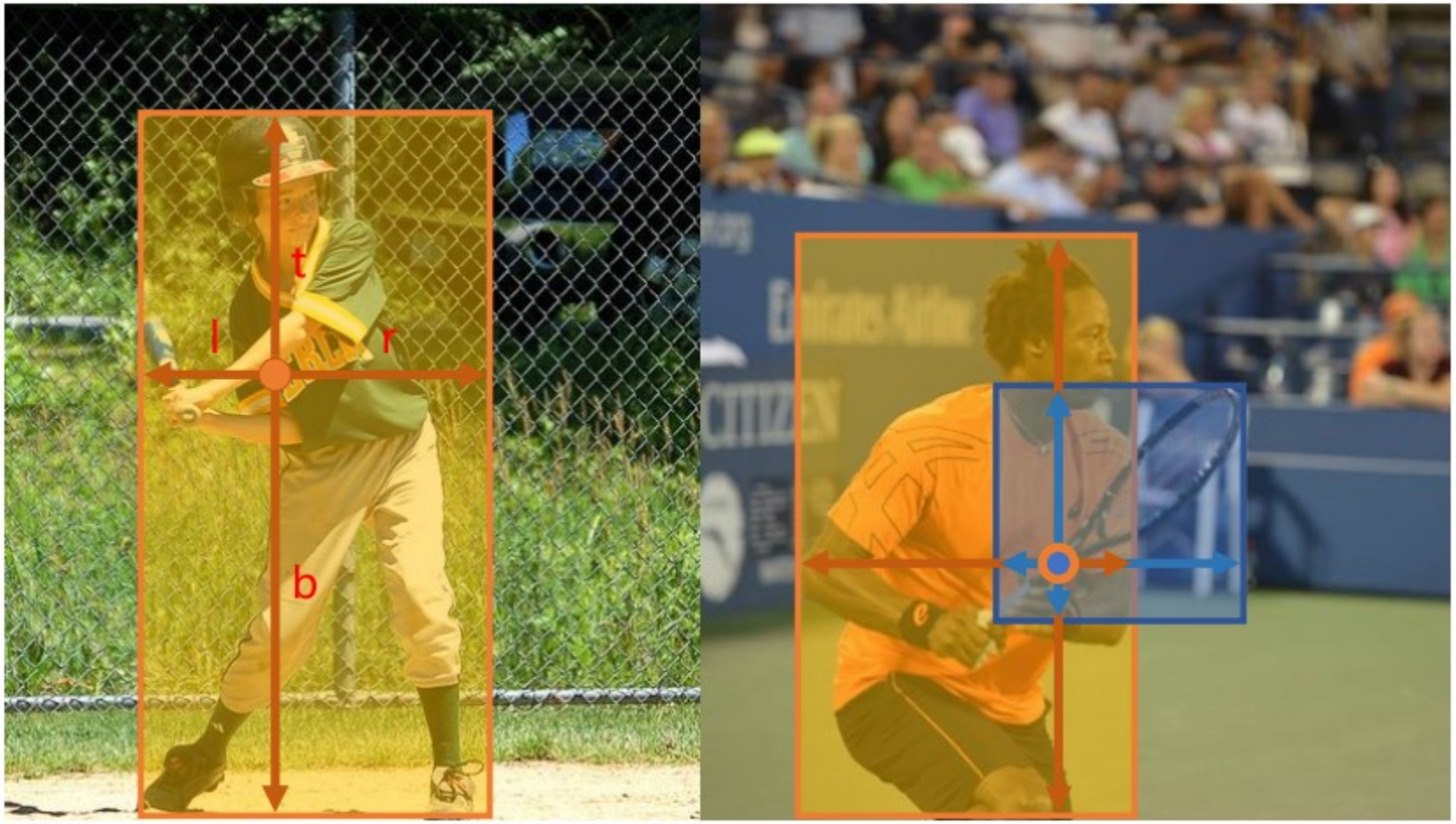
†Not plotted ‡Extrapolated time

Object detection without anchors

- RetinaNet uses *anchors* to define boxes
- The design of anchors must be tailored to a dataset (in order to maximize the recall)
- Dense set of anchors -> reduced efficiency
- What if we don’t use anchors, i.e., an anchor-free object detector?
- Need to find other ways to represent boxes

FCOS: An anchor-free detector

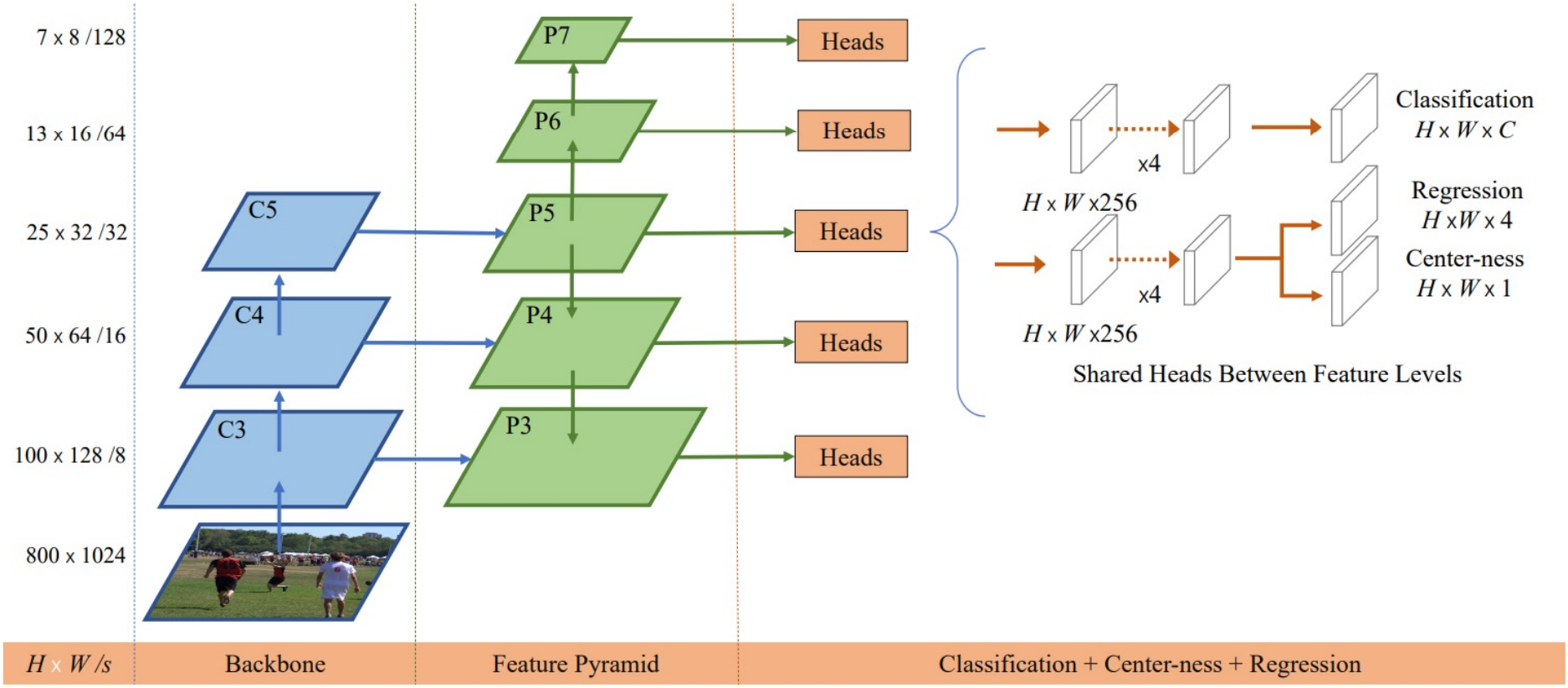
- A box = center (feature spot) + offsets (4)



Tian, Z., Shen, C., Chen, H., & He, T. (2019). FCOS: Fully convolutional one-stage object detection. In *CVPR*.

FCOS: An anchor-free detector

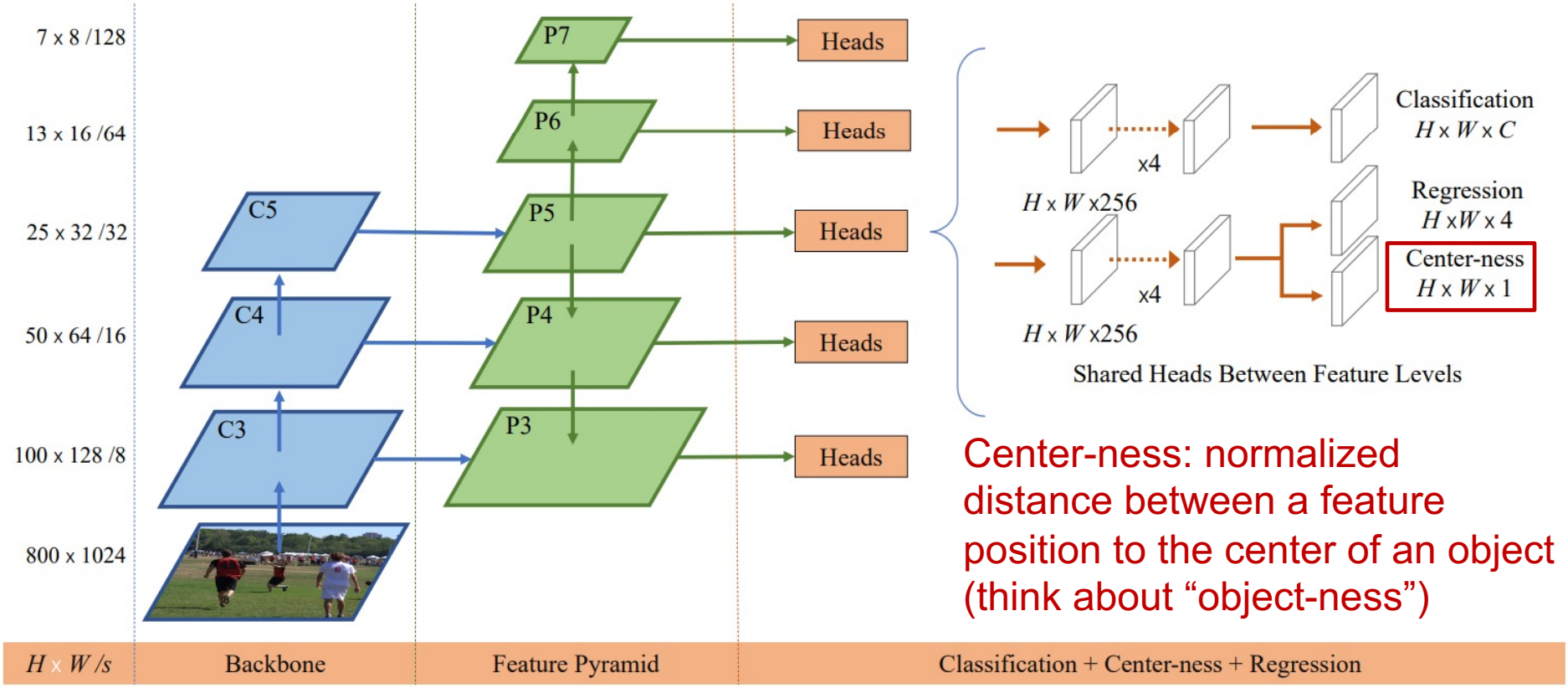
- Classify every feature spot and regress a box (as offsets to the spot) = a single “zero-sized” anchor



Tian, Z., Shen, C., Chen, H., & He, T. (2020). FCOS: A simple and strong anchor-free object detector. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.

FCOS: An anchor-free detector

- Classify every feature spot and regress a box (as offsets to the spot) = a single “zero-sized” anchor
- Also predict a center-ness score



Center-ness: normalized distance between a feature position to the center of an object (think about “object-ness”)

Tian, Z., Shen, C., Chen, H., & He, T. (2020). FCOS: A simple and strong anchor-free object detector. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.

FCOS: An anchor-free detector

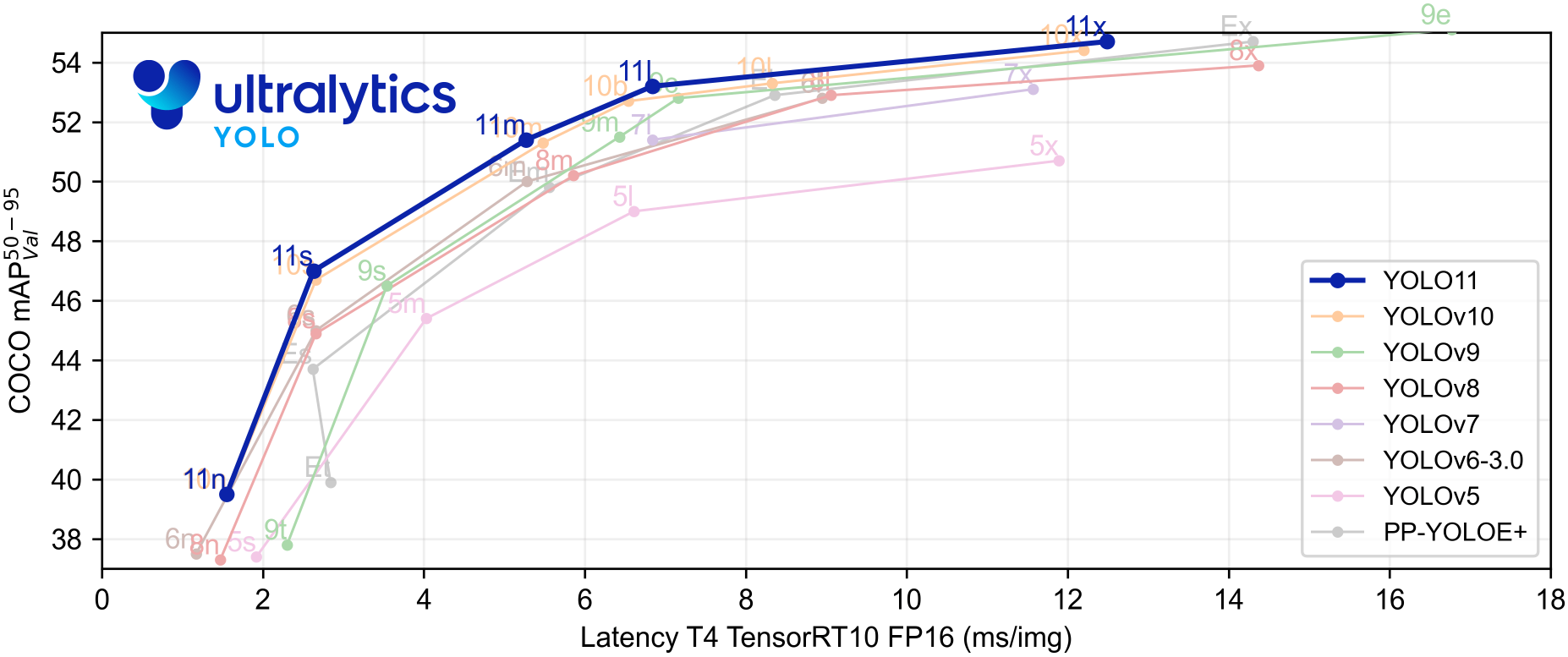
- Classify every feature position (using focal loss) and regress a box (as offsets to the current position)
- Also predict a center-ness score

- Other “tricks” to boost the performance
  - Center sampling: only feature positions close to object centers are considered as positive
  - Group norm at shared heads
  - Generalized IoU loss
  - Learned scalar for decoding boxes at each pyramid level

FCOS: Results

Method	Backbone	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Two-stage methods:							
Faster R-CNN+++ [47]	ResNet-101	34.9	55.7	37.4	15.6	38.7	50.9
Faster R-CNN w/ FPN [9]	ResNet-101-FPN	36.2	59.1	39.0	18.2	39.0	48.2
Faster R-CNN by G-RMI [51]	Inception-ResNet-v2 [52]	34.7	55.5	36.7	13.5	38.1	52.0
Faster R-CNN w/ TDM [53]	Inception-ResNet-v2-TDM	36.8	57.7	39.2	16.2	39.8	52.1
One-stage methods:							
YOLOv2 [30]	DarkNet-19 [30]	21.6	44.0	19.2	5.0	22.4	35.5
SSD513 [5]	ResNet-101-SSD	31.2	50.4	33.3	10.2	34.5	49.8
YOLOv3 608×608 [6]	Darknet-53	33.0	57.9	34.4	18.3	35.4	41.9
DSSD513 [54]	ResNet-101-DSSD	33.2	53.3	35.2	13.0	35.4	51.1
RetinaNet [7]	ResNet-101-FPN	39.1	59.1	42.3	21.8	42.7	50.2
CornerNet [32]	Hourglass-104	40.5	56.5	43.1	19.4	42.7	53.9
FSAF [33]	ResNeXt-64x4d-101-FPN	42.9	63.8	46.3	26.6	46.2	52.7
CenterNet511 [55]	Hourglass-104	44.9	62.4	48.1	25.6	47.4	57.4
FCOS	ResNet-101-FPN	43.2	62.4	46.8	26.1	46.2	52.8
FCOS	ResNeXt-32x8d-101-FPN	44.1	63.7	47.9	27.4	46.8	53.7
FCOS	ResNeXt-64x4d-101-FPN	44.8	64.4	48.5	27.7	47.4	55.0
FCOS w/ deform. conv. v2 [56]	ResNeXt-32x8d-101-FPN	46.6	65.9	50.8	28.6	49.1	58.6
FCOS	ResNet-101-BiFPN [57]	45.0	63.6	48.7	27.0	47.9	55.9
FCOS	ResNeXt-32x8d-101-BiFPN	46.2	65.2	50.0	28.7	49.1	56.5
FCOS w/ deform. conv. v2	ResNeXt-32x8d-101-BiFPN	47.9	66.9	51.9	30.2	50.3	59.9
w/ test-time augmentation:							
FCOS	ResNet-101-FPN	45.9	64.5	50.4	29.4	48.3	56.1
FCOS	ResNeXt-32x8d-101-FPN	47.0	66.0	51.6	30.7	49.4	57.1
FCOS	ResNeXt-64x4d-101-FPN	47.5	66.4	51.9	31.4	49.7	58.2
FCOS w/ deform. conv. v2	ResNeXt-32x8d-101-FPN	49.1	68.0	53.9	31.7	51.6	61.0
FCOS	ResNet-101-BiFPN	47.9	65.9	52.5	31.0	50.7	59.7
FCOS	ResNeXt-32x8d-101-BiFPN	49.0	67.4	53.6	32.0	51.7	60.5
FCOS w/ deform. conv. v2	ResNeXt-32x8d-101-BiFPN	**50.4**	**68.9**	**55.0**	**33.2**	**53.0**	**62.7**

Putting things together: the Yolo series



While these models retain the brand name of Yolo, they integrate many of techniques from other methods, including FPN, focal loss, anchor-free representation.

https://github.com/ultralytics/ultralytics

DETR: A Transformer-based Detector

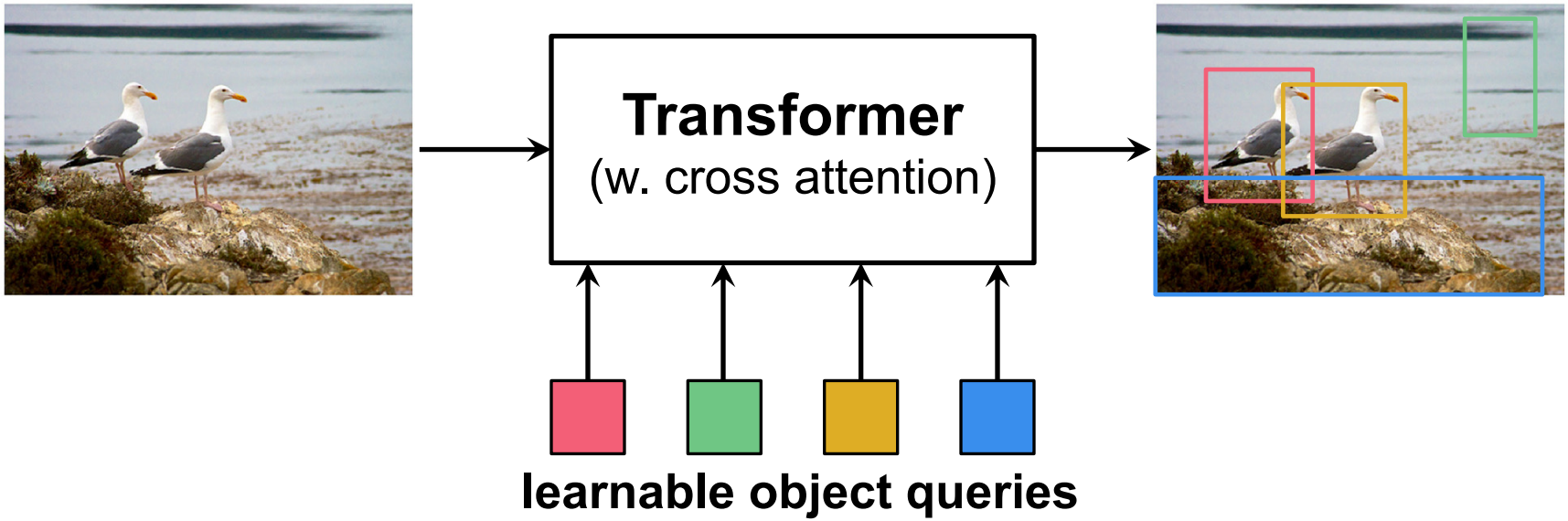
Can we learn to present object locations?

- Learn a set of “object queries” (may not have geometric meanings) from data
- Decode an object (category / box / score) from each “object query”

DETR: A Transformer-based Detector

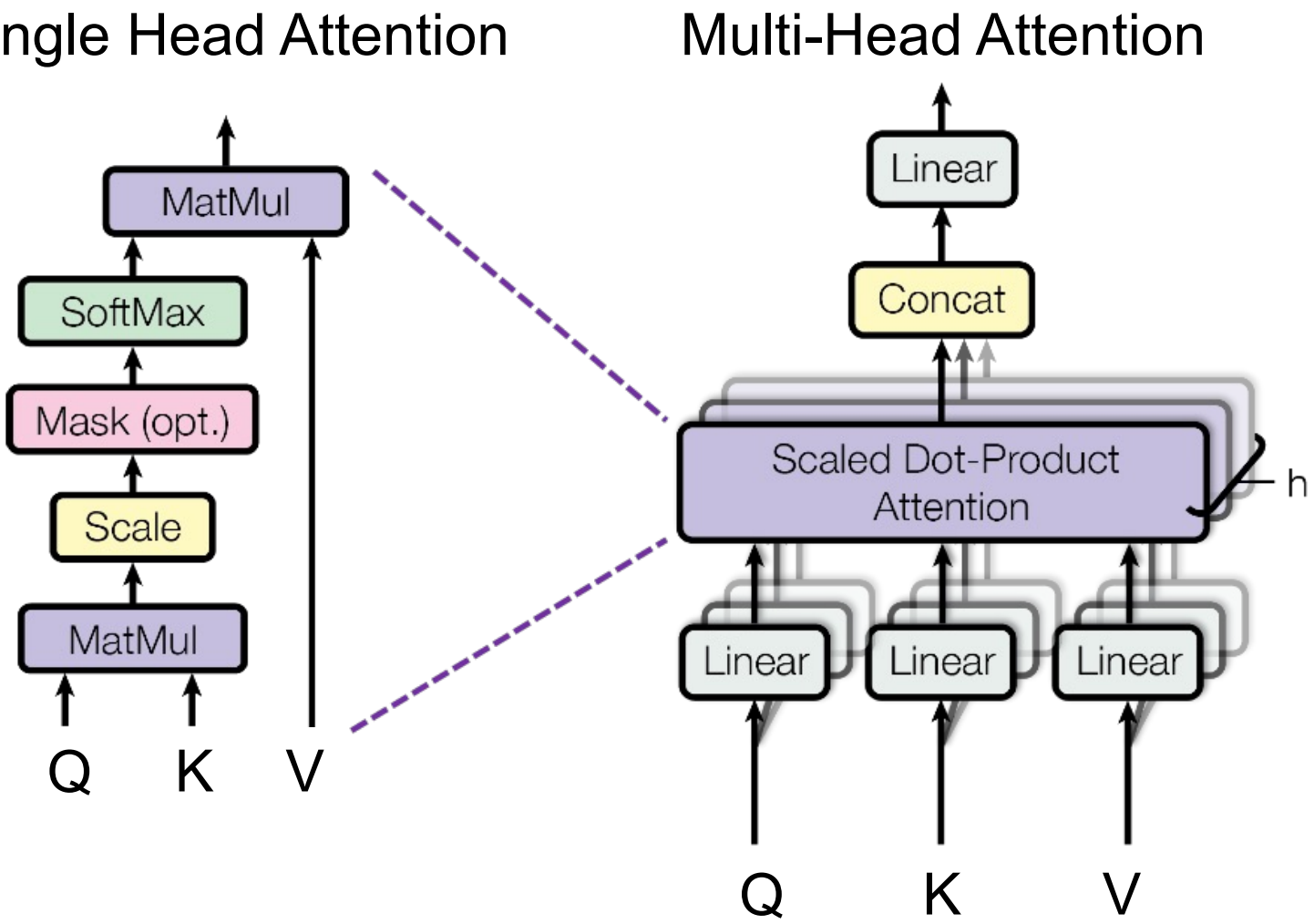
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Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., & Zagoruyko, S. (2020). End-to-end object detection with transformers. In *ECCV*.

DETR: Self Attention & Cross Attention

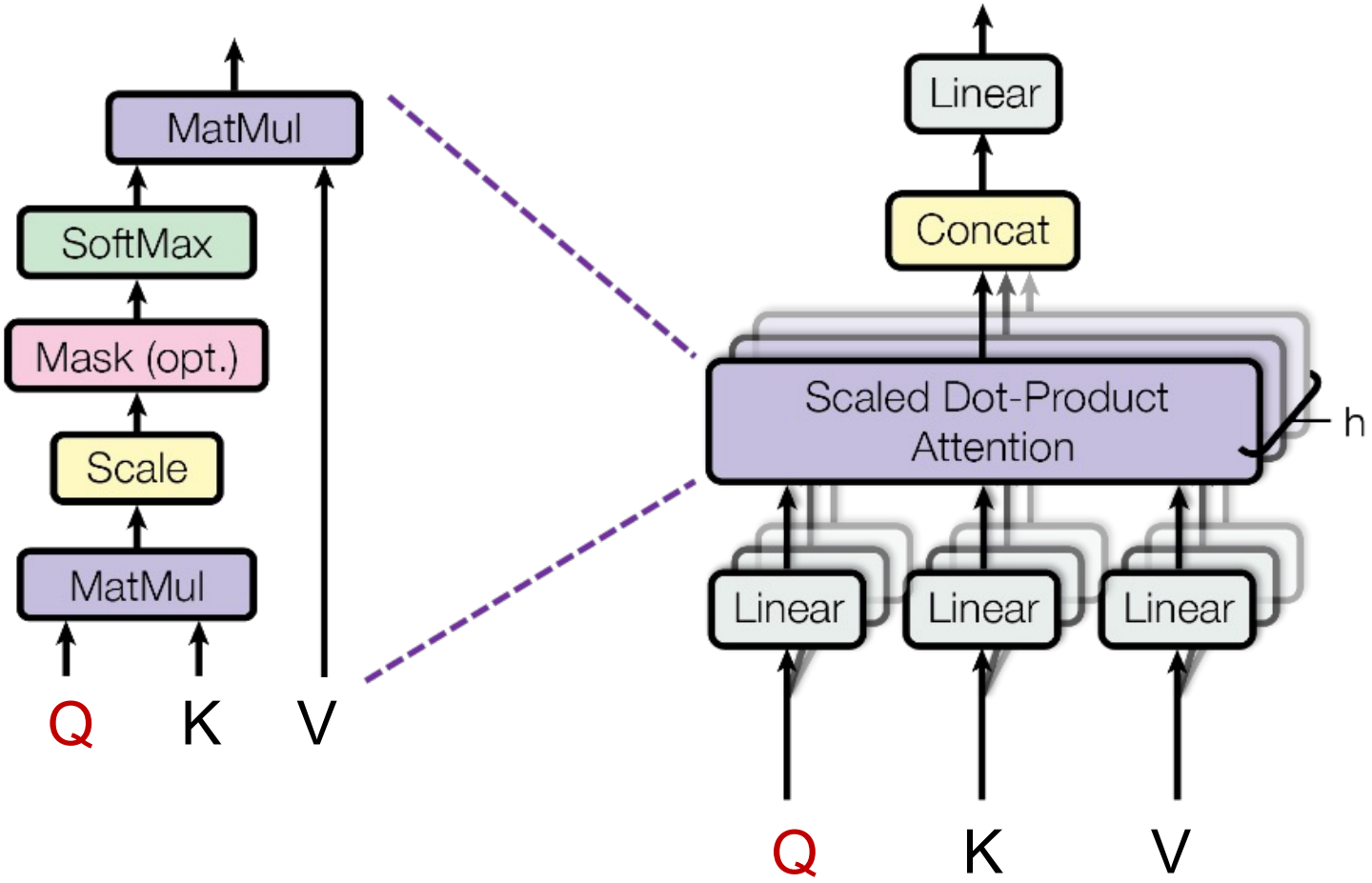


Query $Q = W_Q^T X$ **K**ey $K = W_K^T X$ **V**alue $V = W_V^T X$

DETR: Self Attention & Cross Attention

Single Head Attention

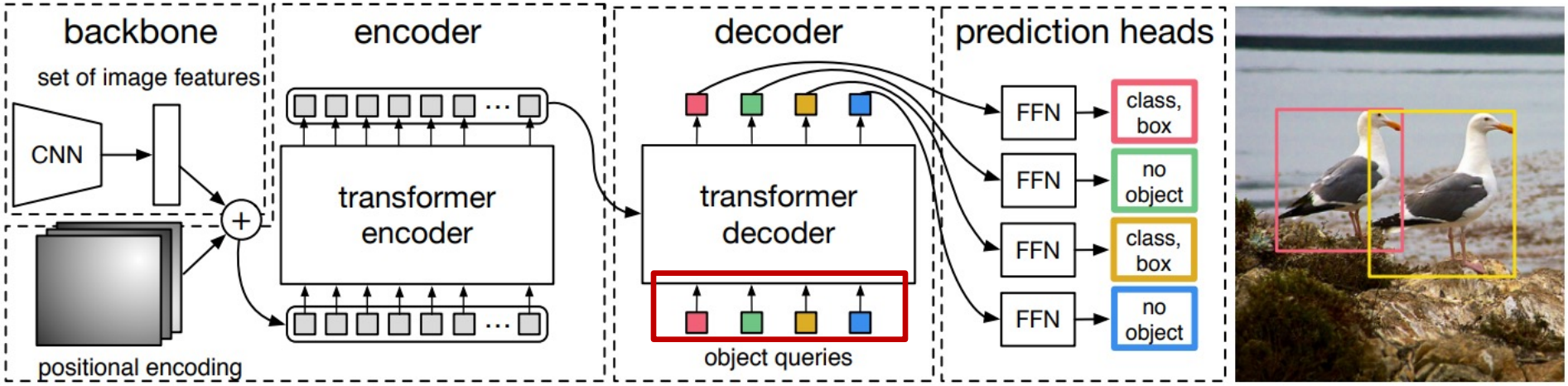
Multi-Head Attention



Query $Q = W_Q^T \hat{X}$ **K**ey $K = W_K^T X$ **V**alue $V = W_V^T X$

DETR: A Transformer-based Detector

- Learn a set of “object queries” (may not have geometric meanings) from data
- Decode an object (category / box / score) from each “object query”
- Training involves a bipartite matching between prediction and ground-truth boxes



Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., & Zagoruyko, S. (2020). End-to-end object detection with transformers. In *ECCV*.

DETR: A Transformer-based Detector

Benefits

- Eliminate the need for hand-crafted representations (e.g., anchors, center-based boxes)
- Can achieve competitive results with modern object detectors (e.g., faster RCNN)
- Can be extended to support instance segmentation

Key Details

- Formulate the problem a set prediction (yet the number of elements are fixed)
- Training requires matching between prediction and ground-truth boxes
- “Queries” capture object location and appearance

Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., & Zagoruyko, S. (2020). End-to-end object detection with transformers. In *ECCV*.

Summary: Object detection with DNNs

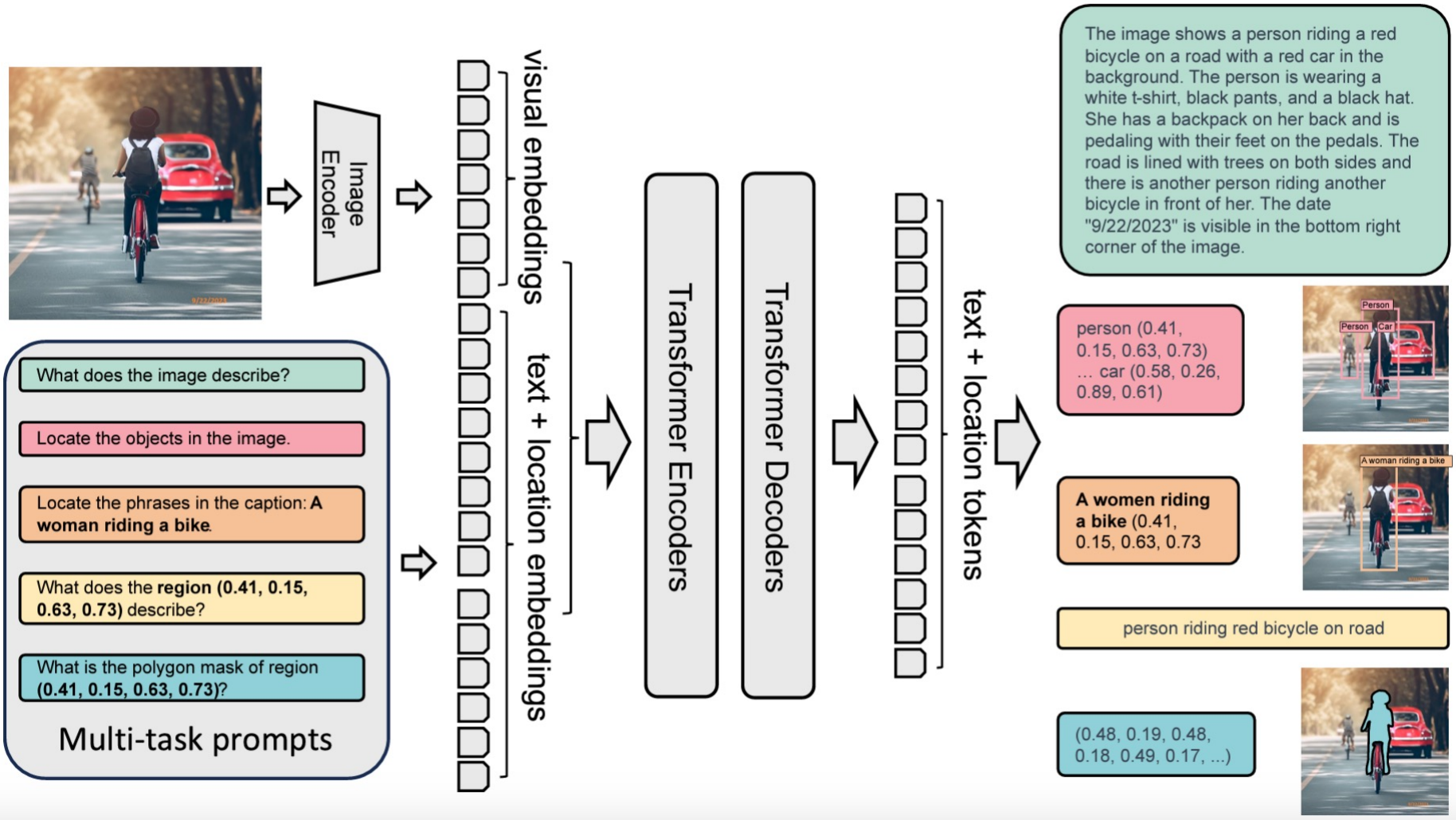
- Two-stage detectors with proposals
  - Image -> proposals (object candidates) -> objects
  - RCNN family (R-CNN, Fast/Faster/Mask R-CNN)

- Single-stage detectors without proposals
  - Image -> objects
  - YOLO / RetinaNet / FCOS (dense labeling of feature positions)
  - DETR (set prediction)
  - See also single-stage instance segmentation [Bolya et al. 2019]

Previous generation of detectors

	AP	AP^{50}	AP^{75}	AP^{S}	AP^{M}	AP^{L}	AR^{1}	AR^{10}	AR^{100}	AR^{S}	AR^{M}	AR^{L}	date
Noah CV Lab (Huawei)	0.588	0.766	0.649	0.407	0.616	0.720	0.418	0.700	0.747	0.591	0.780	0.875	2020-06-11
mmdet	0.578	0.770	0.637	0.399	0.605	0.706	0.414	0.690	0.736	0.577	0.768	0.861	2019-10-04
DeepAR(ETRIxKAIST_AIM)	0.553	0.746	0.609	0.378	0.583	0.668	0.403	0.674	0.724	0.555	0.755	0.859	2019-10-04
DetectoRS	0.550	0.736	0.604	0.377	0.578	0.669	0.401	0.678	0.730	0.565	0.761	0.860	2020-05-28
KiwiDet2	0.547	0.728	0.597	0.362	0.576	0.685	0.401	0.680	0.733	0.552	0.768	0.878	2020-05-29
360 AI Research	0.546	0.737	0.600	0.369	0.583	0.674	0.398	0.665	0.716	0.534	0.749	0.862	2020-04-01
ZFTurbo	0.544	0.728	0.600	0.367	0.579	0.671	0.400	0.657	0.699	0.532	0.735	0.835	2020-06-09
Hyundai Mobis AD Lab	0.538	0.721	0.591	0.343	0.571	0.675	0.398	0.646	0.681	0.494	0.718	0.830	2019-10-17
ByteDance_VC	0.534	0.723	0.589	0.351	0.572	0.663	0.396	0.653	0.697	0.510	0.744	0.849	2019-09-26
ResNeSt	0.533	0.720	0.580	0.351	0.562	0.668	0.397	0.649	0.687	0.512	0.714	0.826	2020-05-11
Innovation	0.532	0.735	0.582	0.360	0.565	0.652	0.397	0.646	0.684	0.513	0.715	0.818	2019-10-02
Chuck	0.527	0.704	0.583	0.358	0.553	0.647	0.393	0.652	0.705	0.536	0.735	0.839	2020-05-27
xiaoming12	0.527	0.715	0.581	0.347	0.558	0.647	0.392	0.649	0.690	0.512	0.726	0.835	2019-10-02
VLAB	0.527	0.713	0.581	0.335	0.565	0.661	0.391	0.651	0.700	0.511	0.734	0.848	2019-10-09
Megvii (Face++)	0.526	0.730	0.585	0.343	0.556	0.660	0.391	0.645	0.689	0.513	0.727	0.827	2017-10-05

Current generation of detectors



Xiao, B., Wu, H., Xu, W., Dai, X., Hu, H., Lu, Y., ... & Yuan, L. (2024). Florence-2: Advancing a unified representation for a variety of vision tasks. In *CVPR*.